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L.E



Fighter

Fight For Your Dream

All What You Need

CONTENT:

- DEPARTMENT DEPARTMENT BANK
- PREVIOUS EXAMS

PHYSIOLOGY BOOKLET

2026



Index

Topics

#	Topic	Page
1	Homeostasis	4
2	Transport across cell membrane (Diffusion)	13
3	Body fluids	13
4	RMP	27
5	Action potential	32
6	Propagation and excitation	38
7	Local action + Excitability factors	39
8	Strength-Duration Curve / Excitability and types of nerves	45
9	Nervous system - reflex action & ANS	61
10	Autonomic ganglia (and sympathetic)	66
11	Sympathetic NS	73
12	Parasympathetic NS	79

Banks

Part	Page
Department Bank (General & Nerve)	51
↳ Dr. Hany Questions	55
Department (Autonomic Nervous System)	82
Dr. Sanad Questions	93
Practical	104

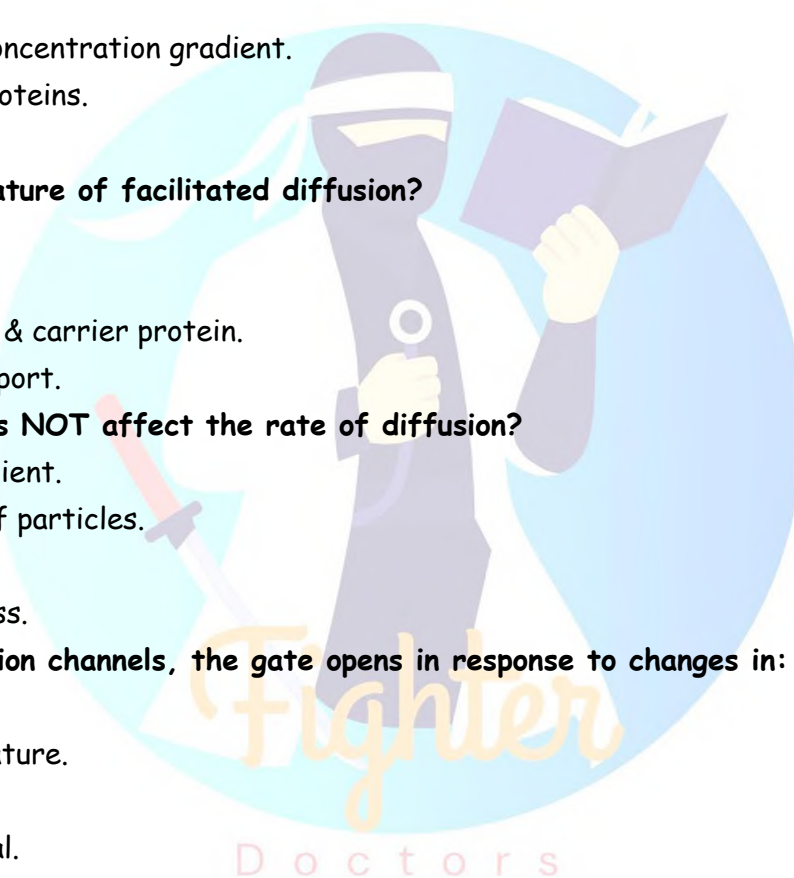
Exams

#	Exam	Page
1	ANS exam 1	110
2	ANS exam 2	112
3	Formative exam (Group A & B)	114
4	Formative exam 2020	114
5	End module 2020 (Group A)	117
6	End module 2020 (Group B)	118
7	End semester 2019 (First round)	119
8	End semester 2019 (Second round)	121
9	End semester 2021	123
10	End semester 2022 (Second round)	124
11	Formative exam 2025	127
12	End module exam 2025	131
13	End semester exam 2025	136
14	End module 2023	138
15	End module 2024	146
16	End semester 2024	151
17	End-module 2025 (أهلية)	153
18	End semester exam KSU 2026	161

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1. Hemostasis

1. Which of the following best describes role of Na^+/K^+ pump maintaining cellular homeostasis?
 - A) Generates energy of ATP.
 - B) Maintaining intracellular Na^+ and K^+ concentration.
 - C) Facilitates passive diffusion of ions.
 - D) Move glucose across the membrane.
 2. Which statement is true about passive transport?
 - A) It requires energy in the form of ATP.
 - B) It is specific to ions only.
 - C) It moves substances down their concentration gradient.
 - D) It always involves the use of carrier protein.
 3. What characterizes simple diffusion?
 - A) Is not saturable.
 - B) Occurs against a concentration gradient.
 - C) Requires carrier proteins.
 - D) Requires ATP.
 4. What is a key feature of facilitated diffusion?
 - A) Active process.
 - B) Requires vesicles.
 - C) Contain ion channel & carrier protein.
 - D) Non-specific transport.
 5. Which factor does NOT affect the rate of diffusion?
 - A) Concentration gradient.
 - B) Molecular weight of particles.
 - C) ATP availability.
 - D) Membrane thickness.
 6. In voltage-gated ion channels, the gate opens in response to changes in:
 - A) Ion concentration.
 - B) Membrane temperature.
 - C) ATP levels.
 - D) Membrane potential.
 7. What does Fick's law of diffusion describe?
 - A) Rate of diffusion.
 - B) Rate of carrier protein binding.
 - C) Rate of active transport.
 - D) Rate of osmosis.
 8. Which ion channel is controlled by chemical binding of a substance?
 - A) Voltage-gated.
 - B) Leak channels.
 - C) Ligand-gated.
 - D) Gated aquaporins.
 9. Which of the following is an example of secondary active transport?
 - A) Glucose transport using Na^+ gradient.
- 



- B) Osmosis.
- C) Ca^{++} pump.
- D) $\text{Na}^{+}\text{-K}^{+}$ pump.

10. What is the scientific term for "cell eating"?

- A) Pinocytosis.
- B) Phagocytosis.
- C) Endocytosis.
- D) Exocytosis.

11. What energy source is used in secondary active transport?

- A) Glucose gradient.
- B) Electrochemical gradient of another ion.
- C) ATP directly.
- D) Light energy.

12. From the following structure (intestinal epithelial cell transport diagram) — which molecule is transported by primary active transport:

- A) 2.
- B) 4.
- C) 3.
- D) 1.

13. From the same structure — which molecule is transported by facilitated diffusion?

- A) Potassium.
- B) Sodium.
- C) Calcium.
- D) Glucose.

14. The cell membrane is made up of proteins and lipids:

- A) The head of the phospholipid molecule is hydrophilic.
- B) The tails are formed of fatty acids which are hydrophobic.
- C) Proteins do not form a continuous layer.
- D) All of the above are correct.

15. Which of the following affects the rate of diffusion?

- A) Pressure difference across the membrane.
- B) Permeability of the membrane.
- C) Molecular weight of the substance.
- D) All of the above are correct.

16. Which of the following increases the rate of diffusion?

- A) Increasing pressure difference across membrane.
- B) Decrease of permeability.
- C) Increase of molecular weight of substance.
- D) All of the above are correct.

17. Simple diffusion:

- A) Has a maximum level of saturation.
- B) Is less sensitive to temperature change.
- C) Needs energy.
- D) Needs a carrier.

18. Concerning facilitated diffusion all the following are correct EXCEPT:

- A) Diffusion of glucose is an example.
- B) Is not affected by fever.
- C) Has a V_{max} level.
- D) Competitive inhibition.

19. Concerning facilitated diffusion, all the following are correct EXCEPT:

- A) For lipid soluble and small lipid insoluble substances.
- B) Needs a carrier.
- C) Has a maximum limit of saturation.
- D) Does not need energy.

**20. Osmosis:**

- A) Occurs against concentration gradient (Passive).
- B) Needs a carrier.
- C) Consumes more energy.
- D) None of the above is correct.

21. Osmosis is one of the special ways of diffusion and can be defined as the difference in:

- A) Gradient concentration of water.
- B) Hydrostatic pressure of solutes.
- C) Pressure.
- D) None of the above is correct.

22. Glucose is transported by:

- A) Simple diffusion.
- B) Facilitated diffusion.
- C) Osmosis.
- D) Both A and B are correct.

23. The rate of diffusion of a particle across a membrane will increase if:

- A) The area of the membrane decreases.
- B) The thickness of the membrane increases.
- C) The size of a particle increases.
- D) The lipid solubility of the particle increases.

24. The rate of diffusion of a substance across the cell membrane is inversely proportional to:

- A) Concentration gradient for the substance.
- B) Diffusion coefficient.
- C) Surface area available for diffusion.
- D) Thickness of the membrane.

25. Facilitated diffusion:

- A) Is more sensitive to temperature than simple diffusion.
- B) Is used for ion transport.
- C) Can be controlled by voltage change of membrane potential.
- D) Has no maximum rate for diffusion.

26. Protein channels are:

- A) Used for glucose and amino acid transport.
- B) Non-gated, non-selective channels.
- C) Ligand gated and have a specific receptor for ligand.
- D) Opened by acetylcholine which binds to inner gate.

27. Ligand-gated channels are characterized by:

- A) They allow diffusion of large protein molecules.
- B) Are continuously opened.
- C) They open when a hormone or neurotransmitter binds to the receptor bound to the channel.
- D) They open or close when the cell membrane potential changes (Voltage).

28. All of the following transport processes need a carrier, EXCEPT:

- A) Na-glucose co-transport.
- B) Simple diffusion.
- C) Facilitated diffusion.
- D) Primary active transport.



29. As regards carrier mediated transport, all are true EXCEPT:

- A) Can be utilized for active transport or facilitated diffusion.
- B) Exhibit competitive inhibition.
- C) Not sensitive to temperature.
- D) Energy can be supplied directly from ATP, indirectly, or not used at all.

30. Concerning the cell membrane, all are true EXCEPT:

- A) The cell membrane is freely permeable.
- B) It is of importance in polarizing the cell membrane.
- C) Maintains size and shape of cell.

31. Osmosis:

- A) Movement of solvent across a semi-permeable membrane from an area where the hydrostatic pressure is high to an area of low hydrostatic pressure.
- B) Movement of a solute across a semi-permeable membrane from an area of high solute concentration to an area of low solute concentration.
- C) Movement of a solvent across a semi-permeable membrane from an area of low solute concentration to an area of high solute concentration.

32. Integral proteins in the cell membrane perform all the following functions EXCEPT:

- A) Some act as carriers for facilitated diffusion.
- B) Act as co-enzyme to modify cellular processes.
- C) Some act as receptors for hormones and neurotransmitters.
- D) Form channels for diffusion of water-soluble substances.

33. Concerning diffusion through the cell membrane:

- A) Occurs from an area of lower concentration to an area of higher concentration till even distribution.
- B) It occurs through lipid bi-layers or protein channels.
- C) A, B.
- D) None of the above.

34. With respect to facilitated diffusion:

- A) It occurs through a carrier system.
- B) It is a type of active transport.
- C) It is for relatively permeable substances.
- D) None of the above.

35. The facilitated diffusion differs from simple diffusion in being:

- A) Absolutely directly related to concentration gradient of the diffusing substances.
- B) Have a maximum rate of diffusion once the carrier is saturated.
- C) (A) and (B) are true.
- D) None of the above.

1-B, 2-C, 3-A, 4-C, 5-C, 6-D, 7-A, 8-C, 9-A, 10-B, 11-B, 12-D, 13-D, 14-D, 15-A, 16-B, 17-B, 18-B, 19-A, 20-D, 21-A, 22-B, 23-D, 24-D, 25-A, 26-C, 27-C, 28-B, 29-C, 30-A, 31-C, 32-B, 33-B, 34-A, 35-B



Source 2

1. The term homeostasis is derived from Greek meaning:

- A) Constant pressure
- B) Same state
- C) Stable metabolism
- D) Internal energy

2. Definition of homeostasis refers to:

- A) Ability to maintain internal stability despite external changes
- B) Maintenance of fixed temperature
- C) Constant energy level
- D) Preventing external stress

3. Internal environmental factors that must be maintained include all EXCEPT:

- A) Concentration of nutrients
- B) O₂ and CO₂ levels
- C) Hydrogen ion concentration
- D) Hair color

4. Concentration of hydrogen ions must be constant to maintain:

- A) Enzyme activity
- B) Bone strength
- C) Blood volume
- D) Nervous reflexes

5. Concentration of salts (NaCl) in internal environment helps:

- A) Maintain cell volume
- B) Regulate hormone secretion
- C) Store glycogen
- D) Produce antibodies

6. Level of body temperature is important because:

- A) Enzyme activity depends on temperature
- B) It affects DNA replication
- C) It changes bone density
- D) It controls digestion only

7. Role of body systems in homeostasis includes:

- A) Providing nutrients
- B) Removing wastes
- C) Regulating body functions
- D) All of the above

8. The lungs primarily:

- A) Provide oxygen and remove CO₂
- B) Digest proteins
- C) Secrete hormones
- D) Filter blood

9. Kidneys help homeostasis by:

- A) Regulating electrolytes and pH





- B) Controlling body movements
- C) Producing antibodies
- D) Secreting enzymes only 1B 2A 3D 4A 5A 6A 7D 8A 9A 10 A

10. The nervous system controls homeostasis by:

- A) Electrical signals for rapid responses
- B) Hormone release
- C) Digestion of nutrients
- D) Blood filtration

11. The endocrine system regulates homeostasis by:

- A) Hormones in blood
- B) Neural signals
- C) Reflex arcs
- D) Digestive enzymes

12. Homeostatic mechanism is:

- A) A process that counters internal changes
- B) A muscular reflex
- C) A type of respiration
- D) A blood clotting process

13. Homeostatic mechanisms operate by:

- A) Counteracting disturbances
- B) Enhancing external changes
- C) Eliminating receptors
- D) Blocking signal pathways

14. Advantages of homeostatic mechanisms include:

- A) Increased survival
- B) Independent life
- C) Brain protection
- D) All of the above

15. The main priority in homeostatic control is:

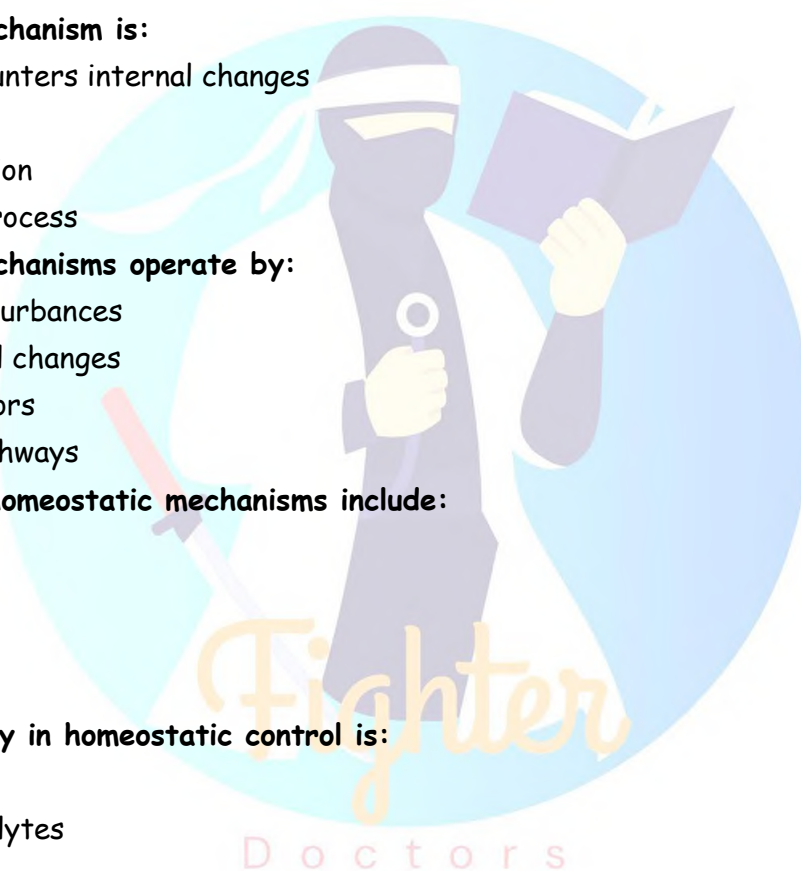
- A) Temperature
- B) Water and electrolytes
- C) Acid-base balance
- D) Blood sugar

16. Excessive sweating in heat illustrates:

- A) Temperature regulation over electrolyte balance
- B) Water retention priority
- C) Positive feedback
- D) Homeostatic failure

17. Efficiency of homeostasis improves with:

- A) Higher evolutionary rank
- B) Lower temperature
- C) Smaller body size
- D) None





18. Infants are more affected by dehydration because:

- A) Their mechanisms are immature
- B) They have more muscle
- C) They sweat less
- D) They store more fat

19. In old age, homeostatic efficiency:

- A) Decreases B) Increases
- C) Unchanged D) Doubles 11 A 12 A 13 A 14 D 15 A 16 A 17 A 18 A 19 A 20 A

20. Sex differences in homeostasis efficiency are:

- A) None
- B) Significant
- C) Temperature-dependent
- D) Genetic only

21. Failure of homeostasis results in:

- A) Disease or death
- B) Growth
- C) Reproduction
- D) Adaptation

22. Homeostatic mechanisms are limited because:

- A) Cannot fully prevent changes
- B) Work infinitely
- C) Never fluctuate
- D) Act only in sleep

23. The receptor in homeostasis:

- A) Detects change and sends signal
- B) Executes response
- C) Stores ions
- D) Generates hormones

24. Afferent pathway carries:

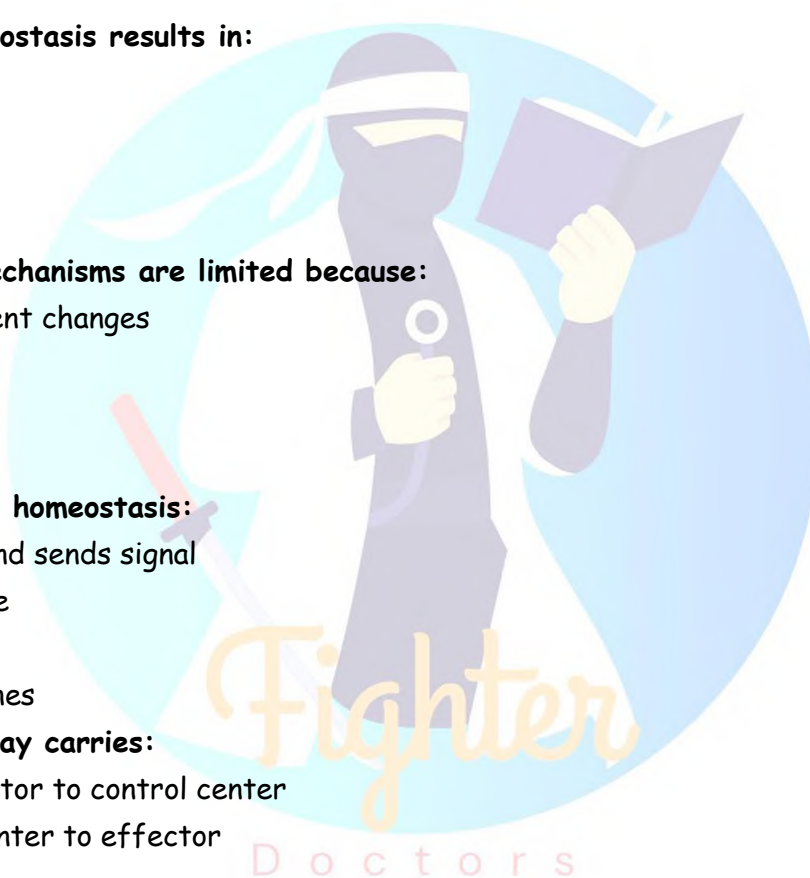
- A) Signal from receptor to control center
- B) Command from center to effector
- C) Waste to kidneys
- D) Hormones to target

25. Control center in homeostasis:

- A) Compares input to set value
- B) Acts as receptor
- C) Acts as effector
- D) Stores water

26. Efferent pathway transmits:

- A) Commands from center to effector
- B) Sensory input
- C) Hormones only
- D) Electrolytes





27. Effector organ function is to:

- A) Carry out corrective response
- B) Detect environmental change
- C) Compare signals
- D) Produce ATP

28. Feedback is:

- A) The response of system to change
- B) Electrical impulse only
- C) Reflex arc
- D) Structural mechanism

29. Negative feedback acts to:

- A) Oppose the stimulus B) Enhance change
- C) Create instability D) Cause destruction 21 A 22 A 23 A 24 A 25 A 26 A 27 A 28 A 29 A 30 A

30. Positive feedback acts to:

- A) Amplify the change B) Reverse stimulus
- C) Stabilize system D) Maintain equilibrium

31. Blood pressure regulation is an example of:

- A) Negative feedback B) Positive feedback
- C) Feed-forward D) None

32. Childbirth contractions are an example of:

- A) Positive feedback B) Negative feedback
- C) Adaptation D) Control failure

33. Most homeostatic systems depend on:

- A) Negative feedback
- B) Positive feedback
- C) Reflex action
- D) None

34. Positive feedback may lead to:

- A) Instability or death
- B) Stability
- C) Reversal of change
- D) Normal balance

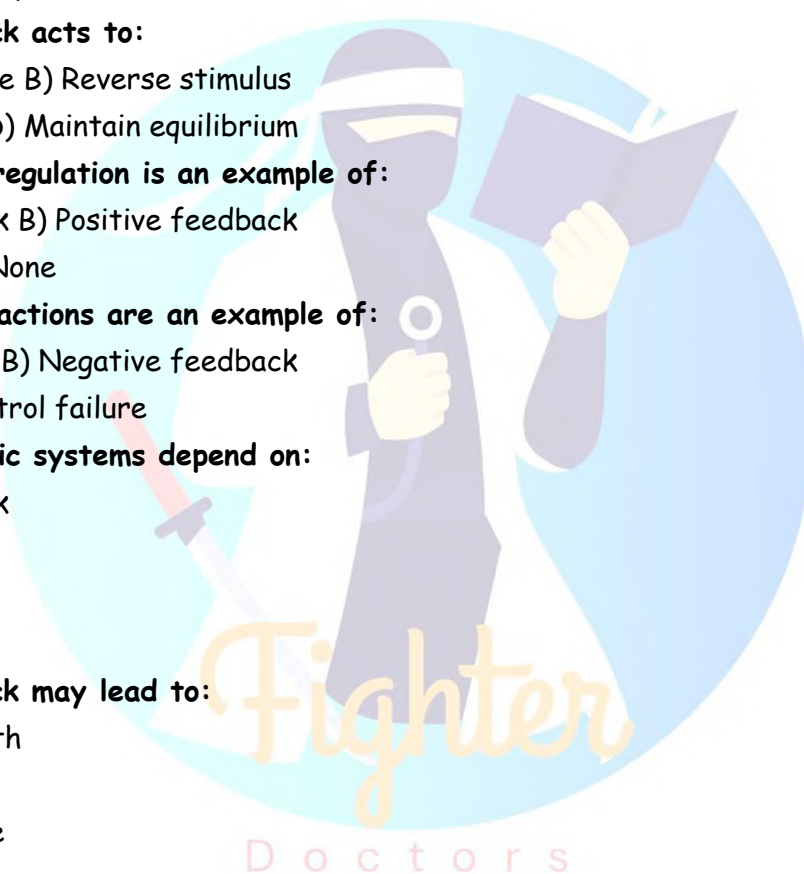
35. Blood clotting is an example of:

- A) Positive feedback
- B) Negative feedback
- C) Neutral mechanism
- D) Transport control

36. Receptor sensitivity affects:

- A) Efficiency of homeostatic response
- B) Strength of effector muscle
- C) Brain temperature
- D) Heart rhythm

37. Set value means:





- A) Optimal or normal reference
- B) Random level
- C) Body size
- D) None

38. Homeostatic fluctuation 'hunting' means:

- A) Small oscillation around set value
- B) Extreme variation
- C) Failure of control
- D) Constant output

39. Each cell contributes to survival by:

- A) Maintaining internal stability B) Producing CO_2
- C) Absorbing nutrients only D) Performing random activity

40. Nearly all diseases represent:

- A) Failure of homeostatic mechanisms
- B) Improved adaptation
- C) Enhanced balance
- D) Stable feedback



31 A 32 A 33 A 34 A 35 A 36 A 37 A 38 A 39 A 40 A

DOCTORS



2. Transport across cell membrane (Diffusion) & Body Fluids

1. Compared to the intracellular fluid, the extracellular fluid has a high concentration of

- A) Protein
- B) Phosphate
- C) Calcium
- D) Magnesium
- E) Potassium

2. Oedema is due to low colloid osmotic pressure of plasma and is caused by

- A) Heart Failure
- B) Arteriolar Dilation
- C) Liver Disease
- D) Venous Obstruction
- E) Inflammation

3. Drinking one liter of pure water is expected to

- A) Induce a greater increase in volume of ECF than ICF
- B) Induce a greater decrease in osmolarity of ECF than ICF
- C) Induce an increase in urine flow
- D) Induce a greater increase in volume of plasma than interstitial fluid
- E) Induce a greater decrease in osmolarity of plasma than interstitial fluid

4. The Interstitial Fluid

- A) Can be measured using insulin
- B) Is lower in volume than plasma
- C) Has a higher sodium concentration than plasma
- D) Has the same protein concentration as plasma
- E) Has slightly a higher concentration of chloride than plasma

5. The osmolarity of extracellular fluid is likely to increase by

- A) Passing hypertonic urine
- B) Infusing half isotonic NaCl solution
- C) Infusing 5% glucose solution
- D) Evaporating water from mucus membrane of respiratory system
- E) Vomitting a hypertonic gastric juice

6. Intravenous infusion of 1 liter of which of the following solutions will lead to the largest increase in ECF volume?

- A) Distilled water
- B) Isotonic NaCl
- C) Hypotonic NaCl
- D) Hypertonic NaCl
- E) Isotonic urea

7. The intracellular osmolarity is mainly due to

- A) Chloride
- B) Proteins
- C) Potassium
- D) ATP
- E) Magnesium

8. The chief difference between plasma and interstitial fluid is that plasma contains more

- A) Na
- B) K⁺
- C) Electrolytes
- D) Water
- E) Protein ions



9. Body Water

- A) Is mostly found in the interstitial compartment
- B) Is heavy water
- C) Moves between compartments by solvent drags
- D) Is the major constituent of the body
- E) Is not lost during expiration in cold weather

10. Osmolality of the extracellular fluid

- A) Is 160-300 msom/L
- B) Is not affected by glucose concentration
- C) Is more than the osmolality of the intracellular fluid
- D) Is mainly due to Na^+ salts
- E) Is greatly increased by hyperkalaemia

11. The following are major constituents of the intracellular fluid except

- A) Ca^{2+}
- B) Phosphates
- C) Chloride
- D) K
- E) Proteins

12. Movement of fluids from the interstitial space to plasma is due to

- A) Blood pressure in the capillaries
- B) Plasma Na^+ concentration
- C) Plasma K^+ concentration
- D) Plasma protein concentration
- E) Plasma pH

13. The substance that contributes most of the intracellular osmolarity is

- A) Protein
- B) Phosphate
- C) Potassium
- D) Urea
- E) Chloride

14. Which of the following combinations of substances is suitable for calculation of the interstitial fluid volume

- A) A radioactive sodium & heavy water 16
- B) Heavy water & Inulin Page
- C) Inulin & Evan's blue dye
- D) Heavy water & Evan's blue dye
- E) Evan's blue dye & Iodine labeled albumin

15. The concentration of Ca^{2+} in the ECF is

- A) 5 mg/dL
- b) 5 mmol/L
- C) 5 meq/L
- d) 2.5 meq/L
- E) 2.5 ug/dL

16. The extracellular fluid differs from the intracellular fluid in that it has

- A) A higher osmolarity
- B) No protein
- C) A lower calcium concentration
- D) A higher chloride concentration
- E) A higher magnesium concentration

**17. Plasma Osmolarity is**

- A) Determined primarily by protein concentration
- B) Determined primarily by potassium concentration
- C) Determined primarily by sodium concentration
- D) Normally higher than that of interstitial fluid osmolarity
- E) About 250 mosm/L

18. Infusion of 1 Liter of which of the following solutions will have the greatest suppressive effect on thirst sensation

- A) Distilled water
- B) Isotonic NaCl
- C) Hypotonic NaCl
- D) Hypertonic NaCl
- E) Tonic urea

19. Which of the following favors the development of oedema?

- A) An increase in colloid osmotic pressure of plasma
- B) A decrease in capillary hydrostatic pressure
- C) A decrease in interstitial fluid colloid pressure
- D) An increase in venous pressure
- E) An increase in arterial blood pressure

20. All of the following substances may be used to measure ECF except

- A) Sucrose
- B) Inulin
- C) Radioactive sodium
- D) Evan's blue dye
- E) Thiosulfate

21. The main difference between plasma and interstitial fluid is the concentration of:

- A) K⁺
- B) Cl⁻
- C) Mg²⁺
- D) Proteins
- E) Phosphates

22. With obstruction of all lymphatics from an extremity

- A) The hydrostatic force favoring filtration is increased
- B) Limit edema formation
- C) The main cause of edema decreases plasma oncotic pressure
- D) Interstitial hydrostatic pressure is markedly increased
- E) Interstitial colloid osmotic pressure is decreased

23. Intravenous infusion of 1 liter of which of the following solutions will lead to the largest increase in ECF volume?

- A) Distilled water
- B) Isotonic NaCl
- C) Hypotonic NaCl
- D) Hypertonic NaCl
- E) Tonic urea

24. The osmolarity of plasma is determined by the concentrations of

- A) Glucose, protein and urea nitrogen
- B) Glucose, sodium and urea nitrogen
- C) Sodium, calcium and protein
- D) Sodium and bicarbonate
- E) Potassium and sodium



25. Low concentration of proteins in the plasma is expected to lead to

- A) Reduced total body water
- B) Increased cardiac output
- C) Increased lymph flow
- D) Shift of fluid from the interstitium to plasma
- E) Decreased interstitial fluid hydrostatic pressure

26. A patient is given a small dose of radioactive sodium, most of this sodium will distribute in

- A) Plasma
- B) Interstitial fluid
- C) Intracellular fluid
- D) Joint fluids
- E) Adipocytes

27. What would be the osmolarity of a solution which is one mole glucose and one mole NaCl

- A) 1
- b) 2
- C) 3
- d) 4
- E) 5

28. The volume of distribution of substance X in an average adult male is found to be

3. 5 liters. This substance could be

- A) Inulin
- B) Radioactive sodium
- C) Tightly bound to plasma proteins
- D) Heavy water
- E) Sucrose

29. The colloid osmotic pressure of plasma is mostly due to

- A) Globulins
- B) Fibrinogen
- C) Albumin
- D) Lipoproteins
- E) Sodium

30. The solution that is expected to induce the greatest increase in plasma volume is:

- A) Normal saline
- B) Hypotonic saline
- C) Albumin solution
- D) 5% glucose
- E) Pure water

31. The volume of distribution of which of the following substances is lower than the extracellular fluid volume

- A) Sucrose
- B) Inulin
- C) Evan's blue dye
- D) Heavy water
- E) Radioactive sodium

32. The concentration of Ca^{2+} in the ECF is

- A) 20 mg/dL
- b) 5 meq/L
- C) 5 mmol/L
- d) 2.5 meq/L
- E) 10 ug/dL

33. In a normal 70 kg adult male

- A) Total body water is approximately 60 liters
- B) About two thirds of total body water is intracellular
- C) Most extracellular water is intravascular
- D) Intracellular fluid is about 12 liters
- E) Interstitial fluid volume is 5% of total body water

34. Oedema may result from all the following except

- A) Arteriolar vasoconstriction
- B) Venous obstruction
- C) Decrease in the oncotic pressure



- D) Increase in the interstitial fluid protein concentration
- E) Increase in capillary hydrostatic pressure

35. The osmolarity of plasma

- A) Is lower than interstitial fluid osmolarity
- B) Is higher than intracellular fluid osmolarity
- C) Is determined mainly by plasma proteins
- D) Is determined mainly by sodium
- E) Ranges from 300 to 310 mosm/l

36. Which of the following decreases plasma osmolarity?

- A) Forming 1 liter of hypertonic urine 19
- B) Secreting 1 liter of sweat (150 mosml/liter) Page
- C) Loosing 1 liter water in expired air
- D) Vomitting 1 liter of gastric fluid (300 mosml/liter)
- E) Bleeding 1 liter of blood Exam no 1 in A.N.S MCQ (15 marks):

37. Which of the following changes in extracellular ion concentrations would expect to hyperpolarize nerve membrane?

- A) Increased K⁺ concentration.
- B) Decreased K⁺ concentration.
- C) Increased Na⁺ concentration.
- D) Decreased Cl⁻ concentration.

38. Which one of the following is true regarding volume of distribution (Vd) of a certain drug?

- A) Apparent volume
- B) It is high for drugs that extensively bind to plasma proteins
- C) It is Low for drugs that highly bind to tissue proteins
- D) Drug distributed in total body fluids if (vd) equals 3L for average individual of body weight 70KG.
- E) Drug distributed in plasma if (Vd) equals 20 L for average individual of body weight 70KG.

39. (:Adrenergic receptors are classified into القصر (

- A) α₁ receptors which cause inhibition of adenylyl cyclase
- B) β₁-receptors which increase the cardiac properties through increased cAMP
- C) α₂- receptors which increase intracellular calcium ions
- D) β₂ receptors which cause contraction of the bronchial muscles

40. (Acting as an antibody b- Contributing -The most important function of albumin is: القصر (Responsible for viscosity of blood

- A) b)
- C) d)

Miscellaneous (Pharmacology/Extra)

1. potassium pump actively transports:

- A) 2Na⁺ in, 3K⁺ out
- B) 3Na⁺ out, 2K⁺ in
- C) 3Na⁺ in, 2K⁺ out
- D) 2Na⁺ out, 2K⁺ in

2. Which of the following molecules is least likely to diffuse directly through the lipid bilayer?



- A) Carbon dioxide (CO_2)
- B) Nitrogen (N_2)
- C) Oxygen (O_2)
- D) Glucose

3. The ratio of body water to body weight is highest among

- A) Average adult males
- B) Obese adult males
- C) Females
- D) Newborn children
- E) Pregnant women

4. A lesion in the cervical symp. chain causes:

- A) Increased sweating.
- B) ptosis.
- C) Vasoconstriction of the skin blood vessels
- D) Mydriasis.

5. The major difference between somatic & autonomic N.S. is: Exam no 2 in A.N.S

- A) Only somatic nerves innervate muscle cells
- B) Both of them release acetylcholine at effector organs
- C) Somatic nerves only originate from thoracolumbar segments of spinal cord
- D) Only autonomic nerves supply blood vessels

6. Which of the following may be produced by parasymp. stimulation?:

- A) Acceleration of the heart.
- B) Contraction of the urinary bladder wall.
- C) V.D. of skeletal muscle blood vessels.
- D) Dilatation of the pupil.

7. Vagal stimulation causes:

- A) Tachycardia.
- B) Dilatation of the bronchi and bronchioles.
- C) Inhibition of plain muscles in stomach and small intestine.
- D) Increase in gastric and pancreatic secretion.

8. The major difference between somatic & autonomic N.S. is: AUTONOMIC NERVOUS SYSTEM

- A) Only somatic nerves innervate muscle cells
- B) Both of them release acetylcholine at effector organs
- C) Somatic nerves only originate from thoracolumbar segments of spinal cord
- D) Only autonomic nerves supply blood vessels

9. Which of the following is actively transported out of the neurones?

- A) K^+ c) Cl^- .
- B) HPO_4^{--} . d) Na^+ .

10. The membrane potential will depolarize by the greatest amount if the membrane permeability increases for:

- A) Potassium.
- B) Sodium & potassium.



C) Potassium & chloride.

D) Sodium.

11. Concerning excitable tissues, all the following is true EXCEPT:

A) They spend energy during rest to maintain their membranes at a polarized state.

B) Become more excitable when partially depolarized.

C) Become more permeable to Na^+ ions when membrane potential decreases to critical level.

D) Usually respond to stimuli that decrease their membrane Na^+ permeability.

12. What is untrue concerning excitable tissues?

A) They spend energy during rest to maintain their membranes at a polarized state.

B) Become more excitable when partially depolarized.

C) Become more permeable to Na^+ ions when membrane potential decreases to firing level.

D) Usually respond to stimuli that decrease their membrane Na^+ permeability.

13. Which one of the following is false regarding dose-response curve?

A) All drugs have affinity

B) Affinity is the ability of drug to bind to the receptor

C) Intrinsic activity is the ability of the drug to produce action after binding to the receptor

D) Affinity of a drug is directly proportional to the (ED_{50})

E) The potency and intrinsic activity are independent

14. Which character of the following belongs to partial agonist?

A) Has maximum intrinsic activity

B) Has submaximal intrinsic activity

C) Produces opposite action to full agonist

D) Binds to the receptor but has no action

E) In practice potency of partial agonist is considered more important than its efficacy

15. Which one of the following is untrue regarding Pharmacokinetics?

A) Absorption c) Metabolism

B) Mechanism of action d) Distribution

E) Excretion

16. Which one of the following reactions occurs in phase- 2 of drug metabolism?

A) Oxidation c) Reduction

B) Hydrolysis d) Glucuronidation

E) Deamination

17. Which one of the following is an essential use of Tamsulosin?

A) Benign prostate hyperplasia c) Pheochromocytoma

B) Tachycardia d) Hypotension

E) Thyrotoxicosis

18. Which is true regarding atropine action?

A) Intestinal smooth muscle spasm c) Miosis

B) Diaphoresis d) Constipation

E) Spasm of accommodation

19. Which of the following can accelerate elimination of a weak acidic drug with a pK_a of 5.1?

A) Urinary excretion would be accelerated by administration of NH_4Cl , an acidifier.

B) Urinary excretion would be accelerated by giving NaHCO_3 , an alkalinizing agent.



- C) Less of the drug would be ionized at blood pH than at stomach pH.
- D) Absorption of the drug would be slower from the stomach than from the small intestine.
- E) More of the drug would be unionized fraction at small intestine than at stomach absorption sites.

20. Which of the following is true regarding Quantal Dose Response curve?

- A) Response is measured against variable drug concentration on continuous scale.
- B) Relate dose to drug concentrations.
- C) Often derived from applying multiple doses on a biological system.
- D) Relate dose and frequency response in a population.
- E) Can be applied for all pharmacological responses.

21. Which of the following can shorten the duration of action of a drug

- A) Renal dysfunction
- B) Concurrent administration of an enzyme inhibitor
- C) Concurrent administration of an enzyme inducer
- D) High tissue binding of the drug
- E) Hepatic dysfunction

22. Which one of the following statements is untrue for Vd of drugs?

- A) It can exceed the volume of water in the body
- B) Drugs with low Vd can be removed by dialysis
- C) Would be expected to be 5L if the drug is confined to the blood.
- D) Drugs with low lipid-solubility would be expected to have large Vd
- E) It can help in the calculation of the total amount of the drug in the body.

23. Which of the following causes down regulation of receptors?

- A) Continuous use of agonists b) Continuous use of antagonists
- C) Chronic use of CNS depressants d) Denervation
- E) Sudden withdrawal of drug was used for long period

24. Which of the following will be a result of biotransformation (metabolism) of a parent compounds to metabolites?

- A) Metabolites with increased polarity
- B) Metabolites with high lipid solubility
- C) Parent drug with shorter duration of action
- D) Parent compounds with increased duration of action.
- E) Parent drug with more rapid onset of action (Mcq) questions sample Excitable Tissues

25. K⁺ pump:

- A) Is inactive at rest because the membrane is absolutely impermeable to sodium ions
- B) Enhanced by hypoxia
- C) Couples Na and K⁺ pump with a ratio 1:1
- D) Causes continuous gain of positive charges outside the membrane

**Additional Questions (from supplementary question bank)**

1. Secondary active transport depends on:

- A) A pre-established ion gradient
- B) Hydrolysis of ATP by transporter protein
- C) Movement of water molecules
- D) Exocytosis of protein

2. glucose transport (SGLT)?

- A) Leak channel
- B) Symporter
- C) Antiporter
- D) Uniporter

3. In secondary active transport, the energy used to move molecules comes from:

- A) The electrochemical gradient of another molecule
- B) Light energy
- C) The breakdown of macromolecules
- D) ATP hydrolysis directly

4. What is the defining characteristic of active transport?

- A) Movement of substances down their concentration gradient
- B) Requires energy in the form of ATP
- C) Always requires channel proteins
- D) Involves movement of only water molecules

5. Which is often transported using secondary active transport?

- A) Chloride "Cl-" c) Potassium "K+"
B) Calcium "Ca+" d) Glucose

6. Active transport is necessary when a substance needs to move:

- A) Through facilitated diffusion
- B) Against concentration gradient
- C) Via simple diffusion
- D) Along electrochemical gradient

7. Which of the following is an example of primary active transport?

- A) Sodium-potassium pump (Na⁺/K⁺ ATPase)
- B) Diffusion of oxygen
- C) Glucose transport via SGLT
- D) Osmosis of water

8. Facilitated diffusion differs from simple diffusion in that it requires:

- A) Vesicles
- B) Carrier or channel protein
- C) ATP
- D) Calcium ions

9. Which of the following is the primary driving force behind passive diffusion transport across cell membranes?

- A) Vesicle formation
- B) Membrane potential



C) ATP hydrolysis

D) Concentration gradient

10. Which statement is true about passive transport?

A) It always involves the use of carrier protein

B) It moves substances down their concentration gradient

C) It requires energy in the form of ATP

D) It is specific to ions only

11. The process of Osmosis is to move water from:

A) High solute concentration to low solute concentration

B) Low water concentration to high water concentration

C) High water concentration to low water concentration

D) Low pressure to high pressure

12. What characterizes simple diffusion? (1 Point)

A) Requires ATP

B) Requires carrier proteins

C) Is not saturable

D) Occurs against a concentration gradient

13. Aquaporins are specialized for the transport of: (1 Point)

A) Water molecules c) Sodium ions

B) Glucose molecules d) Proteins

14. What is a key feature of facilitated diffusion? (1 Point)

A) Non-specific transport

B) Contain ion channel & carrier protein

C) Requires vesicles

D) Active process

15. Which factor does NOT affect the rate of diffusion? (1 Point)

A) ATP availability

B) Concentration gradient

C) Membrane thickness

D) Molecular weight of particles

16. gated ion channels, the gate opens in response to changes in: (1 Point)

A) Membrane potential

B) Membrane temperature

C) Ion concentration

D) ATP levels

17. Which type of ion channel is always open? (1 Point)

A) Leak ion channel

B) Voltage-gated channels

C) All of the above

D) Ligand-gated channels

18. What is the role of carrier proteins in facilitated diffusion? (1 Point)

A) Generate energy

B) Form pores in the membrane



C) Change shape to transport molecules

D) Create ion gradients

19. What does Fick's law of diffusion describe? (1 Point)

A) Rate of carrier protein binding

B) Rate of active transport

C) Rate of osmosis

D) Rate of diffusion

20. Osmosis occurs across a: (1 Point)

A) Non-permeable membrane

B) Fully permeable membrane

C) Partially permeable membrane

D) Rigid membrane

21. Which ion channel is controlled by chemical binding of a substance?

A) Gated aquaporins

B) Ligand-gated

C) Voltage-gated

D) Leak channels

22. What is an example of a substance diffused by a Carrier mediated protein? (1 Point)

A) Glucose

B) Potassium

C) Oxygen

D) Carbon dioxide

23. In osmosis, water moves towards: (1 Point)

A) Equal solute concentration

B) Low solute concentration

C) High solute concentration

D) High water concentration

24. Which of the following does NOT use carrier proteins? (1 Point)

A) All of the above c) Simple diffusion

B) Facilitated diffusion d) Active transport

25. What factor directly increases the diffusion rate according to Fick's law? (1 Point)

A) Increased thickness of the membrane

B) Larger concentration gradient

C) Decreased surface area

D) Smaller diffusion coefficient

26. Which of the following processes requires extra energy? (1 Point)

A) Osmosis

B) Facilitated diffusion

C) Active transport

D) Simple diffusion

27. What is the role of calcium in Exocytosis? (1 Point)

A) Accumulate products from cell

B) Help in formation of vesicle



C) Help to rupture the vesicle

D) Hardening waste products

28. What is the main characteristic of active transport? (1 Point)

A) Relies only on diffusion

B) Moves particles against the concentration or electrical gradient

C) Moves particles along the concentration gradient

D) Does not require energy

29. Which of the following statements is true about ATPase? (1 Point)

A) It helps in passive transport

B) It creates energy from glucose

C) It inhibits active transport

D) It breaks down ATP to release energy

30. What is the main energy source for primary active transport? (1 Point)

A) Glucose

B) Potassium ions

C) Sodium gradient

D) ATP hydrolysis

31. Which type of active transport depends on the concentration gradient of another ion? (1 Point)

A) Passive transport

B) Simple diffusion

C) Primary active transport

D) Secondary active transport

32. What kind of carrier protein moves two substances in opposite directions? (1 Point)

A) Passive transport 10

B) Uniport Page

C) Symport

D) Antiport

33. Vesicular transport includes all of the following EXCEPT: (1 Point)

A) Pinocytosis

B) Phagocytosis

C) Receptor-mediated endocytosis

D) Osmosis

34. What is the scientific term for "cell eating"? (1 Point)

A) Endocytosis

B) Pinocytosis

C) Exocytosis

D) Phagocytosis

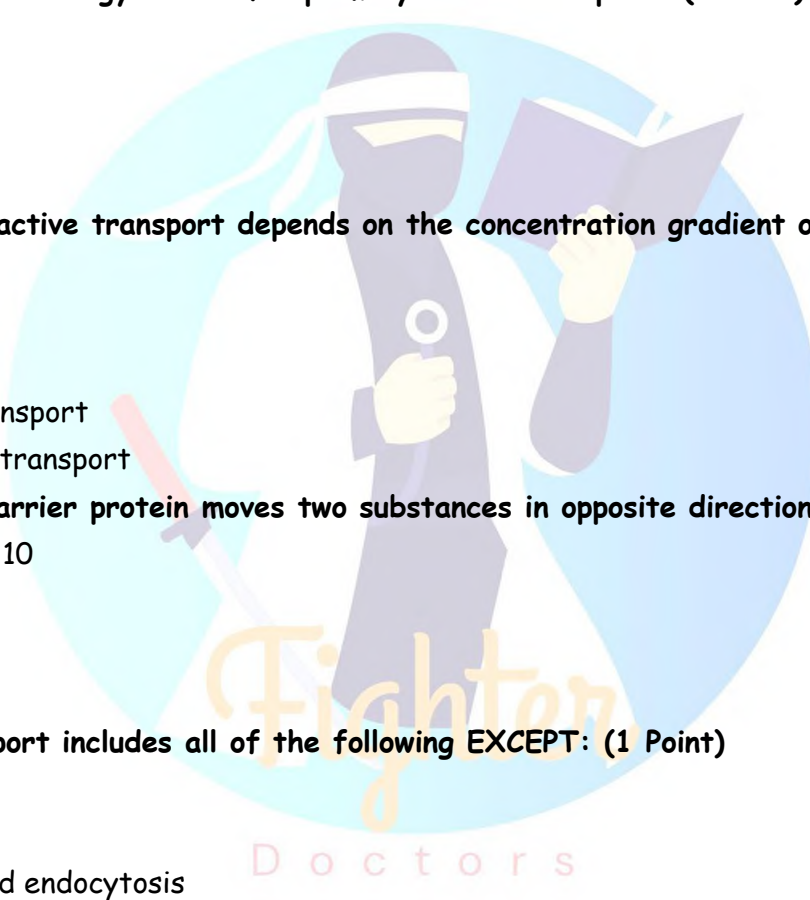
35. What is required for vesicular transport? (1 Point)

A) Calcium and energy

B) Glucose and oxygen

C) Potassium gradient

D) Only ATP





- 36. Which type of transport carrier protein moves a single substance? (1 Point)**
 A) Symport B) Uniport
 C) Antiport D) Multiport
- 37. Which form of endocytosis involves the uptake of extracellular fluid? (1 Point)**
 A) Exocytosis
 B) Phagocytosis
 C) Receptor-mediated endocytosis
 D) Pinocytosis
- 38. What is the process of removing material from the cell called?**
 A) Endocytosis B) Pinocytosis
 C) Exocytosis D) Active transport
- 39. mediated endocytosis depend on?**
 A) Specific receptors in the membrane
 B) Glucose availability
 C) ATP hydrolysis
 D) Na⁺ concentration gradient
- 40. mediated active transport?**
 A) Carrier activity Page
 B) Unlimited transport capacity
 C) Saturation
 D) Presence of competing substances
- 41. What energy source is used in secondary active transport?**
 A) Electrochemical gradient of another ion
 B) Light energy
 C) ATP directly
 D) Glucose gradient
- 42. transport?**
 A) Symport B) Uniport
 C) Both A and B D) Antiport
- 43. Which process is most commonly used by nerve cells for communication?**
 A) Exocytosis B) Pinocytosis
 C) Passive diffusion D) Phagocytosis
- 44. Which form of vesicular transport moves large molecules into the cell?**
 A) Diffusion B) Symport
 C) Endocytosis D) Exocytosis
- 45. Endocytosis & exocytosis are described by all of the following EXCEPT:**
 A) Extend Cell membrane
 B) Renew cell membrane
 C) Active processes
 D) Classified as transport process
- 46. All of the following are features of carrier protein EXCEPT:**
 A) Saturable
 B) Specificity



C) Only found in active transport

D) Show competition

47. which molecule is transported by primary active transport? (1 Point)

A) 3

B) 1

C) 2

D) 4

48. which molecule is transported by facilitated diffusion? (1 Point) Chapter 2 Review Questions

A) 1

c) 3

B) 2

d) 4

49. which molecule is transported by facilitated diffusion? (1 Point) BODY FLUIDS

A) 2

B) 1

C) 3

D) 2 & 4

50. The primary force moving water from the plasma to the interstitial space is

A) Co-transport with sodium

B) Co-transport with H⁺

C) Facilitated diffusion

D) Filtration

E) Active transport

51. In the absence of a hydrostatic pressure gradient, movement of water across a cell plasmalemmal membrane can occur only

A) Up its chemical concentration gradient

B) By primary active transport

C) By secondary active transport

D) Down its chemical concentration gradient

E) Through the lipid matrix of the plasmalemmal membrane

52. The nicotinic cholinergic receptors are

A) Not inhibited by high concentrations of nicotine

B) Inhibited by high concentrations of acetylcholine

C) Blocked by atropine

D) Ligand-gated chloride channels

E) Located in the cell nucleus

53. The termination of acetylcholine action is mainly due to

A) Reuptake by preganglionic neurons

B) Diffusion from synaptic cleft

C) Irreversible combination with cholinergic receptors

D) Uptake by postsynaptic cell

E) Cholinesterase in cholinergic synapses

54. The action of acetylcholine on its receptors is terminated mainly by

A) Choline esterase

B) Reuptake into the terminal buttons

C) Diffusion to the blood

D) Pseudocholine esterase

E) Endocytosis by the target tissue that bears its receptors



3. Resting Membrane Potential

1. The nervous system is divided into two main parts. What are they?

- A) Brain and spinal cord.
- B) Central and peripheral nervous systems.
- C) Sensory and motor nervous systems.
- D) Sympathetic and parasympathetic systems.

2. Which part of the neuron transmits impulses away from the cell body?

- A) Dendrites.
- B) Axon.
- C) Soma.
- D) Nucleus.

3. What is the characteristic feature of excitable tissue?

- A) Presence of resting membrane potential.
- B) Ability to store calcium.
- C) Production of enzymes.
- D) Secretion of hormones.

4. What is the resting membrane potential (RMP) in small nerve fibers?

- A) -70 mV.
- B) +70 mV.
- C) -30 mV.
- D) +30 mV.

5. Which ion is more concentrated on the outer surface of the cell membrane?

- A) Potassium (K⁺).
- B) Sodium (Na⁺).
- C) Proteins.
- D) Calcium (Ca²⁺).

6. Which ion is more concentrated on the inner surface of the cell membrane?

- A) Sodium.
- B) Potassium.
- C) Chloride.
- D) Bicarbonate.

7. What is the function of the sodium-potassium pump?

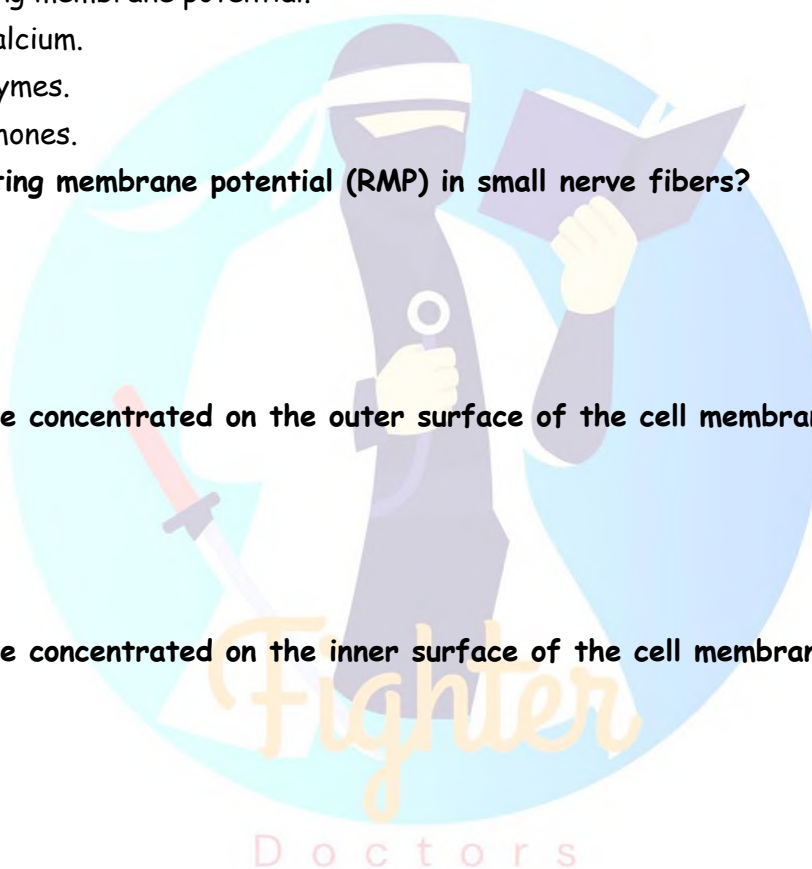
- A) Passively transport ions along their gradient.
- B) Actively transport sodium out and potassium in.
- C) Allow sodium and potassium to flow freely.
- D) Pump calcium into the cell.

8. How many times more permeable is the membrane to potassium ions compared to sodium ions?

- A) 10-20 times.
- B) 30-50 times.
- C) 50-100 times.
- D) 100-200 times.

9. What ensures selective permeability of the cell membrane?

- A) Proteins.





- B) Ion channels.
- C) Cytoplasm.
- D) Myelin sheath.

10. The sodium-potassium pump transports ions in which direction?

- A) Sodium out and potassium out.
- B) Sodium out and potassium in.
- C) Sodium in and potassium in.
- D) Sodium in and potassium out.

11. Out of the cell membrane, the upper hand (predominant ion) is:

- A) Cl^- .
- B) K^+ .
- C) Na^+ .
- D) HCO_3^- .

12. Which of the following potentials is mismatched?

- A) Inner to inner $\rightarrow$ isopotential.
- B) Inner to outer $\rightarrow$ positive potential (+70 mV).
- C) Outer to outer $\rightarrow$ isopotential.
- D) Outer to inner $\rightarrow$ positive potential (+70 mV).

13. Inside the cell membrane, the upper hand (predominant ion) is:

- A) Cl^- .
- B) K^+ .
- C) Na^+ .
- D) HCO_3^- .

14. The most participating cause of RMP:

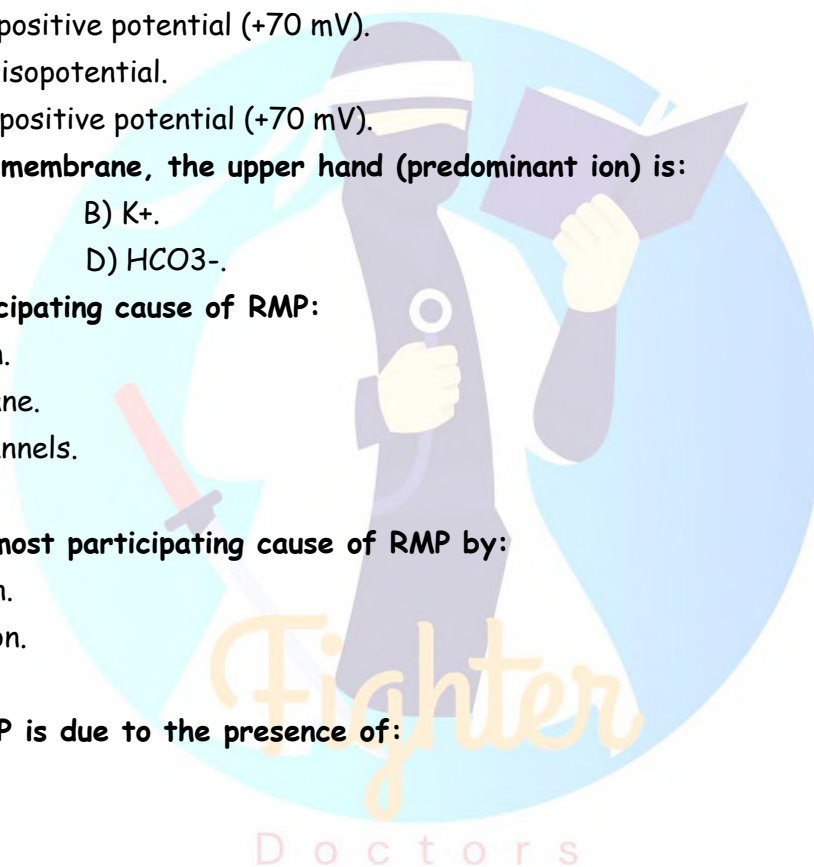
- A) Ion concentration.
- B) Selective membrane.
- C) Voltage gated channels.
- D) Na^+/K^+ pump.

15. We prove the most participating cause of RMP by:

- A) Nernst's equation.
- B) Goldman's equation.
- C) Frick's equation.

16. Maintaining RMP is due to the presence of:

- A) Potassium.
- B) Sodium.
- C) Protein.
- D) Chloride.



**Additional Questions (from supplementary question bank)**

1. What is the typical ratio of ATP used to ion movement in Na⁺/K⁺ pump?
 - A) 1ATP for 4 ion moved
 - B) 1ATP for 2 ion moved
 - C) 1ATP for 5 ion moved
 - D) 1ATP for 3 ion moved
2. Which of the following best describes the role of the Na⁺/K⁺ pump in maintaining cellular homeostasis?
 - A) Generates energy of ATP
 - B) Moves glucose across the membrane
 - C) Maintaining intracellular Na⁺ and K⁺ concentration
 - D) Facilitates passive diffusion of ions
3. Which of the following is an example of secondary active transport? (1 Point)
 - A) Na⁺-K⁺ pump
 - B) Glucose transport using Na⁺ gradient
 - C) Ca⁺⁺ pump
 - D) Osmosis
4. Which pump is considered electrogenic? (1 Point)
 - A) Na⁺-K⁺ pump
 - B) Uniport
 - C) Ca⁺⁺ pump
 - D) H⁺K⁺ pump
5. Secondary active transport and facilitated diffusion are similar in that they
 - A) Use energy
 - B) Use carriers
 - C) Tend to restore equilibrium
 - D) Transport water
 - E) Use channels
6. The sodium pump of an excitable membrane:
 - A) Pumps sodium ions to the outside during depolarization.
 - B) Pumps potassium ions to the outside during depolarization.
 - C) Pumps potassium ions to the inside during depolarization.
 - D) Pumps sodium ions to the outside in the resting state.
7. The resting membrane potential in nerve fibres is:
 - A) Normally about -10mV.
 - B) Due to K⁺ diffusion from outside to inside the nerve fibres.
 - C) Due to diffusion of intracellular protein to outside the nerve fibres.
 - D) None of the above
8. The resting potential of a nerve membrane is primarily dependent on the concentration gradient of:
 - A) Potassium.
 - B) Sodium.
 - C) Calcium.



D) Chloride.

9. As regard resting membrane potential, which of the followings is true?

- A) It is the potential difference across the fibre membrane with negatively charged ions inside and positively charged ions outside.
- B) Its decrease is accompanied with decreased excitability of nerve fibre.
- C) It is due to K^+ diffusion from outside to inside the nerve fibres.
- D) It is due to diffusion of intracellular protein to outside the nerve fibre.

10. About Na^+-K^+ pump, all the following are correct EXCEPT:

- A) Requires high energy phosphate bonds.
- B) Contributes for the creation of resting membrane potential.
- C) Keeps intracellular Na^+ low.
- D) Is explained by facilitated diffusion

11. Na^+-K^+ pump:

- A) Requires energy which is taken by GTP.
- B) Couples Na^+ and K^+ with ratio 3:2.
- C) Helps intracellular accumulation of Na^+
- D) Is a passive process needs no energy

12. Resting membrane potential of a nerve fibre:

- A) Is mainly due to sodium influx.
- B) Is mainly due to potassium influx.
- C) Is mainly due to chloride influx.
- D) Is due to the presence of negative charges on the inner surface in relation to its outer surface.

13. When a nerve fibre is stimulated, depolarization phase stops at +35 mV. This is because:

- A) Activation of the Na^+-K^+ pump that prevent accumulation of Na^+ ions inside the nerve membrane.
- B) Closing of the inactivation gates is faster the opening of activation gates.
- C) The inactivation gates rapidly close to prevent further entry of Na^+ ions.
- D) The activation gates and inactivation gates close at the same time.

14. If the resting membrane potential of a nerve cell falls (becomes less negative), there is a net:

- A) Gain of Na^+ ions into the cell.
- B) Gain of K^+ ions into the cell.
- C) Loss of protein ions from the cell.
- D) Loss of chloride ions from the cell

15. About the nerve excitability:

- A) Chronaxie means the intensity of the stimulus that produces a response
- B) An anelectrotonic response decreases the membrane potential and increases the nerve excitability.
- C) K^+ is the least permeable cation through nerve membranes.
- D) The resting potential becomes less negative if the K concentration outside the nerve membrane is doubled.

16. All the following about the electric activity in nerves is correct EXCEPT:

- A) The self-propagation of nerve impulses is due to successive depolarization to the firing level of the membrane a head of the action potential.



- B) The resting potential keeps constant after blocking of the Na^+ pump.
- C) The after-hyperpolarization phase is the last phase of the action potential.
- D) Repolarization is initiated by K^+ efflux.

17. Which of the following is true Concerning Na^+/K^+ pump?

- A) Pump 3 Na molecules inside & 2 K^+ molecules outside nerve membrane
- B) Needs high energy phosphate bond
- C) Is the main cause of RMP
- D) Pump 3 K^+ molecules inside & 2 Na^+ molecules outside nerve membrane

18. Which statement is true about RMP of a nerve?

- A) Caused by equal distribution of ions along nerve membrane
- B) Caused by selective permeability of membrane to ions
- C) Na^+/K^+ Pump has no role
- D) Caused mainly by inward movement of Na ions

19. What will happen if Na^+ concentration decreases around a nerve?

- A) Decreasing the resting potential
- B) Increases the overshoot
- C) Decreases nerve excitability
- D) Increases action potential

20. What is true about the nerve excitability among the following statements?

- A) During rest, the nerve membrane is depolarized
- B) Chronaxie is time needed to produce response on using rheobase
- C) K^+ is least permeable cation through the nerve membrane
- D) Resting potential becomes less negative if extracellular K^+ concentration is increased.

21. When a nerve fibre is stimulated, depolarization phase stops at +35 mV. This is because:

(Continued in file... Questions 11-70 on Excitable Tissues and 1-67 on A.N.S.) أسئلة من محاضرات

القصر العيني (Nerve) القصر

- A) Activation of the Na^+/K^+ pump that prevent accumulation of Na^+ ions inside the nerve membrane.
- B) Closing of the inactivation gates is faster the opening of activation gates.
- C) The inactivation gates rapidly close to prevent further entry of Na^+ ions. 23
- D) The activation gates and inactivation gates close at the same time. Page

Doctors



4. Action potential

1. Which of the following is untrue about action potential?

- A) Firing level is the level at which the spike begins.
- B) Rapid repolarization is due to increased K^+ efflux.
- C) Rapid depolarization is due to increased Na^+ influx.
- D) Absolute refractory period extends from firing level until repolarization is completed.

2. Which of the following is true concerning the Na^+/K^+ pump?

- A) Pumps 3 Na^+ molecules inside & 2 K^+ molecules outside the nerve.
- B) Membrane needs high energy phosphate bond.
- C) Is the main cause of RMP.
- D) Pumps 3 K^+ molecules inside & 2 Na^+ molecules outside the nerve membrane.

3. Which statement is true about RMP of a nerve?

- A) Caused by equal distribution of ions along the nerve membrane.
- B) Caused by selective permeability of the membrane to ions.
- C) Na^+-K^+ pump has no role.
- D) Caused mainly by inward movement of Na^+ ions.

4. Repolarization:

- A) Occurs gradually at first then becomes fast.
- B) Results from closure of sodium gates and opening of potassium gates.
- C) Is represented by the ascending limb of the spike.
- D) Is followed by appearance of a response.

5. The resting membrane potential is caused by:

- A) Diffusion of K^+ ions outside the nerve fibers.
- B) Diffusion of Na^+ ions inside the nerve fibers.
- C) Opening of the chemically activated ion channels.
- D) Opening of the voltage activated ion channels.

6. During depolarization:

- A) Voltage activated Na^+ channels open.
- B) The membrane becomes impermeable to Na^+ .
- C) When membrane potential reaches -55 mV, Na^+ & K^+ fluxes occur at the same time.
- D) K^+ ions diffuse outside.

7. What name is given to the cells in the nervous system that produce nerve impulses?

- A) Neurotransmitters.
- B) Nerves.
- C) Neurons.
- D) Neuroglia.

8. What is the threshold potential for initiating an action potential?

- A) -70 mV.
- B) -55 mV.
- C) $+35$ mV.
- D) -90 mV.

9. Resting membrane potential of a nerve:

- A) Is caused by equal distribution of ions along both sides of the membrane.
- B) Is caused by selective permeability of the membrane to the ions.
- C) Na^+-K^+ pump has no role in RMP.
- D) Is caused mainly by inward movement of Na^+ ions.

A) Ascending limb. B) Descending limb.
C) After-hyperpolarization. D) Latent period.



19. What event corresponds to the "spike potential"?

- A) Resting potential.
- B) The sum of depolarization and repolarization phases.
- C) After-hyperpolarization.
- D) Slow depolarization only.

20. The slow phase of depolarization corresponds to:

- A) From RMP to the firing level.
- B) From -55 mV to +35 mV.
- C) The descending limb.
- D) After-hyperpolarization.

21. Voltage-gated Na⁺ channels open in response to:

- A) K⁺ influx.
- B) Membrane potential reaching -55 mV.
- C) Resting potential.
- D) Hyperpolarization.

22. What initiates repolarization?

- A) Closure of K⁺ channels.
- B) Opening of K⁺ channels and closure of Na⁺ channels.
- C) Na⁺ influx.
- D) Threshold potential.

23. The latent period depends on:

- A) Distance between electrode & recording area.
- B) Impulse rate.
- C) Stimulus strength.
- D) A & B.

24. What is the purpose of the myelin sheath around an axon?

- A) To control the chemical environment around the nerve cell.
- B) To phagocytose microbes.
- C) To prevent movement of ions through the nerve cell membrane (ion insulator).
- D) To form the blood-brain barrier.

25. Which of the following describes an "action potential"?

- A) The high concentration of Na⁺ and Cl⁻ outside the cell, and of K⁺ inside the cell.
- B) The voltage change that moves along the cell membrane until it reaches the axon hillock.
- C) The movement of a neurotransmitter from the pre-synaptic membrane to the post-synaptic membrane.
- D) The movement of Na⁺ across the cell membrane into the cell, followed by the movement of K⁺ out of the cell.

26. Which of the following would conduct an action potential with the greatest speed?

- A) Myelinated, large diameter fibers.
- B) Myelinated, small diameter fibers.
- C) Unmyelinated, large diameter fibers.
- D) Unmyelinated, small diameter fibers.

27. What is meant by the "absolute refractory period" of a nerve cell membrane? The time during which:

- A) A larger than normal stimulus is required to cause an action potential.
- B) A smaller than normal stimulus will produce an action potential.
- C) No stimulus will produce an action potential.
- D) Two stimuli in quick succession are required to add up to an above-threshold stimulus.

1-D, 2-B, 3-B, 4-B, 5-A, 6-A, 7-C, 8-B, 9-B, 10-A, 11-B, 12-C, 13-B, 14-B, 15-A, 16-B, 17-C, 18-B, 19-B, 20-A, 21-B, 22-B, 23-A, 24-C, 25-D, 26-A, 27-C

**Additional Questions (from supplementary question bank)**

1. The ascending limb of spike of nerve fibre action potential is caused by:
 - A) Rapid sodium influx.
 - B) Closure of activation gates of voltage gated sodium channels.
 - C) Rapid potassium efflux.
 - D) Calcium influx.
2. In a nerve axon, which phase of the action potential is caused by the inactivation of the Na^+ channels?
 - A) Upstroke. c) Downstroke.
 - B) Absolute refractory period. d) Relative refractory period
3. The first step in nerve impulse initiation is:
 - A) Decreased Na^+ permeability. c) Increased K^+ permeability.
 - B) Membrane depolarization. d) Membrane hyperpolarization
4. An action potential has the following characteristics:
 - A) Maximal response to threshold stimulus.
 - B) Slight response to threshold stimulus.
 - C) No response to threshold stimulus.
 - D) Increased response to increased stimulus
5. Regarding the voltage gated Na^+ channels:
 - A) It has two gates, activation gate near the inside of the membrane and inactivation gate near the outside.
 - B) At resting state, the activation gates are closed to prevent influx of Na^+ ions inside the nerve fibre.
 - C) At resting state, the inactivation gates are also closed.
 - D) At the threshold (firing) level, the inactivation gates open.
6. The action potential of nerve:
 - A) Is initiated by Na^+ efflux.
 - B) Is terminated by K^+ efflux.
 - C) Declines in amplitude as it moves along the axon.
 - D) Is not associated with any movement of Na^+ or K^+ across the membrane.
7. In the action potential, all the following is true EXCEPT:
 - A) The firing level is the level at which the spike potential begins.
 - B) The rapid repolarization is due to increased K^+ permeability.
 - C) The duration of the isoelectric interval (latent period) is inversely proportionate to the speed of conduction.
 - D) The absolute refractory period is the period from the firing level until repolarization is complete
8. The after hyperpolarization phase of the action potential is due to:
 - A) Inactivation of the Na^+ channels.
 - B) Increased excitability of the membranes.
 - C) Net Na^+ efflux to the exterior.
 - D) Slow return of the K^+ channels to the closed state
9. During action potential, the rapid depolarization of the membrane is due to:
 - A) Increased sodium permeability. c) Increased chloride permeability.



B) Increased calcium permeability. d) Increased potassium permeability

10. Absolute refractory period (ARP) of a nerve fibre:

- A) Follows the relative refractory period.
- B) Occurs mainly during repolarization.
- C) During which nerve excitability is relatively higher than at rest.
- D) Occurs during the time of increased permeability of the membrane to Na^+ .

11. The absolute refractory period of nerve fibre:

- A) Cover all the phases of action potential.
- B) Is when the fibre is more excitable.
- C) Occurs at the period of repolarization.
- D) Is when a stronger stimulus is required to excite the fibre.

12. Compound action potential:

- A) Obeys all or non-law. b) It can be graded.
- C) It is the action potential recorded from stimulation of a single myelinated nerve fibre by supramaximal stimulus.
- D) It is a biphasic action potential.

13. The absolute refractory period of a nerve fibre:

- A) Lasts throughout an action potential.
- B) Occurs before the relative refractory period.
- C) Is due to a low calcium concentration.
- D) Is when a stronger than normal stimulus is required to excite the fibre.

14. Concerning the voltage gated sodium channels:

- A) It has an activation gate inside and an inactivation gate outside.
- B) Both gates are closed at rest.
- C) At the firing level, the activation gates opens.
- D) At the top of the spike of action potential, the activation gate closes.

15. Saltatory (Jumping) conduction of action potential: Formative Module 3 - First Level

- A) Occurs in unmyelinated nerve fibres.
- B) Is slower than the point to point conduction.
- C) Consumes less energy than point to point conduction. 26
- D) None of the above. Page

16. Which of the following is untrue about action potential?

- A) Firing level is the level at which spike begins
- B) Rapid repolarization is due to increased K^+ efflux
- C) Rapid depolarization is due to increased Na^+ influx
- D) Absolute RF extends from firing level until repolarization is completed

17. What is true regarding The absolute refractory period of the nerve fiber (ARP)?

- A) Lasts throughout the action potential
- B) Follows the relative refractory period
- C) Occurs during increased membrane permeability to Na^+
- D) Occurs mainly during whole repolarization

18. Which statement is true about Repolarization?

- A) occurs at first gradual then becomes fast



- B) Represented by ascending limb of the spike
- C) Results from closure of Na gates and opening of K⁺ gates
- D) terminated by sodium efflux

19. What is the action potential of a neuron?

- A) Is initiated by efflux of Na⁺ ions
- B) Is terminated by efflux of K⁺ ions
- C) Declines in amplitude as it moves along the axons
- D) Is not associated with excitability changes in the axons

20. Which one of the following is true concerning depolarization in the nerve?

- A) Is caused by Na⁺ outflow.
- B) Is caused by K⁺ outflow.
- C) Is due to opening of inactivation gates of voltage gated Na⁺ channels.
- D) Is that during which the nerve cannot respond to any stimulus.

21. What is true about Action potential of a nerve among following statements?

- A) Depolarization only
- b) Repolarization only
- C) Is rapid changes in membrane potential by effective stimulus.
- D) All of the above

22. Which of the following is correct?

- A) The amplitude of action potential depends on stimulus strength.
- B) Graded (local) potentials can cause membrane depolarization without summation.
- C) The amplitude of action potential does not depend on stimulus strength.
- D) Conduction of action potential in the axon occurs only in one direction.

23. (القصر) During depolarization (:

- A) K⁺ ions diffuse outside
- B) The membrane becomes impermeable to Na⁺
- C) When membrane potential reaches -65mV, K⁺ channels open
- D) Activated voltage gated Na⁺ channels open

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5. Ganglia

1. All of the followings are properties of autonomic ganglia EXCEPT:

- A) Conduction is uni-directional
- B) Chemical transmitter is Acetyl-choline (nicotinic like action).
- C) Equal divergence in both lateral & terminal ganglia
- D) Ganglia can be blocked by large dose of nicotine.

2. Preganglionic sympathetic neurons differ from preganglionic parasympathetic neurons in that

- A) They are shorter
- B) They release acetylcholine
- C) They emerge from the lateral horn of the spinal cord
- D) They release noradrenaline
- E) They are myelinated

3. About the nerve impulses, all the following is true EXCEPT:

- A) They can be conducted in both directions, but in the body, the conduction is mainly orthodromic & sometimes antidromic.
- B) They are conducted passively.
- C) Their conduction is an active self-propagating process associated with reversal of polarity across the nerve cell membrane.
- D) They are characterized being constant in amplitude & velocity for a certain nerve fibre.

4. Saltatory conduction:

- A) Occurs only in myelinated fibres. b) Is faster than electronic currents.
- C) Velocity decreases with rise of temperature.
- D) Velocity is inversely proportional to the diameter of the nerve fibre.

5. Which one of the following is true about Sweeping conduction?

- A) More rapid than salutatory conduction
- B) Occurs in myelinated nerves
- C) Consumes less energy than saltatory conduction
- D) Needs more energy than saltatory conduction

6. Which one of the following is untrue About the nerve impulses?

- A) They can be conducted in both directions, but in the body, the conduction is mainly one way.
- B) They are conducted passively.
- C) Their conduction is an active self-propagating process associated with reversal of polarity across the nerve cell membrane.
- D) They are characterized being constant in amplitude & velocity for a certain nerve fiber.

7. (Saltatory conduction: Autonomic Nervous System) (القصر)

- A) Occurs in unmyelinated nerve fibers
- B) Is slower than nonsaltatory conduction
- C) Conserves energy for the axon
- D) The speed of propagation is inversely proportional to the diameter of the nerve 29

1. c 2-a 3-b 4-a 5-d 6-b 7-c



6. Local action excitability factors

1. **Anelectrotonus refers to:**

- A) Changes at the cathode.
- B) Changes at the anode.
- C) Increased excitability.
- D) Depolarization.

2. **During anelectrotonus, the resting membrane potential (RMP):**

- A) Decreases.
- B) Increases.
- C) Remains unchanged.
- D) Is abolished.

3. **Which of the following is true about catelectrotonus?**

- A) It causes hyperpolarization.
- B) It decreases RMP.
- C) It decreases excitability.
- D) It requires stronger stimuli.

4. **Increased excitability at the cathode is due to:**

- A) Addition of positive charges outside the membrane.
- B) Depolarization.
- C) Hyperpolarization.
- D) Strong anelectrotonus.

5. **Which thermal factor decreases nerve excitability?**

- A) Cooling.
- B) Warming.
- C) Stretching.
- D) Increased O₂ concentration.

6. **What happens to nerve fibers when cooled?**

- A) Increased Na-K pump activity.
- B) Increased RMP.
- C) Increased metabolic activity.
- D) Increased conductivity.

7. **Mechanical factors affect nerve excitability by:**

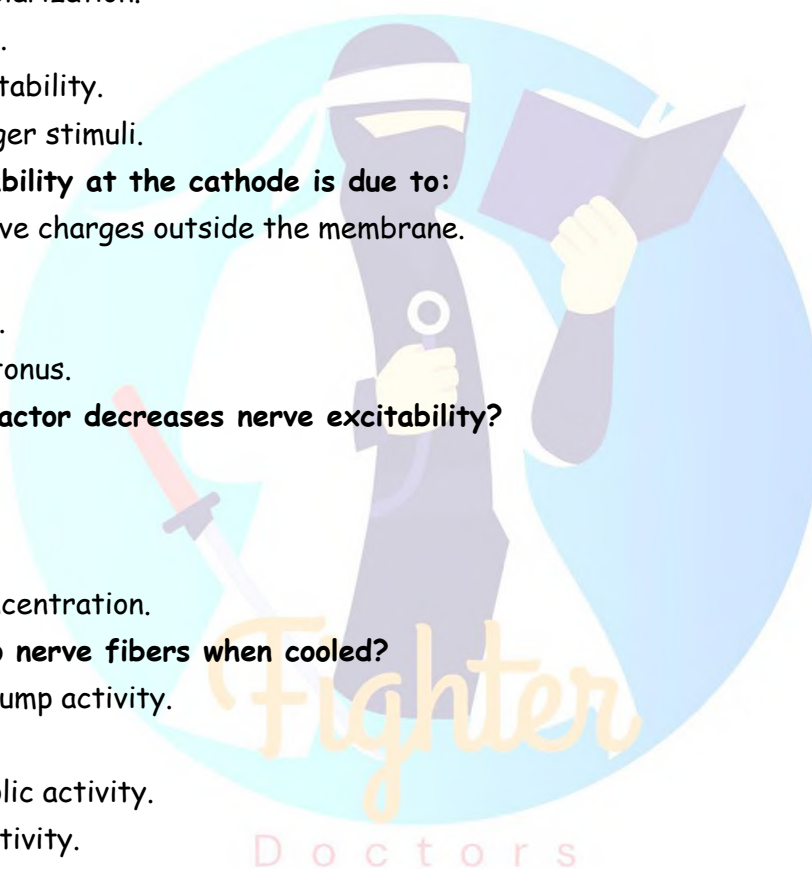
- A) Increasing depolarization.
- B) Applying deep pressure.
- C) Blocking K channels.
- D) Enhancing conductivity.

8. **Local anesthetic drugs like xylocaine work by:**

- A) Blocking K channels.
- B) Blocking Na channels.
- C) Increasing Na permeability.
- D) Hyperpolarizing the membrane.

9. **An increase in extracellular Ca²⁺ levels leads to:**

- A) Increased Na permeability.





B) Decreased excitability.

C) Nerve depolarization.

D) Nerve accommodation.

10. A decrease in K^+ levels in ECF causes:

A) Increased excitability.

B) Hyperpolarization.

C) Nerve excitability.

D) Decreased Na^+ permeability.

11. Lack of oxygen affects nerve excitability by:

A) Enhancing depolarization.

B) Decreasing excitability.

C) Increasing RMP.

D) Blocking K^+ efflux.

12. Increased H^+ ion concentration leads to:

A) Increased excitability.

B) Decreased excitability.

C) No effect on RMP.

D) Increased Na influx.

13. Electrotonus refers to:

A) Chemical changes in the nerve membrane.

B) Electrical changes due to subthreshold galvanic current.

C) Physical stimulation of nerve fibers.

D) Mechanical injury.

14. A strong anelectrotonus can:

A) Increase nerve excitability.

B) Cause nerve block.

C) Decrease depolarization.

D) Increase K efflux.

15. Nerve block is caused by:

A) Increased Na^+ ions.

B) Decreased Ca^{++} ions.

C) Severe cooling or pressure.

D) Weak electrical stimulation.

16. A chemical cause of nerve block includes:

A) Increased Na^+ permeability.

B) Increased extracellular Ca^{++} levels.

C) Reduced RMP.

D) Increased O_2 levels.

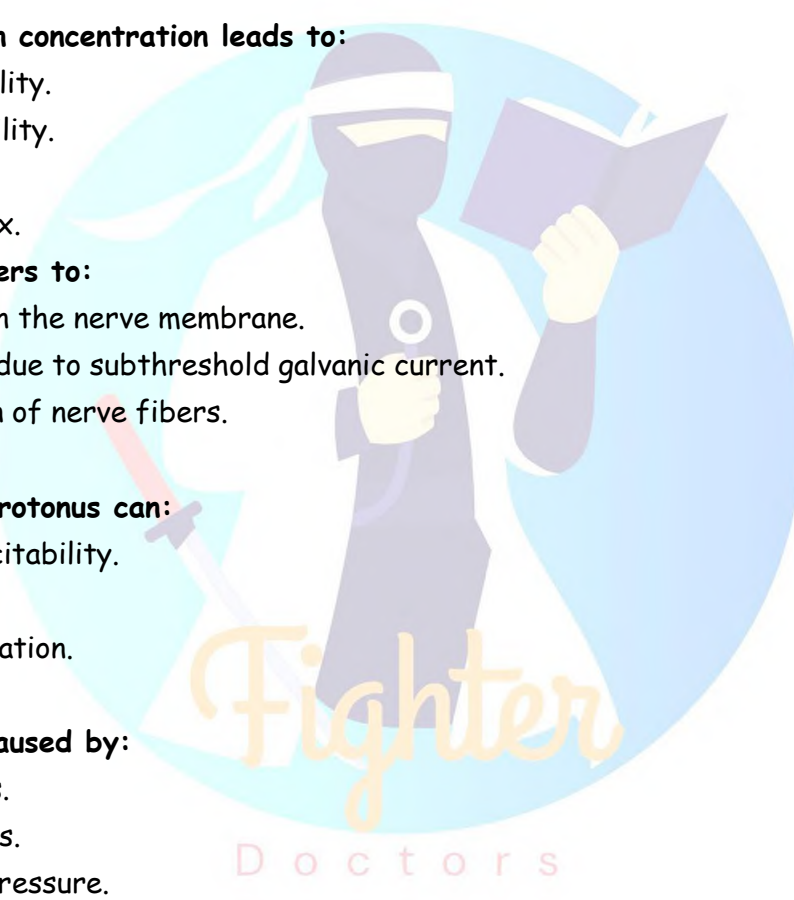
17. Excitability is defined as:

A) Ability to respond to stimuli and convert them into impulses.

B) Ability to propagate impulses.

C) The increase in RMP.

D) The hyperpolarization of nerve fibers.





18. Which ion facilitates nerve depolarization?

- A) K^+ .
- B) Na^+ .
- C) Ca^{2+} .
- D) Cl^- .

19. Rheobase is:

- A) The minimal current strength for a response.
- B) Double the strength of the threshold stimulus.
- C) Time required to stimulate a nerve.
- D) Depolarization caused by a weak stimulus.

20. Chronaxie refers to:

- A) Minimal stimulus needed for excitation.
- B) Time needed by a stimulus double the rheobase strength.
- C) Duration of the refractory period.
- D) The time it takes for depolarization.

21. What causes hyperpolarization in the membrane?

- A) Increased Na influx.
- B) Decreased K influx.
- C) Increased Na efflux.
- D) Increased Ca ions.

22. The $Na-K$ pump contributes to:

- A) Action potential propagation.
- B) Maintaining RMP.
- C) Increasing conductivity.
- D) Blocking nerve impulses.

23. Why does cooling reduce nerve excitability?

- A) It blocks K efflux, leading to hyperpolarization.
- B) It slows metabolic reactions necessary for the $Na-K$ pump.
- C) It increases membrane permeability to Ca ions.
- D) It enhances nerve conductivity.

24. Which chemical factor causes hyperpolarization of the nerve membrane?

- A) Increased extracellular Na ions.
- B) Increased extracellular K ions.
- C) Increased extracellular Ca ions.
- D) Increased O_2 levels.

1-B, 2-B, 3-B, 4-B, 5-A, 6-B, 7-B, 8-B, 9-B, 10-B, 11-B, 12-B, 13-B, 14-B, 15-C, 16-B, 17-A, 18-B, 19-A,
20-B, 21-C, 22-B, 23-B, 24-C

**Additional Questions (from supplementary question bank)**

1. From the strength-duration curve, all the following is true EXCEPT:

- A) Stimuli of extremely short durations fail to excite the nerve.
- B) Within limits, the stronger the current the shorter the duration required to excite.
- C) The threshold galvanic current which can excite the nerve is called the rheobase.
- D) The utilization time is a common measure of excitability. (chronaxie)

2. About cathelectronus, what is the wrong statement?

- A) It results from stimulation of the nerve by a subthreshold galvanic current.
- B) A strong cathelectronus can cause nerve block.
- C) Cathelectronus may be followed by the local response.
- D) Cathelectronus is accompanied by increased excitability of the nerve.

3. During the relative refractory period of nerve:

- A) Some of Na^+ channels are still in the activation state.
- B) Some of Na^+ channels are still in the inactivation state.
- C) Threshold stimuli can excite the nerve.
- D) Inactivation of K^+ channels occurs by opening of the K^+ gates.

4. Chronaxie is:

- A) The strength of threshold galvanic current.
- B) Used to measure the conductivity of nerve fibres.
- C) About 5 m.sec. in smooth muscles.
- D) Shorter in cardiac muscles than in skeletal muscles.

5. From the strength-duration curve, all the following statements are true EXCEPT:

- A) Rheobase is the threshold galvanic current which can excite the nerve.
- B) Chronaxie is the time needed by current of twice the rheobase intensity to excite the nerve.
- C) The stronger the current the shorter the duration required to excite, within limit
- D) The utilization time is the common measure of excitability.

6. The all or none law states that a threshold stimulus produces an impulse which:

- A) Is weaker than that produced by a stronger stimulus.
- B) Is propagated only partially along the cell membrane.
- C) Does not need any energy.
- D) Is not different from that produced by a stronger stimulus.

7. About cathelectronus, all the following is true EXCEPT:

- A) It results from stimulation of the nerve by a subthreshold galvanic current.
- B) Cathelectronus is accompanied by increased excitability of the nerve.
- C) A strong cathelectronus can cause nerve block.
- D) Cathelectronus may be followed by the local response.

8. Chronaxie:

- A) Is twice the rheobase.
- B) Is the utilization time.
- C) Is the time taken for the rheobase to stimulate the nerve fibre.
- D) Is used to determine the degree of nerve excitability.

9. The Chronaxie:

- A) Is the threshold stimulus.



- B) Is twice the rheobase.
- C) Can be used as a measure of excitability.
- D) Is the time needed to excite a nerve by a current strength equal to the rheobase.

10. Excitability of nerve fibres is increased in all the following conditions EXCEPT:

- A) When extracellular Ca^{++} is decreased.
- B) When extracellular fluid K^+ is decreased.
- C) With any condition that increase the membrane permeability for Na^+ .
- D) During catelectrotonic state.

11. Excitability of nerve fibres is decreased in all the following conditions EXCEPT:

- A) With decreased ECF K^+ .
- B) During anelectronic state.
- C) With any conditions that decrease membrane permeability to Na^+ .
- D) During the local excitatory state.

12. From the strength-duration curve one can observe all of the following EXCEPT:

- A) Stimuli of extreme short duration will not excite.
- B) Within limits the stronger the current the longer the duration required to produce its effect.
- C) When the current rises too slowly, the nerve accommodates itself to the rise of the current.
- D) The threshold intensity of a galvanic current which can excite is the Rheobase.

13. Concerning local response, the following statements are true, EXCEPT:

- A) Magnitude and duration vary with the strength of the stimulus.
- B) It can be summated.
- C) It does not obey all or non-law.
- D) It is accompanied by decreased excitability.

14. What is chronaxie?

- A) Is a threshold stimulus
- B) Twice the rheobase
- C) Can be used as a measure of excitability
- D) Time needed to excite the nerve on using current strength equal rheobase

15. Which statement is untrue About electrotonus?

- A) It results from stimulation of nerve by sub-threshold galvanic current
- B) Catelectrotonus is accompanied by increased nerve excitability
- C) Strong catelectrotonus can lead to nerve block
- D) Catelectrotonus may be followed by local response

16. Which one of the following can produce Nerve block?

- A) Sever cooling
- B) Decreased Ca^{++} ions
- C) strong catelectrotonus
- D) increased Na^+ ions

17. Which one of the following is true regarding The local response?

- A) Is characterized by depressed excitability.
- B) Obeys the all or none law.
- C) Has an absolute refractory period.
- D) None of the above.



18. Which one of the following causes decrease in nerve fibres excitability?

- A) When extracellular Ca^{++} is decreased.
- B) When extracellular fluid K^{+} is decreased.
- C) With any condition that increase the membrane permeability for Na^{+}
- D) During catelectrotonic state.

19. Which impulse a threshold stimulus can produce according to the all or none law?

- A) Is weaker than that produced by a stronger stimulus.
- B) Is propagated only partially along the cell membrane.
- C) Does not need any energy.
- D) Is not different from that produced by a stronger stimulus

20. duration curve: a- Within limits the stronger the stimulus the longer its duration needed to excite nerve b- The excitability of the nerve is inversely proportional to the chronaxie c- Rheobase is the maximum amount of current needed to produce nerve activity d- Utilization time is the time needed by double rheobase to give a response

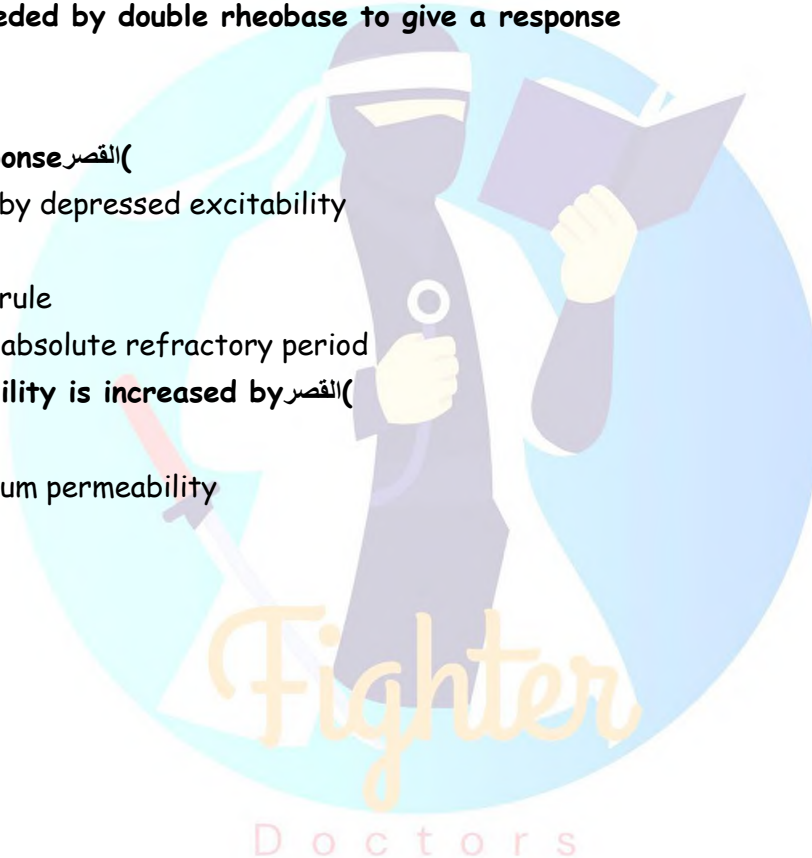
- A) b)
- C) d)

21. (:The local response القصر(

- A) Is characterized by depressed excitability
- B) Can be summated
- C) Obeys all or none rule
- D) Is followed by an absolute refractory period

22. (:Nerve excitability is increased by القصر(

- A) Hypercalcemia
- B) Increased potassium permeability
- C) Cooling
- D) Hypocalcaemia





7. Excitability and types of nerves

1. Continuous conduction:

- A) Occurs in myelinated nerve fibers.
- B) Occurs by jumping of charges from one node of Ranvier to another.
- C) Is relatively slow, 0.5-2.0 meter/second.
- D) Occurs in the neuro-muscular junction.

2. The supernormal phase of excitability in an activated nerve fiber is due to:

- A) Rapid Na^+ influx during the spread of action potential.
- B) Slow K^+ outflux during the after-depolarization.
- C) Pumping out of excess Na^+ by the pump.
- D) Cl^- influx.

3. The velocity of conduction is faster in:

- A) Myelinated nerve fiber.
- B) Small diameter nerve fiber.
- C) Cooled nerve fiber.
- D) Unmyelinated nerve fiber.

4. The absolute refractory period (ARP) of a nerve fiber:

- A) Lasts throughout the action potential.
- B) Follows the relative refractory period.
- C) Occurs during the time of increased permeability of the membrane to Na^+ .
- D) Occurs mainly during repolarization.

5. As regards conduction of action potential in a nerve:

- A) In thick myelinated nerve fibers can reach up to 120 meter/second.
- B) Can be increased by increasing calcium.
- C) Can be increased by cooling.
- D) Is conducted with decrement.

6. Slow rise in stimulus strength leads to:

- A) Increased excitability.
- B) Nerve accommodation.
- C) Depolarization.
- D) Nerve block.

7. Increased extracellular potassium concentration:

- A) Increases resting membrane potential.
- B) Decreases resting membrane potential.
- C) Causes depolarization.
- D) Decreases nerve excitability.

8. Which factor does NOT affect nerve excitability?

- A) Cooling.
- B) Mechanical injury.
- C) Sodium concentration.
- D) Nitrogen concentration.

9. Which of the following best describes excitability?

- A) The ability of a nerve fiber to transmit an impulse.

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- B) The ability of a nerve fiber to respond to a stimulus and generate an action potential.
- C) The ability of a nerve to block external stimuli.
- D) The depolarization of the resting membrane potential.

10. Conductivity is defined as:

- A) The ability to maintain the resting potential.
- B) The transfer of electrical impulses along a nerve fiber.
- C) The resistance of nerve fibers to external stimuli.
- D) The failure of nerve impulses to propagate.

11. Why is saltatory conduction faster than continuous conduction?

- A) It requires less energy for depolarization.
- B) It involves direct passage through unmyelinated axons.
- C) It occurs via the jumping of impulses between nodes of Ranvier.
- D) It does not depend on sodium channels.

12. Saltatory conduction can reach speeds of:

- A) 5 m/s.
- B) 10 m/s.
- C) 50 m/s.
- D) 120 m/s.

13. During the second section of the absolute refractory period, what happens to sodium channels?

- A) They remain open.
- B) They close and cannot reopen temporarily.
- C) They allow increased efflux of sodium.
- D) They become hyperactive.

14. What happens to nerve excitability when calcium concentration increases?

- A) Excitability decreases due to blocked sodium channels.
- B) Excitability increases due to enhanced depolarization.
- C) Excitability remains unchanged.
- D) Sodium influx is facilitated.

15. Excess of oxygen and lack of CO₂ in nerve fibers lead to:

- A) Increased excitability and faster conduction.
- B) Reduced excitability due to metabolic disruption.
- C) Hyperpolarization and nerve block.
- D) Enhanced sodium-potassium pump activity.

1-C, 2-B, 3-A, 4-C, 5-A, 6-B, 7-C, 8-D, 9-B, 10-B, 11-C, 12-D, 13-B, 14-A, 15-A

**Additional Questions (from supplementary question bank)****1. Chronaxie:**

- A) Is the minimal stimulus which produces response after a relatively long time.
- B) Is the minimal time needed by the strong stimulus to produce a response.
- C) Is the time required to stimulate the nerve by a minimal stimulus.
- D) Is the time required to stimulate the nerve by a stimulus which is double rheobase.

2. Saltatory conduction:

- A) Occurs in unmyelinated nerve fibers.
- B) May reach up to 120 meters/second.
- C) Occurs by jumping from one neuron to another.
- D) Decreases gradually with distance till it disappears.

3. Continuous conduction:

- A) Occurs in myelinated nerve fibers.
- B) Occurs by jumping of charges from one node of Ranvier to another.
- C) Is relatively slow, 0.5-2.0 meters/second.
- D) Occurs in the neuromuscular junction.

4. Rheobase is:

- A) The minimal time needed by a stimulus to give a response.
- B) The excitability during which it equals zero.
- C) The minimal current strength that gives a response regardless of its duration.
- D) A chemical transmitter released by the nerve.

5. The supernormal phase of excitability in an activated nerve fiber is due to:

- A) Rapid Na^+ influx during the spread of action potential.
- B) Slow K^+ outflux during the after-depolarization.
- C) Pumping out of excess Na^+ by the pump.
- D) Cl^- influx.

6. The velocity of conduction is faster in:

- A) Myelinated nerve fiber.
- B) Small diameter nerve fiber.
- C) Cooled nerve fiber.
- D) Unmyelinated nerve fiber.

7. The absolute refractory period (ARP) of a nerve fiber:

- A) Lasts throughout the action potential.
- B) Follows the relative refractory period.
- C) Occurs during the time of increased permeability of the membrane to Na^+ .
- D) Occurs mainly during repolarization.

8. As regards conduction of action potential in a nerve:

- A) In thick myelinated nerve fibers, it can reach up to 120 meters/second.
- B) It can be increased by an increase in calcium.
- C) It can be increased by cooling.
- D) It is conducted with decrement.

9. Chronaxie:

- A) Is the minimal time needed by the strong stimulus to produce a response.



- B) Is the minimal stimulus which produces a response after a relatively long time.
- C) Is the time required to stimulate the nerve by a minimal stimulus.
- D) Is the time required to stimulate the nerve by a stimulus which is double rheobase.

10. Rheobase:

- A) Is the minimal stimulus which produces a response after a relatively long time.
- B) Is the minimal time needed by the strong stimulus to produce a response.
- C) Is the time required to stimulate the nerve by a minimal stimulus.
- D) Is the time required to stimulate the nerve by a stimulus which is double rheobase.

11. Slow rise in stimulus strength leads to:

- A) Increased excitability
- B) Nerve accommodation
- C) Depolarization
- D) Nerve block

12. Increased extracellular potassium concentration:

- A) Increases resting membrane potential.
- B) Decreases resting membrane potential.
- C) Causes depolarization.
- D) Decreases nerve excitability.

13. Which factor does NOT affect nerve excitability?

- A) Cooling
- B) Mechanical injury
- C) Sodium concentration
- D) Nitrogen concentration

14. Lack of oxygen and excess carbon dioxide result in:

- A) Increased excitability
- B) Decreased excitability
- C) No change in excitability
- D) Increased conductivity

15. Which of the following best describes excitability?

- A) The ability of a nerve fiber to transmit an impulse.
- B) The ability of a nerve fiber to respond to a stimulus and generate an action potential.
- C) The ability of a nerve to block external stimuli.
- D) The depolarization of the resting membrane potential.

16. Conductivity is defined as:

- A) The ability to maintain the resting potential.
- B) The transfer of electrical impulses along a nerve fiber.
- C) The resistance of nerve fibers to external stimuli.
- D) The failure of nerve impulses to propagate.

17. Which physical factor increases both excitability and conductivity of nerve fibers?

- A) Severe cooling
- B) Moderate warming
- C) Deep pressure
- D) Hyperpolarization



18. What characterizes galvanic currents used in nerve stimulation?

- A) High intensity and short duration
- B) Low intensity and long duration
- C) High intensity and long duration
- D) Low intensity and short duration

19. Which of the following stimuli are NOT classified as chemical stimuli?

- A) Acetylcholine and noradrenaline
- B) Hormones like adrenaline
- C) Ions such as calcium and sodium
- D) Electromagnetic radiation

20. Rheobase is best defined as:

- A) The minimal time needed to stimulate a nerve with double the rheobasic strength.
- B) The lowest current intensity required to excite a nerve, regardless of stimulus duration.
- C) The ability of a stimulus to generate a subthreshold depolarization.
- D) The time required to block sodium channels.

21. Chronaxie measures:

- A) The minimal stimulus required to depolarize a nerve membrane.
- B) The duration of time needed to excite a nerve using double the rheobase intensity.
- C) The strength of a stimulus needed for an action potential.
- D) The effectiveness of a chemical neurotransmitter.

22. What does the strength-duration curve illustrate?

- A) The relationship between resting membrane potential and sodium concentration.
- B) The relationship between stimulus intensity and stimulus duration required to excite a nerve.
- C) The relationship between conductivity and nerve diameter.
- D) The effect of physical factors on excitability.

23. Why is saltatory conduction faster than continuous conduction?

- A) It requires less energy for depolarization.
- B) It involves direct passage through unmyelinated axons.
- C) It occurs via the jumping of impulses between nodes of Ranvier.
- D) It does not depend on sodium channels.

24. What is the speed range of continuous conduction in unmyelinated fibers?

- A) 0.5-2.0 m/s
- B) 10-50 m/s
- C) 20-100 m/s
- D) 100-120 m/s

25. Saltatory conduction can reach speeds of:

- A) 5 m/s
- B) 10 m/s
- C) 50 m/s
- D) 120 m/s

26. The first phase of the absolute refractory period is characterized by:

- A) Complete inability to generate a second action potential.
- B) Partial excitability of the nerve.



C) Increased membrane permeability to potassium.

D) Resting state of the nerve membrane.

27. During the second section of the absolute refractory period, what happens to sodium channels?

A) They remain open.

B) They close and cannot reopen temporarily.

C) They allow increased efflux of sodium.

D) They become hyperactive.

28. How does increased extracellular potassium affect resting membrane potential (RMP)?

A) It lowers the RMP, making the nerve more excitable.

B) It increases the RMP, inhibiting excitability.

C) It causes hyperpolarization.

D) It has no effect on RMP.

29. What happens to nerve excitability when calcium concentration increases?

A) Excitability decreases due to blocked sodium channels.

B) Excitability increases due to enhanced depolarization.

C) Excitability remains unchanged.

D) Sodium influx is facilitated.

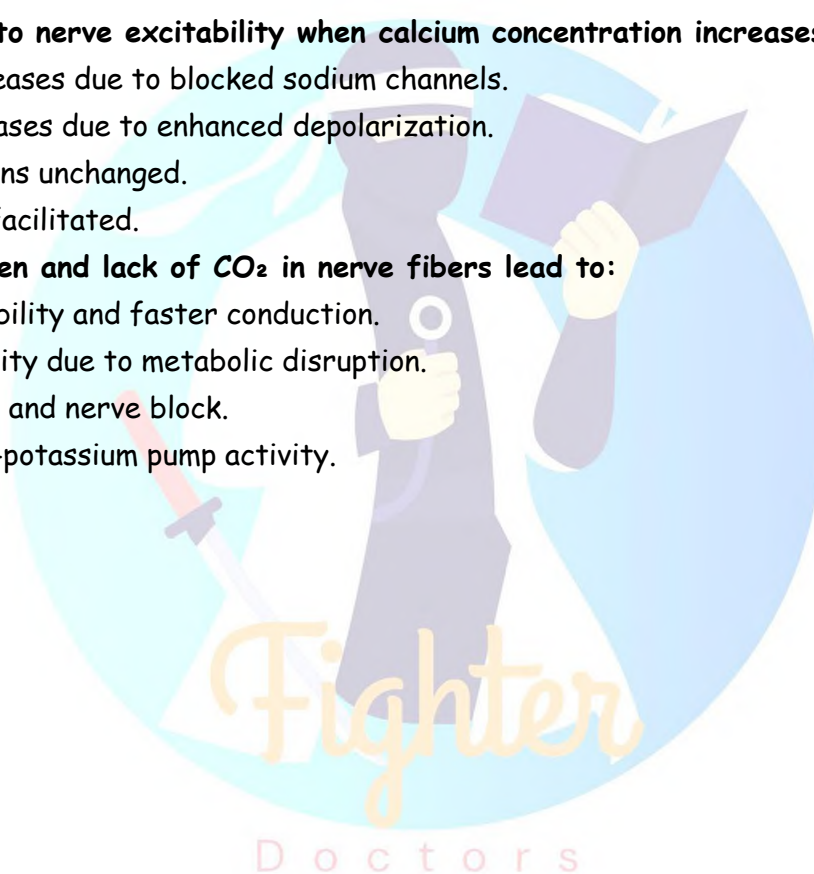
30. Excess of oxygen and lack of CO₂ in nerve fibers lead to:

A) Increased excitability and faster conduction.

B) Reduced excitability due to metabolic disruption.

C) Hyperpolarization and nerve block.

D) Enhanced sodium-potassium pump activity.





Department Bank (General & Nerve)

1. Which of the following is the primary driving force behind passive transport across cell membranes?
 - A) ATP hydrolysis.
 - B) Concentration gradient.
 - C) Vesicle formation.
 - D) Membrane potential.
2. Facilitated diffusion differs from simple diffusion in that it requires:
 - A) ATP.
 - B) Carrier or channel proteins.
 - C) Vesicles.
 - D) Calcium ions.
3. Which of the following molecules is least likely to diffuse directly through the lipid bilayer?
 - A) Oxygen (O₂).
 - B) Carbon dioxide (CO₂).
 - C) Glucose.
 - D) Nitrogen (N₂).
4. The process of osmosis moves water from:
 - A) High solute concentration to low solute concentration.
 - B) Low water concentration to high water concentration.
 - C) High water concentration to low water concentration.
 - D) Low pressure to high pressure.
5. Which statement is TRUE about passive transport?
 - A) It requires energy in the form of ATP.
 - B) It always involves the use of carrier proteins.
 - C) It moves substances down their concentration gradient.
 - D) It is specific to ions only.
6. What is the defining characteristic of active transport?
 - A) Movement of substances down their concentration gradient.
 - B) Requires energy in the form of ATP.
 - C) Involves the movement of only water molecules.
 - D) Always requires channel proteins.
7. Which of the following is an example of primary active transport?
 - A) Sodium-potassium pump (Na⁺/K⁺-ATPase).
 - B) Glucose transport via SGLT.
 - C) Diffusion of oxygen.
 - D) Osmosis of water.
8. Active transport is necessary when substances need to be moved:
 - A) Along the electrochemical gradient.
 - B) Against the concentration gradient.
 - C) Through facilitated diffusion.
 - D) Via simple diffusion.
9. The sodium-potassium pump actively transports:
 - A) 2 Na⁺ in, 3 K⁺ out.
 - B) 3 Na⁺ in, 2 K⁺ out.
 - C) 3 Na⁺ out, 2 K⁺ in.
 - D) 2 Na⁺ out, 3 K⁺ in.



10. Which is often transported using secondary active transport?

- A) Chloride (Cl^-).
- B) Potassium (K^+).
- C) Glucose.
- D) Calcium (Ca^{2+}).

11. In secondary active transport, the energy used to move molecules comes from:

- A) ATP hydrolysis directly.
- B) The breakdown of macromolecules.
- C) The electrochemical gradient of another molecule.
- D) Light energy.

12. What type of carrier protein is used in co-transport mechanisms like the sodium-glucose transporter (SGLT)?

- A) Symporter.
- B) Antiporter.
- C) Uniporter.
- D) Leak channel.

13. Which of the following best describes the role of the Na^+/K^+ pump in maintaining cellular homeostasis?

- A) Generates energy for ATP synthesis.
- B) Maintains intracellular Na^+ and K^+ concentrations.
- C) Facilitates passive diffusion of ions.
- D) Moves glucose across the membrane.

14. What is the typical ratio of ATP usage to ion movement in the Na^+/K^+ pump?

- A) 1 ATP for 2 ions moved.
- B) 1 ATP for 3 ions moved.
- C) 1 ATP for 4 ions moved.
- D) 1 ATP for 5 ions moved.

15. Secondary active transport depends on:

- A) Hydrolysis of ATP by the transporter protein.
- B) A pre-established ion gradient.
- C) Movement of water molecules.
- D) Exocytosis of proteins.

16. Concerning the Na^+/K^+ pump:

- A) Pumps 3 Na^+ molecules inside & 2 K^+ molecules outside the nerve membrane.
- B) Needs high energy phosphate bond.
- C) Is the main cause of RMP.
- D) Pumps 3 K^+ molecules inside & 2 Na^+ molecules outside the nerve membrane.

17. RMP of a nerve:

- A) Caused by equal distribution of ions along the nerve membrane.
- B) Caused by selective permeability of the membrane to ions.
- C) Na^+/K^+ pump has no role.
- D) Caused mainly by inward movement of Na^+ ions.

18. The resting membrane potential in nerve fibers is:

- A) Normally about -10 mV.
- B) Due to K^+ diffusion from outside to inside the nerve fibers.
- C) Due to diffusion of intracellular protein to outside the nerve fibers.
- D) None of the above.

19. The resting potential of a nerve membrane is primarily dependent on the concentration gradient of:



A) Potassium.

B) Sodium.

C) Calcium.

D) Chloride.

20. Which of the following is actively transported out of the neurons?

A) K^+ .

B) HPO_4^{--} .

C) Cl^- .

D) Na^+ .

21. The resting membrane potential is caused by:

A) Diffusion of K^+ ions outside the nerve fibers.

B) Diffusion of Na^+ ions inside the nerve fibers.

C) Opening of the chemically activated ion channels.

D) Opening of the voltage activated ion channels.

22. During depolarization:

A) Voltage activated Na^+ channels open.

B) The membrane becomes impermeable to Na^+ .

C) When membrane potential reaches -55 mV, Na^+ & K^+ fluxes occur at the same time.

D) K^+ ions diffuse outside.

23. Repolarization:

A) Occurs gradually at first then becomes fast.

B) Results from closure of sodium gates and opening of potassium gates.

C) Is represented by the ascending limb of the spike.

D) Is followed by appearance of a response.

24. The firing level of a stimulated nerve fiber:

A) Is reached when the membrane potential reaches -55 mV.

B) Is reached at the end of the spike potential.

C) Is the membrane potential when the permeability to K^+ increases.

D) Can be reached using a subthreshold stimulus.

25. In the all-or-none rule:

A) Minimal stimulus produces a maximal response.

B) The response in a single nerve fiber increases with increased intensity of stimulus.

C) The nerve trunk either responds maximally or does not respond at all.

D) Minimal stimulus produces minimal response.

26. Repolarization of nerve fiber:

A) Occurs gradually at first then becomes fast.

B) Results from closure of sodium gates and opening of potassium gates.

C) Is represented by the ascending limb of the spike.

D) Is followed by appearance of a response.

27. Saltatory conduction:

A) Occurs in unmyelinated nerve fibers.

B) May reach up to 120 meter/second.

C) Occurs by jumping from one neuron to another.

D) Decreases gradually with distance till it disappears.

28. What is the primary function of Schwann cells in the nervous system?

A) Release hormones into the bloodstream.

B) Wrap around axons and form insulating myelin sheaths.



- C) Facilitate the absorption of nutrients in the gut.
- D) Connect neurons to muscle fibers for movement.

29. What is saltatory conduction?

- A) Movement of neurotransmitters across the synaptic cleft.
- B) Transmission of signals through chemical synapses.
- C) Propagation of action potentials along myelinated axons, jumping from one node of Ranvier to the next.
- D) Continuous flow of ions through unmyelinated axons.

30. What is the difference between absolute and relative refractory periods?

- A) Absolute refractory period: a weak stimulus can trigger an action potential; Relative refractory period: no stimulus can trigger an action potential.
- B) Absolute refractory period: the neuron is at rest; Relative refractory period: the neuron is fully activated.
- C) Absolute refractory period: no stimulus can trigger another action potential; Relative refractory period: only a larger-than-normal stimulus can initiate a new action potential.
- D) Absolute refractory period: the neuron is inactive; Relative refractory period: the neuron can only respond to external stimuli.

31. During which phase of the action potential is a second action potential not generated, regardless of the strength of the stimulus?

- A) Spike phase.
- B) Hyperpolarization phase.
- C) Absolute refractory period.
- D) Relative refractory period.

32. Chronaxie:

- A) Is the minimal stimulus which produces a response after a relatively long time.
- B) Is the minimal time needed by the strong stimulus to produce a response.
- C) Is the time required to stimulate the nerve by a minimal stimulus.
- D) Is the time required to stimulate the nerve by a stimulus which is double the rheobase.

33. Saltatory conduction:

- A) Occurs in unmyelinated nerve fibers.
- B) May reach up to 120 meter/second.
- C) Occurs by jumping from one neuron to another.
- D) Decreases gradually with distance till it disappears.

34. Continuous conduction:

- A) Occurs in myelinated nerve fibers.
- B) Occurs by jumping of charges from one node of Ranvier to another.
- C) Is relatively slow, 0.5-2.0 meter/second.
- D) Occurs in the neuro-muscular junction.

35. Rheobase is:

- A) The minimal time needed by a stimulus to give a response.
- B) The excitability during which is equal to zero.
- C) The minimal current strength that gives a response regardless of its duration.
- D) A chemical transmitter released by the nerve.

36. The supernormal phase of excitability in an activated nerve fiber is due to:



- A) Rapid Na^+ influx during the spread of action potential.
- B) Slow K^+ outflux during the after-depolarization.
- C) Pumping out of excess Na^+ by the pump.
- D) Cl^- influx.

37. The velocity of conduction is faster in:

- A) Myelinated nerve fiber.
- B) Small diameter nerve fiber.
- C) Cooled nerve fiber.
- D) Unmyelinated nerve fiber.

38. The absolute refractory period (ARP) of a nerve fiber:

- A) Lasts throughout the action potential.
- B) Follows the relative refractory period.
- C) Occurs during the time of increased permeability of the membrane to Na^+ .
- D) Occurs mainly during repolarization.

39. As regards conduction of action potential in a nerve:

- A) In thick myelinated nerve fibers can reach up to 120 meter/second.
- B) Can be increased by increasing calcium.
- C) Can be increased by cooling.
- D) Is conducted with decrement.

1-B, 2-B, 3-C, 4-C, 5-C, 6-B, 7-A, 8-B, 9-C, 10-C, 11-C, 12-A, 13-B, 14-B, 15-B, 16-B, 17-B, 18-D, 19-A, 20-D, 21-A, 22-A, 23-B, 24-A, 25-A, 26-B, 27-B, 28-B, 29-C, 30-C, 31-C, 32-D, 33-B, 34-C, 35-C, 36-B, 37-A, 38-C, 39-A

Dr. Hany Questions

1. The sodium pump of an excitable membrane:

- A) Pumps sodium ions to the outside during depolarization.
- B) Pumps potassium ions to the outside during depolarization.
- C) Pumps potassium ions to the inside during depolarization.
- D) Pumps sodium ions to the outside in the resting state.

2. The resting membrane potential in nerve fibers is:

- A) Normally about -10 mV.
- B) Due to K^+ diffusion from outside to inside the nerve fibers.
- C) Due to diffusion of intracellular protein to outside the nerve fibers.
- D) None of the above.

3. As regards resting membrane potential, which of the following is true?

- A) Is the potential difference across the fiber membrane with negatively charged ions inside & positively charged ions outside.
- B) Its decrease is accompanied with decreased excitability of the nerve fiber.
- C) Is due to K^+ diffusion from outside to inside the nerve fibers.
- D) It is due to diffusion of intracellular protein to outside the nerve fiber.

4. About the Na^+-K^+ pump, all the following are correct EXCEPT:

- A) Requires high energy phosphate bonds.



- B) Contributes to the creation of resting membrane potential.
- C) Keeps intracellular Na^+ low.
- D) Is explained by facilitated diffusion.

5. Na^+-K^+ pump:

- A) Requires energy which is taken from GTP.
- B) Couples Na^+ and K^+ with ratio 3:2.
- C) Helps intracellular accumulation of Na^+ .
- D) Is a passive process, needs no energy.

6. Resting membrane potential of a nerve fiber:

- A) Is mainly due to sodium influx.
- B) Is mainly due to potassium influx.
- C) Is mainly due to chloride influx.
- D) Is due to the presence of negative charges on the inner surface in relation to its outer surface.

7. The ascending limb of the spike of a nerve fiber action potential is caused by:

- A) Rapid sodium influx.
- B) Closure of activation gates of voltage gated sodium channels.
- C) Rapid potassium efflux.
- D) Calcium influx.

8. When a nerve fiber is stimulated, depolarization phase stops at +35 mV. This is because:

- A) Activation of the Na^+-K^+ pump that prevents accumulation of Na^+ ions inside the nerve membrane.
- B) Closing of the inactivation gates is faster than the opening of activation gates.
- C) The inactivation gates rapidly close to prevent further entry of Na^+ ions.
- D) The activation gates and inactivation gates close at the same time.

9. In a nerve axon, which phase of the action potential is caused by the inactivation of the Na^+ channels?

- A) Upstroke.
- B) Absolute refractory period.
- C) Downstroke.
- D) Relative refractory period.

10. The first step in nerve impulse initiation is:

- A) Decreased Na^+ permeability.
- B) Membrane depolarization.
- C) Increased K^+ permeability.
- D) Membrane hyperpolarization.

11. An action potential has the following characteristics:

- A) Maximal response to threshold stimulus.
- B) Slight response to threshold stimulus.
- C) No response to threshold stimulus.
- D) Increased response to increased stimulus.

12. Regarding the voltage gated Na^+ channels:

- A) It has two gates, activation gate near the inside of the membrane and inactivation gate near the outside.



B) At resting state, the activation gates are closed to prevent influx of Na^+ ions inside the nerve fiber.

C) At resting state, the inactivation gates are also closed.

D) At the threshold (firing) level, the inactivation gates open.

13. If the resting membrane potential of a nerve cell falls (becomes less negative), there is a net:

A) Gain of Na^+ ions into the cell.

B) Gain of K^+ ions into the cell.

C) Loss of protein ions from the cell.

D) Loss of chloride ions from the cell.

14. The action potential of a nerve:

A) Is initiated by Na^+ efflux.

B) Is terminated by K^+ efflux.

C) Declines in amplitude as it moves along the axon.

D) Is not associated with any movement of Na^+ or K^+ across the membrane.

15. The membrane potential will depolarize by the greatest amount if the membrane permeability increases for:

A) Potassium.

B) Sodium & potassium.

C) Potassium & chloride.

D) Sodium.

16. In the action potential, all the following is true EXCEPT:

A) The firing level is the level at which the spike potential begins.

B) The rapid repolarization is due to increased K^+ permeability.

C) The duration of the isoelectric interval (latent period) is inversely proportionate to the speed of conduction.

D) The absolute refractory period is the period from the firing level until repolarization is complete.

17. The after-hyperpolarization phase of the action potential is due to:

A) Inactivation of the Na^+ channels.

B) Increased excitability of the membranes.

C) Net Na^+ efflux to the exterior.

D) Slow return of the K^+ channels to the closed state.

18. During action potential, the rapid depolarization of the membrane is due to:

A) Increased sodium permeability.

B) Increased calcium permeability.

C) Increased chloride permeability.

D) Increased potassium permeability.

19. About nerve excitability:

A) Chronaxie means the intensity of the stimulus that produces a response.

B) An anelectrotonic response decreases the membrane potential and increases the nerve excitability.

C) K^+ is the least permeable cation through nerve membranes.

D) The resting potential becomes less negative if the K^+ concentration outside the nerve membrane is doubled.

20. From the strength-duration curve, all the following is true EXCEPT:

A) Stimuli of extremely short durations fail to excite the nerve.

B) Within limits, the stronger the current the shorter the duration required to excite.



C) The threshold galvanic current which can excite the nerve is called the rheobase.

D) The utilization time is a common measure of excitability.

21. All the following about the electric activity in nerves is correct EXCEPT:

A) The self-propagation of nerve impulses is due to successive depolarization to the firing level of the membrane ahead of the action potential.

B) The resting potential keeps constant after blocking of the Na^+ pump.

C) The after-hyperpolarization phase is the last phase of the action potential.

D) Repolarization is initiated by K^+ efflux.

22. About electrotonus, what is the wrong statement?

A) It results from stimulation of the nerve by a subthreshold galvanic current.

B) A strong catelectrotonus can cause nerve block.

C) Catelectrotonus may be followed by the local response.

D) Catelectrotonus is accompanied by increased excitability of the nerve.

23. Concerning excitable tissues, all the following is true EXCEPT:

A) They spend energy during rest to maintain their membranes at a polarized state.

B) Become more excitable when partially depolarized.

C) Become more permeable to Na^+ ions when membrane potential decreases to a critical level.

D) Usually respond to stimuli that decrease their membrane Na^+ permeability.

24. About nerve impulses, all the following is true EXCEPT:

A) They can be conducted in both directions, but in the body, conduction is mainly orthodromic & sometimes antidromic.

B) They are conducted passively.

C) Their conduction is an active self-propagating process associated with reversal of polarity across the nerve cell membrane.

D) They are characterized by being constant in amplitude & velocity for a certain nerve fiber.

25. During the relative refractory period of a nerve:

A) Some of the Na^+ channels are still in the activation state.

B) Some of the Na^+ channels are still in the inactivation state.

C) Threshold stimuli can excite the nerve.

D) Inactivation of K^+ channels occurs by opening of the K^+ gates.

26. From the strength-duration curve, all the following statements are true EXCEPT:

A) Rheobase is the threshold galvanic current which can excite the nerve.

B) Chronaxie is the time needed by a current of twice the rheobase intensity to excite the nerve.

C) The stronger the current the shorter the duration required to excite, within limits.

D) The utilization time is the common measure of excitability.

27. Absolute refractory period (ARP) of a nerve fiber:

A) Follows the relative refractory period.

B) Occurs mainly during repolarization.

C) During which nerve excitability is relatively higher than at rest.

D) Occurs during the time of increased permeability of the membrane to Na^+ .

28. The all-or-none law states that a threshold stimulus produces an impulse which:

A) Is weaker than that produced by a stronger stimulus.

B) Is propagated only partially along the cell membrane.



- C) Does not need any energy.
- D) Is not different from that produced by a stronger stimulus.

29. About electrotonus, all the following is true EXCEPT:

- A) It results from stimulation of the nerve by a subthreshold galvanic current.
- B) Catelectrotonus is accompanied by increased excitability of the nerve.
- C) A strong catelectrotonus can cause nerve block.
- D) Catelectrotonus may be followed by the local response.

30. Chronaxie:

- A) Is twice the rheobase.
- B) Is the utilization time.
- C) Is the time taken for the rheobase to stimulate the nerve fiber.
- D) Is used to determine the degree of nerve excitability.

31. The chronaxie:

- A) The threshold stimulus.
- B) Is twice the rheobase.
- C) Can be used as a measure of excitability.
- D) Is the time needed to excite a nerve by a current strength equal to the rheobase.

32. Excitability of nerve fibers is increased in all the following conditions EXCEPT:

- A) When extracellular Ca^{++} is decreased.
- B) When extracellular fluid K^{+} is decreased.
- C) With any condition that increases the membrane permeability for Na^{+} .
- D) During catelectrotonic state.

33. Excitability of nerve fibers is decreased in all the following conditions EXCEPT:

- A) With decreased ECF K^{+} .
- B) During anelectrotonic state.
- C) With any condition that decreases membrane permeability to Na^{+} .
- D) During the local excitatory state.

34. Saltatory conduction:

- A) Occurs only in myelinated fibers.
- B) Is faster than electrotonic currents.
- C) Velocity decreases with rise of temperature.
- D) Velocity is inversely proportional to the diameter of the nerve fiber.

35. From the strength-duration curve one can observe all of the following EXCEPT:

- A) Stimuli of extreme short duration will not excite.
- B) Within limits, the stronger the current the longer the duration required to produce its effect.
- C) When the current rises too slowly, the nerve accommodates itself to the rise of the current.
- D) The threshold intensity of a galvanic current which can excite is the rheobase.

36. Concerning local response, the following statements are true EXCEPT:

- A) Magnitude and duration vary with the strength of the stimulus.
- B) It can be summated.
- C) It does not obey the all-or-none law.
- D) It is accompanied by decreased excitability.

37. Compound action potential:



- A) Obeys the all-or-none law.
- B) It can be graded.
- C) It is the action potential recorded from stimulation of a single myelinated nerve fiber by a supramaximal stimulus.
- D) It is a biphasic action potential.

38. Regarding nerve fibers:

- A) A fibers are most sensitive to local anaesthetics.
- B) B fibers are not affected by hypoxia.
- C) C fibers are affected by local anaesthetics more than A fibers.
- D) Postganglionic fibers belong to group B.

39. Which of the following changes in extracellular ion concentrations would you expect to hyperpolarize the nerve membrane?

- A) Increased K^+ concentration.
- B) Decreased K^+ concentration.
- C) Increased Na^+ concentration.
- D) Decreased Cl^- concentration.

40. The absolute refractory period of a nerve fiber:

- A) Lasts throughout an action potential.
- B) Occurs before the relative refractory period.
- C) Is due to a low calcium concentration.
- D) Is when a stronger than normal stimulus is required to excite the fiber.

41. Chronaxie:

- A) Is the time taken by a rheobase stimulus to excite a nerve fiber.
- B) Is the time taken by a double rheobase stimulus to excite a nerve fiber.
- C) Is the minimum duration of any stimulus to excite a nerve fiber.
- D) Is directly proportional to excitability.

42. Concerning the voltage gated sodium channels:

- A) It has an activation gate inside and an inactivation gate outside.
- B) Both gates are closed at rest.
- C) At the firing level, the activation gates open.
- D) At the top of the spike of action potential, the activation gate closes.

43. Saltatory (jumping) conduction of action potential:

- A) Occurs in unmyelinated nerve fibers.
- B) Is slower than the point to point conduction.
- C) Consumes less energy than point to point conduction.
- D) None of the above.

1-D, 2-D, 3-A, 4-D, 5-B, 6-D, 7-A, 8-C, 9-C, 10-B, 11-A, 12-B, 13-A, 14-B, 15-D, 16-D, 17-D, 18-A, 19-D, 20-D, 21-B, 22-B, 23-D, 24-B, 25-B, 26-B, 27-D, 28-D, 29-C, 30-D, 31-C, 32-B, 33-D, 34-A, 35-B, 36-D, 37-B, 38-C, 39-B, 40-A, 41-B, 42-C, 43-C



Autonomic Nervous System

8. Introduction of ANS

1. The parasympathetic nervous system:

- A) Is called the anabolic system.
- B) Originates in the thoracolumbar region.
- C) Regulates "fight or flight" responses.
- D) Has short preganglionic fibers.

2. Example for autonomic reflex arc, EXCEPT:

- A) Micturition reflex.
- B) Withdrawal reflex arc.
- C) Vision or smell of food.
- D) Reflex secretion of saliva.

3. How many segments are in the coccygeal region?

- A) 1.
- B) 12.
- C) 8.
- D) 5.

4. An example of an unconditioned reflex is:

- A) Salivating at the sight of food.
- B) Salivating when food is in the mouth.
- C) Reflexive withdrawal from heat after learning.
- D) Memorizing motor responses.

5. The system that controls smooth muscle, cardiac muscle, and gland activity is the:

- A) Sensory nervous system.
- B) Skeletal division.
- C) Autonomic nervous system.
- D) Somatic nervous system.

6. The somatic nervous system is part of the:

- A) Sympathetic nervous system.
- B) Parasympathetic nervous system.
- C) Autonomic nervous system.
- D) None of the above.

7. The nervous system is classified into:

- A) Somatic and autonomic systems.
- B) Central nervous system and peripheral nervous system.
- C) Brain and spinal cord.
- D) Peripheral nervous system and autonomic nervous system.

8. An unconditioned reflex:

- A) Is acquired through learning.
- B) Requires the integrity of the cerebral cortex.
- C) Is inborn and instinctive.
- D) Involves voluntary actions.

9. Which root carries autonomic fibers?

- A) Lateral horn.
- B) Posterior root (sensory).
- C) Afferent nerve.
- D) Anterior horn (motor).

10. The following reflexes are all autonomic, EXCEPT:

- A) Salivary secretion.
- B) Micturition.
- C) Flexion of the arm following a painful stimulus.
- D) Acceleration of the heart.



11. What distinguishes an autonomic motor neuron from a somatic motor neuron?

- A) Somatic motor neurons are exclusively unmyelinated.
- B) Somatic motor neurons influence glandular activity.
- C) Autonomic motor neurons have ganglionic relay points.
- D) Autonomic motor neurons are always myelinated.

12. Sympathetic activation typically results in:

- A) Decreased respiratory rate.
- B) Increased activity of the digestive system.
- C) Increased heart rate and blood pressure.
- D) Constriction of the pupils.

13. Which of the following is a characteristic of the somatic nervous system?

- A) Uses norepinephrine as a neurotransmitter.
- B) Directly innervates skeletal muscles.
- C) Consists of a two-neuron pathway.
- D) Regulates involuntary activities.

14. The autonomic nervous system controls:

- A) Reflex arcs.
- B) Sensory neurons.
- C) Cardiac and smooth muscles and glands.
- D) Skeletal muscles.

15. The efferent limb of the ANS generally contains ... neuron(s):

- A) 1.
- B) 2.
- C) 3.
- D) 4.

16. The dorsal root of a spinal nerve contains:

- A) Motor fibers.
- B) Sensory fibers.
- C) Both sensory and motor fibers.
- D) Ganglionic fibers.

17. A reflex that requires prior learning and cerebral cortex involvement is:

- A) Visceral reflex.
- B) Conditioned reflex.
- C) Unconditioned reflex.
- D) Local axon reflex.

18. Effector organs of the ANS include all of the following EXCEPT:

- A) Heart.
- B) Skeletal muscle.
- C) Visceral smooth muscles.
- D) Arteriolar smooth muscle.

19. The integration center of a reflex arc is located in:

- A) Peripheral nerves.
- B) Effector organ.
- C) Spinal cord or brainstem.
- D) Sensory receptor.

Answer key (1-19):

1-A, 2-B, 3-A, 4-B, 5-C, 6-D, 7-B, 8-C, 9-A, 10-C, 11-C, 12-C, 13-B, 14-C, 15-B, 16-B, 17-B, 18-B, 19-C

**Additional Questions (from supplementary question bank)**

1. Which root carries autonomic fibers?

- A) Lateral horn
- B) Posterior root (sensory)
- C) Afferent nerve
- D) Anterior horn (Motor)

2. The system that controls smooth muscle, cardiac muscle, and gland activity is the:

- A) Sensory nervous system
- B) Skeletal division
- C) Autonomic nervous system
- D) Somatic nervous system

3. An unconditioned reflex:

- A) Is acquired through learning
- B) Requires the integrity of the cerebral cortex
- C) Is inborn and instinctive
- D) Involves voluntary actions

4. A reflex that requires prior learning and cerebral cortex involvement is:

- A) Visceral reflex
- B) Conditioned reflex
- C) Unconditioned reflex
- D) Local axon reflex

5. The autonomic nervous system controls:

- A) Reflex arcs
- B) Sensory neurons
- C) Cardiac and smooth muscles, and glands
- D) Skeletal muscles

6. All of the following are examples of an autonomic reflex arc, EXCEPT:

- A) Micturition reflex
- B) Withdrawal reflex arc
- C) Reflex Secretion of Saliva
- D) Vision or smell of Food

7. The following reflexes are all autonomic, EXCEPT:

- A) Salivary secretion
- B) Micturition
- C) Flexion of the arm following a painful stimulus
- D) Acceleration of the heart

8. Cell bodies of preganglionic neurons of the sympathetic nervous system are found in:

- A) The dorsal root ganglion
- B) The sympathetic chain
- C) Collateral ganglia
- D) The lateral horn cells of the spinal cord
- E) The dorsal horn cells of the spinal cord

9. Conditioned reflexes require the integrity of:

- A) Cerebral cortex
- B) Peripheral nerves



C) Spinal cord

D) Brainstem

10. The parasympathetic nervous system:

A) Is called the Anabolic system

B) Originates in the thoracolumbar region

C) Regulates "fight or flight" responses

D) Has short preganglionic fibers

11. Sympathetic activation typically results in:

A) Decreased respiratory rate

B) Increased activity of the digestive system

C) Increased heart rate and blood pressure

D) Constriction of the pupils

12. The somatic nervous system is part of the:

A) Sympathetic nervous system

B) Parasympathetic nervous system

C) Autonomic nervous system

D) None of the above

13. What distinguishes an autonomic motor neuron from a somatic motor neuron?

A) Somatic motor neurons are exclusively unmyelinated.

B) Somatic motor neurons influence glandular activity.

C) Autonomic motor neurons have ganglionic relay points.

D) Autonomic motor neurons are always myelinated.

14. Which of the following statements about spinal nerves is correct?

A) Spinal nerves contain only sensory fibers.

B) Each spinal nerve is connected to the spinal cord by two roots.

C) Spinal nerves arise only from the thoracic region.

D) There are 12 pairs of spinal nerves.

15. Which of the following structures is a component of a reflex arc?

A) Effector organ

B) Sensory receptor

C) Efferent neuron

D) Afferent neuron

E) All of the above

16. The integration center of a reflex arc is located in the:

A) Peripheral nerves

B) Effector organ

C) Spinal cord or brainstem

D) Sensory receptor

17. An example of an unconditioned reflex is:

A) Salivating at the sight of food

B) Salivating when food is in the mouth

C) Reflexive withdrawal from heat after learning

D) Memorizing motor responses



18. The nervous system is classified into:

- A) Somatic and Autonomic Systems
- B) Central Nervous System and Peripheral Nervous System
- C) Brain and Spinal Cord
- D) Peripheral Nervous System and Autonomic Nervous System

19. The efferent pathway of the ANS generally contains neuron(s).

- A) 1
- B) 2
- C) 3
- D) 4

20. How many segments are in the coccygeal region?

- A) 1
- B) 8
- C) 12
- D) 5

21. Which of the following is a characteristic of the somatic nervous system?

- A) Uses norepinephrine as a neurotransmitter
- B) Directly innervates skeletal muscles
- C) Consists of a two-neuron pathway
- D) Regulates involuntary activities

22. Effector organs of the ANS include all of the following EXCEPT:

- A) Heart
- B) Skeletal muscle
- C) Visceral smooth muscles
- D) Arteriolar smooth muscle

23. The dorsal root of a spinal nerve contains:

- A) Motor fibers
- B) Sensory fibers
- C) Both sensory and motor fibers
- D) Ganglionic fibers

24. The sympathetic nervous system originates from the:

- A) Brainstem
- B) Sacral region only
- C) Thoracolumbar region
- D) Craniosacral region



9. Autonomic ganglia and sympathetic

1. The autonomic ganglia have all the following characters EXCEPT:

- A) Are three types: lateral, collateral and terminal.
- B) Are sites of relay of autonomic preganglionic and postganglionic fibers.
- C) Are all present adjacent to the spinal cord.
- D) Are functioning as distributing centers.

2. Among symptoms of Horner's syndrome there is:

- A) Increased sweating.
- B) Dilatation of pupil.
- C) Dropping of the upper eye lid.
- D) Vasoconstriction of skin blood vessels.

3. Regarding the sympathetic nervous system:

- A) Preganglionic fibers release noradrenalin.
- B) Postganglionic fibers release noradrenalin.
- C) It is stimulated by food intake.
- D) Release of acetylcholine by the preganglionic fibers.
- E) Has longer preganglionic fibers than postganglionic fibers.

4. Preganglionic sympathetic neurons differ from preganglionic parasympathetic neurons in that:

- A) They are shorter.
- B) They release acetylcholine.
- C) They emerge from the lateral horn of the spinal cord.
- D) They release noradrenaline.
- E) They are myelinated.

5. The sympathetic division of the autonomic nervous system is characterized by:

- A) Localized actions affecting few systems.
- B) Thoracolumbar outflow from the spinal cord.
- C) Short postganglionic fibers.
- D) Adrenergic preganglionic fibers.

6. The autonomic ganglia:

- A) Are 5 types.
- B) Are the sites of relay of afferent neurons.
- C) Function as distributing centers.
- D) Are located inside the CNS.
- E) Are relay stations for all preganglionic fibers passing.

7. Collateral ganglia of the sympathetic nervous system include:

- A) Adrenal medullary cells.
- B) Coeliac ganglion.
- C) Paravertebral ganglia.
- D) The basal ganglia.
- E) Nucleus of the tractus solitarius.

8. The autonomic nervous system differs from the somatic nervous system in:

- A) Having one efferent neuron.
- B) Preganglionic neurons are located in the anterior horn of the spinal cord.



- C) Efferent nerves have either excitatory or inhibitory effects on the effector organs.
- D) Postganglionic neurons are located in the lateral horn of spinal cord.

9. Sympathetic stimulation of the iris causes:

- A) Astigmatism.
- B) Pupillary constriction.
- C) Pupillary dilation.
- D) Vitreous secretion.

10. Drooping of the eyelid (ptosis) is caused by damage to:

- A) The third cranial nerve.
- B) The parasympathetic supply to the eye.
- C) The sympathetic supply to the eye.
- D) Lesion in lumbar 2, 3, 4.

11. The sympathetic system:

- A) Has short postganglionic neurons.
- B) Fibers are carried in the vagus nerve.
- C) Produces nicotine at its nerve endings.
- D) Has a thoraco-lumbar outflow from the spinal cord.

12. Nerve cell body inside the CNS is called:

- A) Ganglia.
- B) Center.
- C) Nucleus.
- D) B & C.

13. A synapse is composed of all of the following EXCEPT:

- A) Synapsid.
- B) Presynapse.
- C) Postsynapse.
- D) Synaptic cleft.

14. The sympathetic chain is localized:

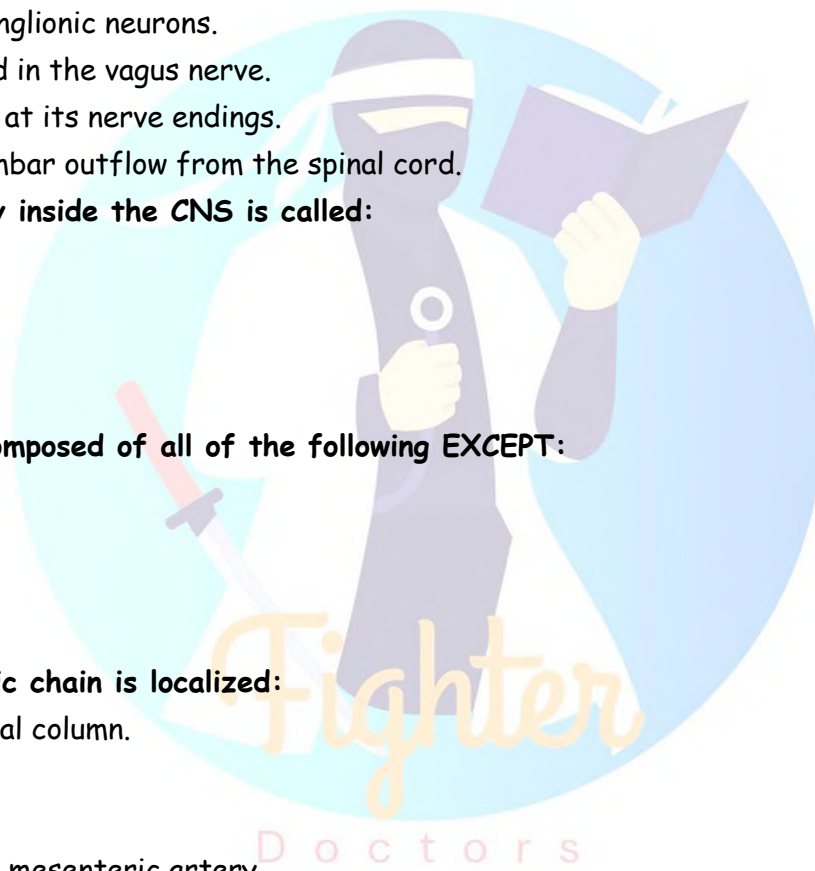
- A) Near the vertebral column.
- B) In the viscera.
- C) Near the organ.
- D) Near the inferior mesenteric artery.

15. The relation between the postganglionic nerve fiber & preganglionic nerve fiber in terminal ganglia is:

- A) Preganglionic fibers are longer than postganglionic fibers.
- B) Postganglionic fibers are longer than preganglionic fibers.
- C) Both have equal length.
- D) None of the above.

16. Distributing centers give rise to ____ nerve fiber(s):

- A) 3:10.
- B) 5:18.
- C) 8:12.
- D) 10:20.





17. All of the following are properties of autonomic ganglia EXCEPT:

- A) Unidirectional (transmission).
- B) Delay of transmission.
- C) Nicotinic-like acting.
- D) May have one or more relays.

18. In trophic stimulation to salivary glands:

- A) Big in volume.
- B) Saliva is rich in minerals.
- C) Blood vessels are extended.
- D) Saliva is viscid.

19. All of the following have low blood flow in a frightening situation, EXCEPT:

- A) Eye.
- B) Cerebral.
- C) Salivary glands.
- D) Skin.

20. Which of the following is true about local axon reflexes?

- A) They have a central integration center.
- B) They involve afferent and efferent branches of the same fiber.
- C) They require the cerebral cortex.
- D) They are conditioned reflexes.

21. In Horner's syndrome, the patient is characterized by:

- A) Mydriasis.
- B) Exophthalmos.
- C) Endophthalmos.
- D) Pallor.

22. Conditioned reflexes require:

- A) No prior learning.
- B) Integrity of the cerebral cortex.
- C) Spinal cord centers.
- D) Primitive reflex centers.

23. Terminal ganglia are located in:

- A) The abdominal cavity.
- B) The spinal cord.
- C) The wall of effector organs.
- D) The sympathetic chain.

24. A single sympathetic preganglionic fiber can activate:

- A) Only 1 postganglionic fiber.
- B) 2 postganglionic fibers.
- C) 8-12 postganglionic fibers.
- D) 20-30 postganglionic fibers.

25. Sweat gland stimulation is mediated by:

- A) Adrenergic fibers.
- B) Cholinergic sympathetic fibers.
- C) Parasympathetic fibers.
- D) Somatic fibers.

1-C, 2-C, 3-B&D, 4-A, 5-B, 6-C, 7-B, 8-C, 9-B, 10-C, 11-D, 12-C, 13-A, 14-A, 15-A, 16-C, 17-A, 18-B, 19-D, 20-B, 21-C, 22-B, 23-C, 24-D, 25-B

**Additional Questions (from supplementary question bank)****1. The autonomic ganglia:**

- A) Are 5 types
- B) are site of relay of afferent neuron
- C) Located inside the CNS
- D) act as distributing centers

2. About suprarenal medulla, all of the following is true EXCEPT:

- A) Secrete 80% adrenalin & 20% noradrenalin
- B) Is modified sympathetic ganglia
- C) Receives preganglionic nerve supply from the anterior horn cells of the lower 6 thoracic segments
- D) No postganglionic fibers arise from it.

3. Concerning the sympathetic nervous system:

- A) Consists only of cranial nerves.
- B) Its efferent pathway is made of one neuron.
- C) All postganglionic sympathetic fibers secrete adrenalin.
- D) Its stimulation causes dilation of the airways.

4. The sympathetic division of the autonomic nervous system:

- A) generally has short postganglionic fibers.
- B) Has its cell of origin in the lateral portion of the thoracic and lumbar segments of spinal cord.
- C) Is concerned primarily with energy conservation and Secretory activity.
- D) Has its cells originating from the brain stem

5. Which of the following structure is innervated by preganglionic sympathetic nerve fibers?

- A) lung.
- B) Pancreas.
- C) Heart.
- D) Adrenal medulla.

6. The preganglionic sympathetic nerve that innervates the eye originates from:

- A) Superior colliculi.
- B) Superior cervical ganglion.
- C) Thoracic segments of spinal cord.
- D) Ciliary ganglion

7. Postganglionic nerves from superior cervical ganglia cause:

- A) Pupillary constriction (miosis)
- B) Trophic secretion from salivary glands
- C) Decreased sweat secretion
- D) Cutaneous vasodilatation

8. All of the following about sacral autonomic outflow is true EXCEPT:

- A) It is the nerve of defecation.
- B) It is the nerve of ejaculation.
- C) It is the nerve of micturition.
- D) It relays in terminal ganglia.

9. Facial nerve:

- A) Stimulation increases the power of the lens of the eye.
- B) Supplies the lacrimal gland.



- C) Supplies the parotid salivary gland.
- D) Contains sympathetic postganglionic fibres.

10. Concerning the cranial parasympathetic outflow:

- A) preganglionic outflow in the oculomotor nerve arises from the ciliary ganglion.
- B) preganglionic outflow in the glossopharyngeal nerve arises from superior salivary nucleus.
- C) They relax the sphincter of gall bladder.
- D) They cause bronchodilatation.

11. Acetylcholine

- A) Is synthesized by the enzyme acetylcholine esterase
- B) Action in synapses is terminated by re-uptake of pre-synaptic neurons
- C) Is released by some postganglionic sympathetic neurons
- D) Is destroyed by monoamine oxidase
- E) Reduces the heart rate by acting on nicotinic receptors in the SA node

12. Nicotinic Receptors are blocked by

- A) Atropine
- B) Curare
- C) Suxmethonium
- D) Propranolol
- E) Acetylcholine esterase

13. Acetylcholine

- A) Is blocked by propanolol
- B) Is blocked by curare in the autonomic ganglion
- C) Is the neurotransmitter at all sites that are blocked by atropine
- D) Is secreted by postganglionic sympathetic to the heart
- E) Is degraded by the enzyme choline transferase

14. The sympathetic system

- A) Has short postganglionic neurons
- B) Fibers are carried in the vagus nerve
- C) Produces nicotine at its nerve endings
- D) Has a thoraco-lumbar outflow from the spinal cord
- E) Action is antagonized by atropine

15. The parasympathetic system

- A) Has short preganglionic fibers
- B) Secretes dopamine
- C) Controls most of the movements and secretions of the gut
- D) Brings increase in heart rate during exercise
- E) Action is blocked by Beta receptor blocker

16. Cell bodies of preganglionic neurons of the sympathetic nervous system are found in

- A) The dorsal root ganglion
- B) The sympathetic chain
- C) Collateral ganglia
- D) The dorsal horn cells of the spinal cord
- E) The lateral horn cells of the spinal cord

17. Collateral ganglia of the sympathetic nervous system include the

- A) Adrenal medullary cells
- B) Coeliac ganglion
- C) Paravertebral ganglia
- D) The basal ganglia
- E) Nucleus of the tractus solitaries

18. Dopamine is

- A) Found in all autonomic ganglia
- B) An inhibitory transmitter



- C) An acetylcholine derivative
- D) Responsible for facilitating autonomic transmission
- E) Released by the interneurons of the parasympathetic ganglia

19. Sweating is served by which of the following receptors?

- A) Nicotinic cholinergic receptors c) Beta- adrenergic receptors
- B) Alpha- adrenergic receptors d) Muscarine cholinergic receptors
- E) Dopaminergic receptors

20. Blocking of the nicotinic ganglionic receptors in the ANS supplying the heart leads to

- A) Inhibition of the sympathetic but not the parasympathetic
- B) Inhibition of the parasympathetic but not the sympathetic
- C) Inhibition of both sympathetic and parasympathetic
- D) Decrease in the heart rate
- E) Decrease in the force of contraction

21. Regarding the sympathetic nervous system

- A) Preganglionic fibers release noradrenalin
- B) Postganglionic fibers release noradrenalin
- C) It is stimulated by food intake
- D) Release of acetylcholine by the pre-ganglionic fibers
- E) Has longer pre-ganglionic fibers than post-ganglionic fibers

22. Muscarinic cholinergic receptors are found

- A) On postganglionic sympathetic neurons
- B) On postganglionic parasympathetic neurons
- C) In the sweat glands
- D) At the end-plate
- E) None of the above

23. Transmitters of the autonomic nervous system include

- A) Acetylcholine at all postganglionic parasympathetic
- B) Dopamine in sympathetic ganglia
- C) Adrenaline in most sympathetic postganglionic neurons to the adrenal medulla 43
- D) Adrenalin and nor-adrenalin in preganglionic neurons to the adrenal medulla Page
- E) Dopamine in postganglionic sympathetic neurons

24. Which of the following is a characteristic of postganglionic parasympathetic neurons that make them different from postganglionic sympathetic neurons

- A) They can be excited by dopamine
- B) Postganglionic parasympathetic nerve bodies are located in ganglia close to the spinal cord
- C) Most postganglionic parasympathetic nerves release the neurotransmitter acetylcholine
- D) Some postganglionic parasympathetic nerves release the neurotransmitter norepinephrine
- E) They are most active under conditions of stress, such as fight/flight

25. A selective beta adrenergic antagonist

- A) Binds to and activates adrenergic receptors
- B) Does not inactivate all beta adrenergic receptors
- C) Prevents the binding of all beta adrenergic antagonists
- D) Is released from some postganglionic sympathetic nerve terminals
- E) May be released from the adrenal medulla

26. Acetylcholine



- A) Is blocked by Propranolol
- B) Is blocked by curare in the autonomic ganglion
- C) Is the neurotransmitter at all sites that are blocked by atropine
- D) Is secreted by postganglionic sympathetic to sweat glands
- E) Is degraded by the enzyme COMT

27. Tachycardia due to sympathetic stimulation is mediated by

- A) B1 adrenergic receptors c) Alpha adrenergic receptors
- B) B2 adrenergic receptors d) Both Alpha and Beta receptors
- E) Nicotinic receptors Cover Excitable Tissues

28. Regarding nerve fibres:

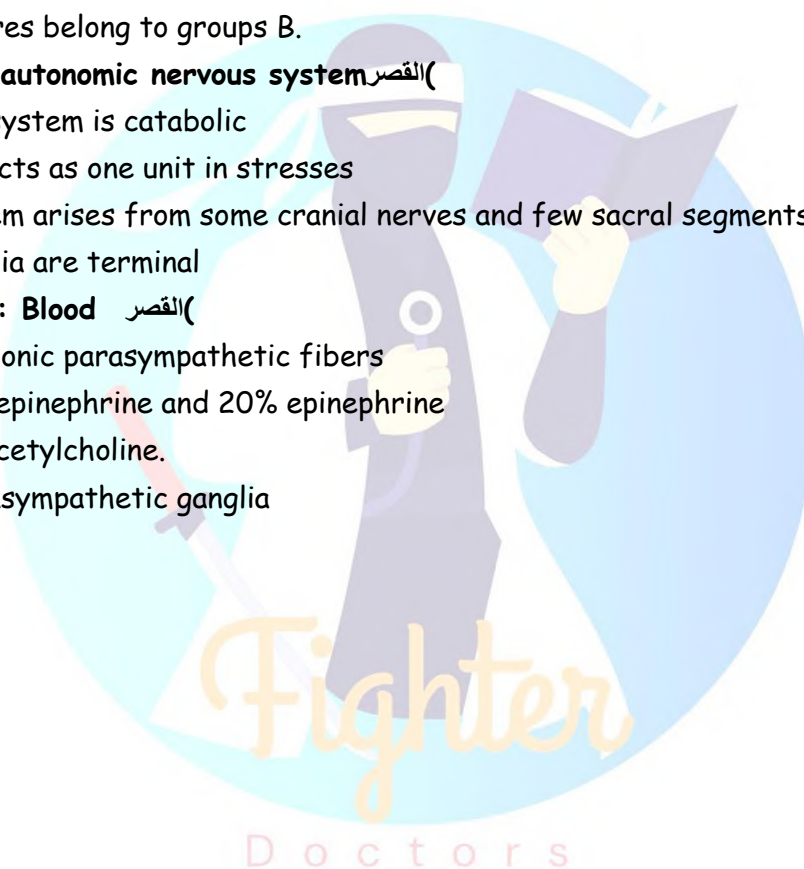
- A) A fibres are most sensitive to local anaesthetics.
- B) B fibres are not affected by hypoxia.
- C) C fibres are affected by local anaesthetics more than A fibres.
- D) Postganglionic fibres belong to groups B.

29. (:Regarding the autonomic nervous system (القصر

- A) The sympathetic system is catabolic
- B) Parasympathetic acts as one unit in stresses
- C) Sympathetic system arises from some cranial nerves and few sacral segments
- D) Sympathetic ganglia are terminal

30. Adrenal medulla: Blood (القصر

- A) Receives preganglionic parasympathetic fibers
- B) Secretes 80% norepinephrine and 20% epinephrine
- C) Is stimulated by acetylcholine.
- D) Is a modified parasympathetic ganglia





10. Sympathetic Nervous System

1. Stimulation of the greater splanchnic nerve produces all the following EXCEPT:

- A) Secretion of catecholamines from the suprarenal medulla.
- B) Increased blood glucose level.
- C) Ejaculation.
- D) Decreased gastrointestinal motility.

2. Stimulation of the greater splanchnic nerve produces:

- A) Increased blood glucose level.
- B) Decreased blood volume.
- C) Secretion of epinephrine 20% & norepinephrine 80% from the adrenal medulla.
- D) Increased gastrointestinal motility.

3. Activation of the sympathetic nervous system produces the following EXCEPT:

- A) Bronchodilatation.
- B) Tachycardia.
- C) Mydriasis (dilatation of pupil).
- D) Contraction of detrusor muscle of urinary bladder wall.

4. Choose the correct statement:

- A) Postganglionic fibers starting from the superior cervical ganglion cause constriction of the pupils.
- B) The sympathetic nervous system supplies every organ in the body.
- C) Sympathetic supply to the lungs causes bronchoconstriction.
- D) Stimulation of the greater splanchnic nerve results in adrenomedullary secretion.

5. Generalized sympathetic stimulation causes:

- A) Inhibition of sweating.
- B) Increased gastric motility.
- C) Decreased blood sugar levels.
- D) Rise in arterial blood pressure.

6. Concerning the adrenal medulla, all the following is true EXCEPT:

- A) Its secretion is increased in emergency conditions.
- B) It secretes epinephrine 80% & norepinephrine 20%.
- C) It augments and prolongs the effects of sympathetic stimulation.
- D) It is a modified sympathetic ganglion.
- E) It receives preganglionic nerve supply from the anterior horn cells of the lower 6 thoracic segments of the spinal cord.

7. Sympathetic supply to the pelvic viscera causes:

- A) Micturition.
- B) Defecation.
- C) Ejaculation.
- D) Dilatation of vessels in external genitalia.

8. The sympathetic nervous system is:

- A) Activated by exposure to cold weather.
- B) Inhibited by exposure to hot weather.
- C) Concerned with initiation of micturition.
- D) Involved in erection of external genital organs.



9. All of the following about the adrenal medulla is true EXCEPT:

- A) It is a sympathetic ganglion.
- B) Secretes more noradrenaline than adrenaline.
- C) It releases epinephrine into circulating blood.
- D) It's innervated by sympathetic cholinergic fibers.

10. Increased activity of the sympathetic nervous system causes:

- A) Penile erection.
- B) Pupillary constriction.
- C) Accommodation for near vision.
- D) Bronchiolar dilatation.

11. Stimulation of the lesser splanchnic nerve causes:

- A) Micturition.
- B) Defecation.
- C) Ejaculation.
- D) Erection.

12. Stimulation of the sympathetic nervous system causes:

- A) Contraction of the ciliary muscle for near vision.
- B) Generalized actions affecting many systems.
- C) Bronchoconstriction.
- D) Decreased glycogenolysis & lipolysis.

13. Generalized sympathetic activity is characterized by:

- A) Bile retention.
- B) Contraction of ciliary muscle.
- C) Vasodilatation of pulmonary blood vessels.
- D) No effect on the spleen.

14. True or false: bronchial obstruction is a contraindication for the injection of catecholamines.

- A) True.
- B) False.

15. All of the following are increased in sympathetic activity EXCEPT:

- A) Fatty acid.
- B) Catecholamines.
- C) Acetylcholine.
- D) Salivary secretion.

16. Which of the following leads to increased cerebral blood flow?

- A) Sympathetic to head & neck (LHC of 1st, 2nd thoracic segment).
- B) Sympathetic to thorax (LHC of upper 4 thoracic segments).
- C) Sympathetic of abdomen.
- D) Sympathetic of pelvis.

17. Sympathetic stimulation causes the following effect on the air passages:

- A) Constriction of bronchi.
- B) Relaxation of smooth muscles leading to bronchodilation.
- C) Increased mucus secretion.



D) Vasodilation of pulmonary vessels.

18. Sympathetic stimulation of the liver leads to:

- A) Glycogenesis.
- B) Glycogenolysis.
- C) Decreased glucose levels.
- D) Fat synthesis.

19. The Orbeli phenomenon refers to:

- A) Decreased sensitivity of motor end plates.
- B) Vasodilation of skeletal muscle blood vessels.
- C) Delayed fatigue and early recovery in skeletal muscles.
- D) Vasoconstriction in skeletal muscles.

20. Which of the following cardiac properties is NOT directly stimulated by the sympathetic system?

- A) Conductivity.
- B) Rhythmicity.
- C) Contractility.
- D) Glycogenesis.

21. Sympathetic stimulation leads to:

- A) Vasodilation of pulmonary vessels.
- B) Vasoconstriction of pulmonary vessels.
- C) Increased mucus secretion.
- D) Contraction of bronchial smooth muscles.

22. What is the role of the adrenal medulla in the sympathetic system?

- A) Acts as a postganglionic fiber.
- B) Secretes adrenaline and noradrenaline into the bloodstream.
- C) Controls parasympathetic responses.
- D) Regulates glycogen synthesis.

23. The sympathetic nervous system influences the urinary bladder by:

- A) Contracting the wall and relaxing the sphincter.
- B) Relaxing the wall and contracting the internal sphincter.
- C) Promoting urination.
- D) Increasing bladder sensitivity.

24. Which of the following is a result of sympathetic stimulation in the skin?

- A) Increased heat dissipation through vasodilation.
- B) Vasoconstriction leading to reduced blood flow.
- C) Relaxation of sweat glands.
- D) Relaxation of sweat glands.

25. Which of the following best describes sympathetic tone?

- A) Continuous parasympathetic activity.
- B) Basal level of sympathetic discharge maintaining vascular tone.
- C) Sudden activation of the sympathetic system.
- D) Decreased blood pressure.

26. The "alarm or stress response" involves all of the following EXCEPT:

- A) Decreased heart rate.
- B) Increased blood glucose level.
- C) Pupil dilation.
- D) Redistribution of blood flow to muscles.

1-C, 2-A, 3-D, 4-D, 5-D, 6-E, 7-C, 8-A, 9-B, 10-D, 11-C, 12-B, 13-D, 14-A, 15-C, 16-A, 17-B, 18-B, 19-C, 20-D, 21-B, 22-B, 23-B, 24-B, 25-B, 26-A

**Additional Questions (from supplementary question bank)****1. Stimulation of greater splanchnic nerve produces:**

- A) Increase motility of plain muscles of stomach
- B) Decrease levels of blood glucose & free fatty acids
- C) Secretion of adrenaline & noradrenalin by adrenal medulla
- D) Inhibition of micturition

2. Paralysis of sympathetic to head causes all of the following EXCEPT:

- A) Dropping of the upper eye lid
- B) pupillary constriction
- C) Inability of the eye to accommodate
- D) cutaneous vasodilatation

3. Stimulation of lesser splanchnic nerve produces:

- A) Increased blood glucose level
- B) Increased adrenalin secretion from adrenal medulla
- C) Shrinkage of external genitalia
- D) Erection

4. Stimulation of lesser splanchnic nerve causes:

- A) Micturition c) defecation
- B) Erection d) ejaculation

5. All of the following about Horner syndrome is true EXCEPT:

- A) Manifested by dryness & redness of skin.
- B) Group of signs including ptosis & miosis.
- C) Is due to injury of L.H.Cs. present in thoracic 1-2 spinal segments.
- D) Causes pallor of skin due to V.C.

6. The parasympathetic nerves:

- A) Are important in emergency conditions e.g. fight & flight
- B) Tend to have longer postgang. fibres than pregang. fibres
- C) Are catabolic in their actions.
- D) Generally exert opposite effects to those of the symp. Nerves.

7. Vagus parasympathetic nerve supplies all the following structures EXCEPT:

- A) Gall bladder. c) Rectum.
- B) Stomach. d) Pancreas.

8. All of the following organs receive their parasympathetic innervation via the vagus nerves EXCEPT:

- A) Gall bladder.
- B) Atria of the heart.
- C) Proximal half of large intestine.
- D) Rectum.

9. Motility of the intestine is increased by: Exam no 1 in A.N.S (Solved Version) MCQ (15 marks):

- A) Adrenaline.
- B) Parasympathetic stimulation.
- C) Sympathetic blockers.



D) Adrenal medulla secretions.

10. Generalized sympathetic stimulation causes all the following except

- A) Increases the heart rate
- B) Increases sweating
- C) Relaxes bronchial smooth muscles
- D) Tightens GIT sphincters
- E) Constriction of pupil

11. adrenergic receptors

- A) Constriction of cutaneous blood vessels
- B) Increase of myocardial contractility
- C) Dilation of arterioles to the viscera
- D) Relaxation of urinary bladder
- E) Dilation of systemic veins

12. Noreadrenaline is released by sympathetic nervous system at all the following except

- A) Cutaneous blood vessels
- B) SA Node
- C) Sweat glands
- D) Cardiac muscles
- E) Renal nerves

13. The effects of parasympathetic stimulation include

- A) Ejaculation
- B) Secretion of sweat
- C) Contraction of the rectum
- D) Constriction of gastrointestinal sphincters
- E) Dilation of the bronchioles

14. Adrenal medullary cells are stimulated by

- A) Dopamine c) Adrenaline
- B) GABA d) Acetylcholine
- E) Nor-adrenaline

15. 1 adrenergic receptor stimulation causes

- A) Coronary vasodilation c) Bronchodilation
- B) Cardiac acceleration d) Skin vasoconstriction
- E) Constriction of anal sphincter

16. Stimulation of the parasympathetic nervous system leads to

- A) Constriction of pupils
- B) Constriction of skin vessels
- C) Constriction of gastrointestinal sphincters 47
- D) Constriction of brain vessels Page
- E) Increased blood flow in coronary vessels

17. Sympathetic stimulation is expected to lead to

- A) Constriction of the pupil
- B) Dilation of the bronchioles
- C) Inhibition of sweating



D) Venodilation

E) Copious salivary secretion

18. Atropine will block all the following except

A) The action of the parasympathetic nervous system on the heart

B) The action of the sympathetic nervous system on the sweat glands

C) The action of the dilator sympathetic nervous system on the muscle blood vessels

D) The action of the parasympathetic nervous system on the gastric secretion

E) The action of acetylcholine on skeletal muscles

19. The sympathetic nerves supply

A) The external sphincters of the gastrointestinal & urinary tracts

B) Smooth muscle of the constrictor pupillae

C) Smooth muscle of the bronchioles

D) Vasoconstrictor fibers to coronary vessels

E) Vasodilator fibers to skin vessels

20. A drug that increases the heart rate from 70 to 100 beats per minute could be a:

A) B1 adrenergic receptor agonist

B) Cholinergic receptor antagonist

C) Dopaminergic receptor agonist

D) B2 adrenergic agonist

E) Non-cholinergic, non-adrenergic agonist

21. Sympathetic stimulation results in

A) Increased residual volume in the ventricles

B) More rapid peristaltic waves

C) Constriction of the internal anal sphincter

D) Dilation of external urinary sphincter

E) Contraction in bronchiolar smooth muscle

22. Which one of the following is untrue regarding clinical uses of beta-adrenergic blockers?

A) Hypertension

B) Cardiac arrhythmias

C) Variant angina

D) Heart failure

E) Myocardial infarction

23. Which one of the following is mainly alpha-1 agonist?

A) Clonidine

B) Oxymetazoline

C) Salbutamol

D) isoproterenol

E) Dopamine

24. (:Stimulation of the cervical sympathetic division causes الفص

A) Contraction of Pupil dilator muscle

B) Vasodilation of skin blood vessels

C) Trophic salivary secretion

D) a and c are correct

25. (:In Horner's syndrome, all correct, except الفص

A) The upper eye lid is elevated

B) The skin is dry and red

C) Miosis

D) Enophthalmos



11. Parasympathetic Nervous System

1. The effects of parasympathetic stimulation include:

- A) Secretion of sweat.
- B) Contraction of the rectum.
- C) Ejaculation.
- D) Constriction of gastrointestinal sphincters.
- E) Dilation of the bronchioles.

2. Which of the following is a characteristic of postganglionic parasympathetic neurons that make them different from postganglionic sympathetic neurons?

- A) They can be excited by dopamine.
- B) Postganglionic parasympathetic nerve cell bodies are located in ganglia close to the spinal cord.
- C) Most postganglionic parasympathetic nerves release the neurotransmitter acetylcholine.
- D) Some postganglionic parasympathetic nerves release the neurotransmitter norepinephrine.
- E) They are most active under conditions of stress, such as fight/flight.

3. The parasympathetic system:

- A) Has a short preganglionic fiber.
- B) Secretes dopamine.
- C) Controls most of the movements and secretions of the gut.
- D) Brings about an increase in heart rate during exercise.

4. The efferent impulses that stimulate salivary secretion proceed along all the following nerves EXCEPT the:

- A) Chorda tympani.
- B) Sympathetic nerves.
- C) Oculomotor nerve.
- D) Glossopharyngeal nerve.

5. The parasympathetic nervous system affects all of these organs EXCEPT:

- A) Heart.
- B) Pupillary smooth muscles.
- C) Salivary glands.
- D) Adrenal glands.

6. Which of the following structures is an example where double innervation does NOT apply:

- A) The heart.
- B) The stomach.
- C) Splenic capsule.
- D) The bladder.
- E) The intestines.

7. An injury to the first 4 sacral segments of the spinal cord produces:

- A) Vasodilatation in the cutaneous blood vessels.
- B) Hyperglycaemia.
- C) Disturbance of sweat secretion.
- D) Lack of erection of the penis in males & clitoris in females.
- E) Relaxation of the internal anal & urethral sphincters.

8. Stimulation of the pelvic nerve (sacral autonomic) produces:



- A) Relaxation of the wall of the rectum & contraction of the internal anal sphincter.
- B) Vasoconstriction of the blood vessels in the external genitalia causing erection of the penis.
- C) Contraction of smooth muscles in the male sexual passages causing ejaculation.
- D) Relaxation of the internal urethral and anal sphincters.

9. Parasympathetic tone is mostly responsible for:

- A) Decreasing the blood flow to the gut (GIT/alimentary canal).
- B) Maintenance of the systemic arterial blood pressure constant.
- C) Keeping the fasting blood glucose level constant.
- D) Maintaining resting heart rate within normal.
- E) Causing bronchodilatation.

10. Parasympathetic innervation is antagonistic to sympathetic innervation on:

- A) Constrictor pupillae muscle.
- B) Cutaneous blood vessels.
- C) Salivary glands secretion.
- D) The act of micturition.

11. Parasympathetic preganglionic neurons:

- A) Are much shorter than the postganglionic neurons.
- B) Are more numerous than the postganglionic neurons.
- C) Secrete the same transmitter as the postganglionic neurons.
- D) Originate from all the sacral segments of the spinal cord.

12. The vagus parasympathetic nerve supplies all the following structures EXCEPT:

- A) Gall bladder.
- B) Rectum.
- C) Stomach.
- D) Pancreas.

13. Motility of the intestine is increased by:

- A) Adrenaline.
- B) Parasympathetic stimulation.
- C) Sympathetic blockers.
- D) Adrenal medulla secretions.

14. Concerning the cranial parasympathetic outflow:

- A) Preganglionic outflow in the oculomotor nerve arises from the ciliary ganglion.
- B) Preganglionic outflow in the glossopharyngeal nerve arises from the superior salivary nucleus.
- C) They relax the sphincter of Oddi.
- D) They cause bronchodilatation.

15. The parasympathetic nervous system affects all of these organs EXCEPT:

- A) Heart.
- B) Salivary glands.
- C) Pupillary smooth muscles.
- D) Adrenal glands.

16. Salivary secretion increases on stimulation of:

- A) Trigeminal nerve.
- B) Glossopharyngeal nerve.
- C) Vagus nerve.
- D) Hypoglossal nerve.



Additional Questions (from supplementary question bank)

1. Stimulation of the pelvic nerve causes:

- A) Contraction of the bronchial muscle.
- B) Contraction of the internal urethral & rectal (anal) sphincters.
- C) V.D. of the blood vessels in the external genitalia.
- D) Stoppage of prostatic secretion.

2. Stimulation of the pelvic nerve (sacral autonomic) produces:

- A) Relaxation of the wall of rectum & contraction of the internal anal sphincter.
- B) V.C. of the blood vessels in the external genitalia causing erection of penis.
- C) Contraction of the urinary bladder wall & relaxation of the internal urethral sphincter.
- D) Contraction of smooth muscles in the male sexual passages causing ejaculation.

3. Concerning the pelvic nerve, all the following is true EXCEPT:

- A) It is the nerve of erection (=nervus erigentes).
- B) It is motor to the urinary bladder & distal half of the large intestine.
- C) It is inhibitory to both the internal anal & urethral sphincters.
- D) It is motor to the wall of gall bladder.

4. The functions of parasymp. innervation to the salivary glands is:

- A) V.C. & secretion. b) V.D. & secretion.
- C) V.D. & inhibition of secretion. d) Secretion of scanty viscous saliva.

5. Salivary secretion increases on stimulation of:

- A) Trigeminal nerve. c) Vagus nerve.
- B) Glossopharyngeal nerve. d) Hypoglossal nerve.

6. Stimulation of pelvic nerve causes:

- A) Contraction of the internal urethral sphincter.
- B) Contraction of the internal anal sphincter.
- C) Contraction of bronchial muscle.
- D) Dilatation of the blood vessels of the external genitalia.

7. Muscarinic effects of acetylcholine can be inhibited by

- A) Nicotine c) Cholinesterase inhibitors
- B) Muscarine d) Physostigmine
- E) Atropine

8. Which of the following is an expected symptom of poisoning with Organophosphorous compound?

- A) Bronchospasm b) Mydriasis
- C) Tachycardia d) Constipation
- E) Dry mouth

9. (:Stimulation of the vagus nerve producesالقصر(

- A) Contraction of the spleen b) Decrease the ventricular contraction
- C) Decrease the heart rate d) Dilatation of the bronchioles

10. (:Stimulation of the pelvic nerve may lead toالقصر(

- A) Decrease heart rate b) Relaxation of smooth muscles of bronchi and bronchioles
- C) Ejaculation of semen d) Micturition

1. c 2. C 3. D 4. B 5. B 6. D 7. E 8. A 9. C 10. D

**Department (Autonomic Nervous System)**

1. The two major divisions of the autonomic system are the sympathetic and parasympathetic. Where do parasympathetic fibers originate?
 - A) The thoraco-lumbar spinal region.
 - B) The cranial and sacral regions.
2. Concerning the ganglia, all the following is true EXCEPT:
 - A) They act as distributing centers.
 - B) They are 3 types.
 - C) Their chemical transmitter is acetylcholine.
 - D) Their receptors are blocked by atropine.
3. Collateral ganglia of the sympathetic nervous system include the:
 - A) Adrenal medullary cells.
 - B) Coeliac ganglion.
 - C) Paravertebral ganglia.
 - D) The basal ganglia.
4. Regarding the sympathetic nervous system:
 - A) Preganglionic fibers release noradrenalin.
 - B) Postganglionic fibers release noradrenalin.
 - C) It is stimulated by food intake.
 - D) Release of acetylcholine by the preganglionic fibers.
 - E) Has longer preganglionic fibers than postganglionic fibers.
5. Preganglionic sympathetic neurons differ from preganglionic parasympathetic neurons in that:
 - A) They are shorter.
 - B) They release acetylcholine.
 - C) They emerge from the lateral horn of the spinal cord.
 - D) They release noradrenaline.
6. The major difference between the autonomic & somatic division of the nervous system is:
 - A) Only the somatic nerves innervate skeletal muscle.
 - B) The somatic nerves only originate from the thoraco-lumbar segments of the spinal cord.
 - C) Autonomic nerves only originate from the thoraco-lumbar segments of the spinal cord.
 - D) Only the autonomic nerves innervate blood vessels.
7. The sympathetic division of the autonomic nervous system is characterized by:
 - A) Localized actions affecting few systems.
 - B) Thoracolumbar outflow from the spinal cord.
 - C) Short postganglionic fibers.
 - D) Adrenergic preganglionic fibers.
8. The autonomic nervous system differs from the somatic nervous system in:
 - A) Having one efferent neuron.
 - B) Preganglionic neurons are located in the anterior horn of the spinal cord.
 - C) Efferent nerves have either excitatory or inhibitory effects on the effector organs.
 - D) Postganglionic neurons are located in the lateral horn of the spinal cord.
9. Drooping of the eyelid (ptosis) is caused by damage to:
 - A) The third cranial nerve.
 - B) The parasympathetic supply to the eye.
 - C) The sympathetic supply to the eye.
 - D) Lesion in lumbar 2, 3, 4.

**10. Sympathetic stimulation of the iris causes:**

- A) Astigmatism.
- B) Pupillary constriction.
- C) Pupillary dilation.
- D) Vitreous secretion.

11. Horner's syndrome causes:

- A) Increased sweating.
- B) Drooping of the upper eyelid (= ptosis).
- C) Vasoconstriction of the skin blood vessels (leading to pallor).
- D) Mydriasis (= dilatation of the pupil).
- E) Inhibition of salivary gland secretion.

12. The functions of sympathetic innervation to the salivary glands are:

- A) Vasodilatation & secretion.
- B) Vasodilatation & secretion.
- C) Vasodilatation & inhibition of secretion.
- D) Secretion of scanty viscous saliva.

13. All preganglionic autonomic neurons secrete:

- A) Epinephrine.
- B) Dopamine.
- C) Acetylcholine.
- D) Nicotine.

14. Sympathetic system:

- A) Has short postganglionic neurons.
- B) Fibers are carried in the vagus nerve.
- C) Produces nicotine at its nerve endings.
- D) Has a thoraco-lumbar outflow from the spinal cord.

15. Preganglionic sympathetic nerve that innervates the eye:

- A) Superior colliculi.
- B) Thoracic segments of the spinal cord.
- C) Superior cervical ganglion.
- D) Ciliary ganglion.

16. Stimulation of the cervical sympathetic nerve causes:

- A) Drooping of the upper eyelid.
- B) Small-sized pupil.
- C) Exophthalmos.
- D) Relaxation of the ciliary muscle of the eye.

17. All of the following about Horner's syndrome is true EXCEPT:

- A) Manifested by dryness & redness of skin.
- B) Group of signs including ptosis & miosis.
- C) Is due to injury of LHCS present in thoracic 1-2 spinal segments.
- D) Causes pallor of skin due to vasoconstriction.

18. Among symptoms of Horner's syndrome there is:

- A) Increased sweating.
- B) Dilatation of pupil.
- C) Drooping of the upper eyelid.
- D) Vasoconstriction of skin blood vessels.



19. The autonomic ganglia have all the following characters EXCEPT:

- A) Are three types: lateral, collateral and terminal.
- B) Are sites of relay of autonomic preganglionic and postganglionic fibers.
- C) Are all present adjacent to the spinal cord.
- D) Are functioning as distributing centers.

20. Stimulation of the greater splanchnic nerve produces:

- A) Increased motility of the stomach & upper half of the small intestine.
- B) Inhibition of micturition.
- C) Increased blood levels of both glucose & catecholamines.
- D) Vasoconstriction of blood vessels in the erectile tissue of both males & females.
- E) Increased secretion by both the stomach & pancreas.

21. Stimulation of the sympathetic nervous system causes:

- A) Contraction of the ciliary muscle for near vision.
- B) Mostly generalized actions affecting many systems.
- C) Bronchoconstriction.
- D) Decreased glycogenolysis & lipolysis.
- E) Increased gastric HCl & pepsin secretion.

22. Concerning the adrenal medulla, all the following is true EXCEPT:

- A) Its secretion is increased in emergency conditions.
- B) It secretes epinephrine 80% & norepinephrine 20%.
- C) It augments and prolongs the effects of sympathetic stimulation.
- D) It is a modified sympathetic ganglion.
- E) It receives preganglionic nerve supply from the anterior horn cells of the lower 6 thoracic segments of the spinal cord.

23. Sympathetic tone serves the following autonomic functions:

- A) Maintenance of basal cardiac rhythm.
- B) Maintenance of basal gastrointestinal motility.
- C) Maintenance of normal arterial blood pressure.
- D) Maintenance of basal sleep rhythm.

24. The sympathetic nervous system is:

- A) Activated by exposure to cold weather.
- B) Inhibited by exposure to hot weather.
- C) Concerned with initiation of micturition.
- D) Involved in erection of external genital organs.

25. Mass discharge of the sympathetic nervous system is:

- A) Regulated by a center in the thoracic and lumbar segments of the cord.
- B) Initiated and coordinated by higher centers in the hypothalamus.
- C) Associated with increased activity of the gastrointestinal tract.
- D) Harmful and may cause failure of the cardiovascular & respiratory system.

26. All of the following about the adrenal medulla is true EXCEPT:

- A) It is a sympathetic ganglion.
- B) Secretes more noradrenaline than adrenaline.
- C) It releases epinephrine into circulating blood.



D) It's innervated by sympathetic cholinergic fibers.

27. Sympathetic supply to the pelvic viscera causes:

A) Micturition.

B) Dilatation of vessels in external genitalia.

C) Defecation.

D) Ejaculation.

28. Postganglionic sympathetic fibers starting from the cervical and upper four thoracic ganglia cause:

A) Secretion of adrenomedullary hormones.

B) Dilatation of the coronary vessels.

C) Contraction of the splenic capsule.

D) Narrowing of the respiratory passages.

29. All the following structures are supplied by the sympathetic system only, EXCEPT:

A) Ventricles of the heart.

B) Constrictor pupillae muscle.

C) Adrenal medulla.

D) Skin.

30. Increased activity of the sympathetic nervous system causes:

A) Penile erection.

B) Pupillary constriction.

C) Accommodation for near vision.

D) Bronchiolar dilatation.

31. Stimulation of the lesser splanchnic nerve causes:

A) Micturition.

B) Defecation.

C) Ejaculation.

D) Erection.

32. Stimulation of the greater splanchnic nerve produces:

A) Increased blood glucose level.

B) Decreased blood volume.

C) Secretion of epinephrine 20% & norepinephrine 80% from the adrenal medulla.

D) Increased gastrointestinal motility.

33. Activation of the sympathetic nervous system produces the following EXCEPT:

A) Bronchodilatation.

B) Tachycardia.

C) Mydriasis (dilatation of pupil).

D) Contraction of the detrusor muscle of the urinary bladder wall.

34. Choose the correct statement:

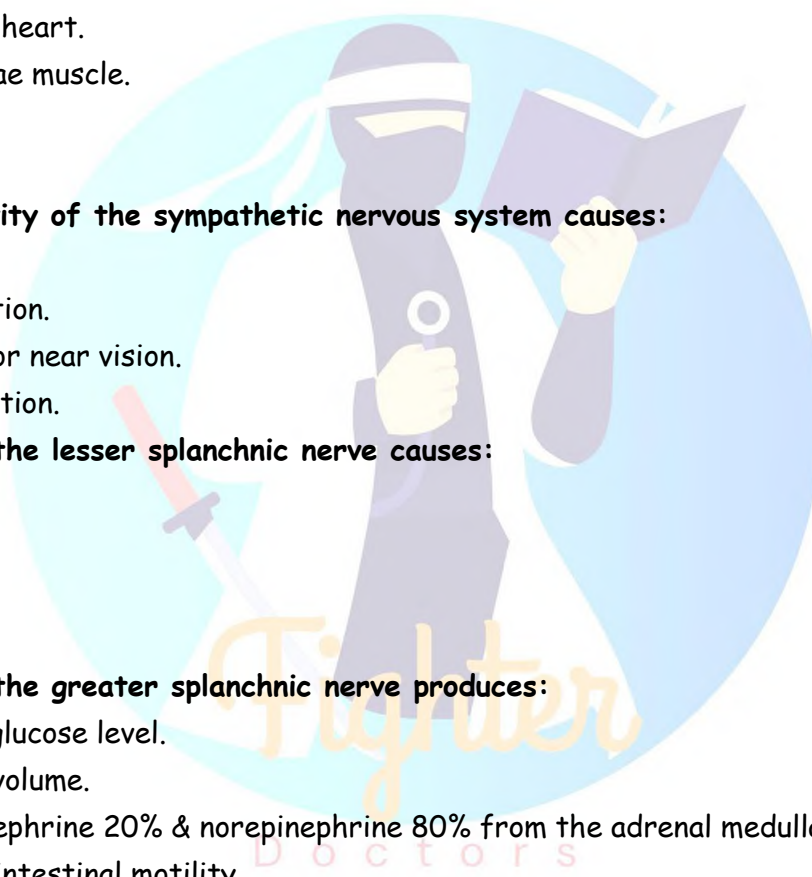
A) Postganglionic fibers starting from the superior cervical ganglion cause constriction of the pupils.

B) The sympathetic nervous system supplies every organ in the body.

C) Sympathetic supply to the lungs causes bronchoconstriction.

D) Stimulation of the greater splanchnic nerve results in adrenomedullary secretion.

35. Stimulation of the greater splanchnic nerve produces all the following EXCEPT:





- A) Secretion of catecholamines from the suprarenal medulla.
- B) Ejaculation.
- C) Increased blood glucose level.
- D) Decreased gastrointestinal motility.

36. Generalized sympathetic stimulation causes:

- A) Inhibition of sweating.
- B) Increased gastric motility.
- C) Decreased blood sugar levels.
- D) Raised arterial blood pressure.

37. Stimulation of the pelvic nerve causes:

- A) Contraction of the bronchial muscle.
- B) Contraction of the internal urethral & rectal (anal) sphincters.
- C) Vasodilatation of the blood vessels in the external genitalia.
- D) Stoppage of prostatic secretions.

38. Which of the following may be produced by parasympathetic stimulation:

- A) Acceleration of heart.
- B) Contraction of urinary bladder wall.
- C) Vasodilatation of skeletal muscle blood vessels.
- D) Dilatation of pupil.

39. The efferent impulses that stimulate salivary secretion proceed along all the following nerves EXCEPT the:

- A) Chorda tympani.
- B) Sympathetic nerves.
- C) Oculomotor nerve.
- D) Glossopharyngeal nerve.

40. The parasympathetic nervous system affects all of these organs EXCEPT:

- A) Heart.
- B) Pupillary smooth muscles.
- C) Salivary glands.
- D) Adrenal glands.

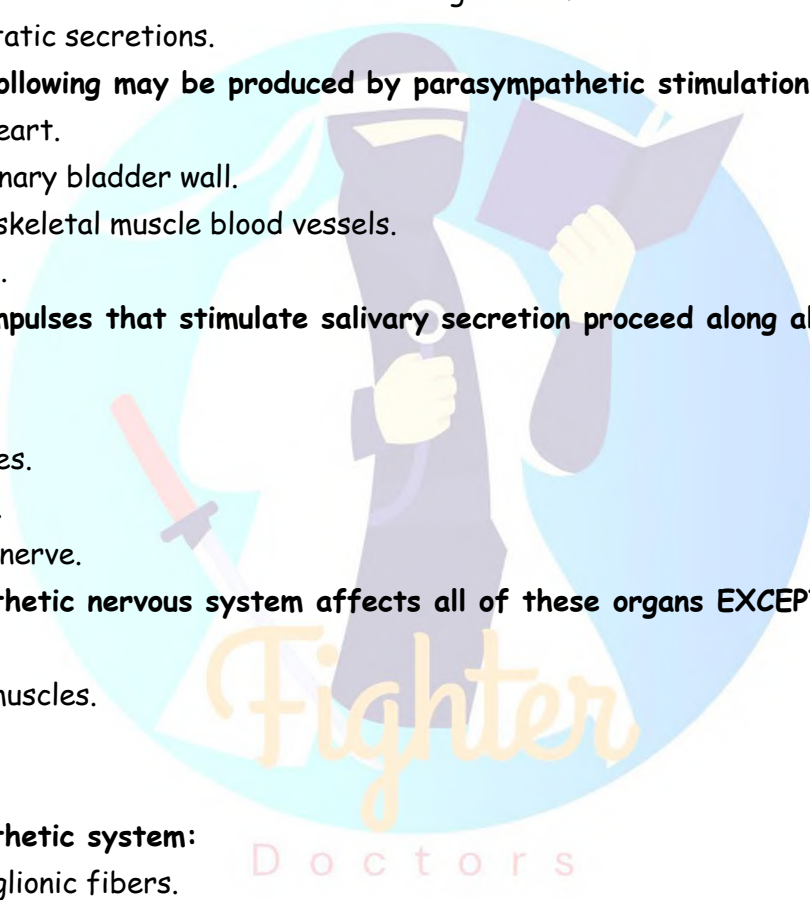
41. The parasympathetic system:

- A) Has short preganglionic fibers.
- B) Secretes dopamine.
- C) Controls most of the movements and secretions of the gut.
- D) Brings about an increase in heart rate during exercise.

42. The effects of parasympathetic stimulation include:

- A) Ejaculation.
- B) Secretion of sweat.
- C) Contraction of the rectum.
- D) Constriction of gastrointestinal sphincters.
- E) Dilation of the bronchioles.

43. Which of the following is a characteristic of postganglionic parasympathetic neurons that make them different from postganglionic sympathetic neurons?





- A) They can be excited by dopamine.
- B) Postganglionic parasympathetic nerve bodies are located in ganglia close to the spinal cord.
- C) Most postganglionic parasympathetic nerves release the neurotransmitter acetylcholine.
- D) Some postganglionic parasympathetic nerves release the neurotransmitter norepinephrine.
- E) They are most active under conditions of stress, such as fight/flight.

44. Vagal stimulation causes:

- A) Tachycardia.
- B) Dilatation of the bronchi and bronchioles.
- C) Inhibition of plain muscles in the stomach and small intestine.
- D) Increase in gastric and pancreatic secretion.

45. Motility of the intestine is increased by:

- A) Adrenaline.
- B) Parasympathetic stimulation.
- C) Sympathetic blockers.
- D) Adrenal medulla secretions.

46. Concerning the cranial parasympathetic outflow:

- A) Preganglionic outflow in the oculomotor nerve arises from the ciliary ganglion.
- B) Preganglionic outflow in the glossopharyngeal nerve arises from the superior salivary nucleus.
- C) They relax the sphincter of Oddi.
- D) They cause bronchodilatation.

47. Which of the following structures is an example where double innervation does NOT apply?

- A) The heart.
- B) The bladder.
- C) The stomach.
- D) The intestines.
- E) Splenic capsule.

48. Generalized sympathetic activity is characterized by all the following EXCEPT:

- A) Contraction of the radial muscle in the iris.
- B) A raised blood level of catecholamines.
- C) Constriction of blood vessels in the skin.
- D) A fall in the blood glucose level.

49. All the following statements are true EXCEPT:

- A) Accommodation to near vision is mediated by increased acetylcholine secreted in the ciliary muscle of the eye.
- B) Vagotomy may cause the heart rate to increase significantly.
- C) Arteriolar vasoconstriction present at rest may be due to a basal secretion of catecholamines from the adrenal medulla.
- D) Noradrenaline is not secreted by the adrenal medulla.

50. Stimulation of the pelvic nerve (sacral autonomic) produces:

- A) Relaxation of the wall of the rectum & contraction of the internal anal sphincter.
- B) Vasoconstriction of the blood vessels in the external genitalia causing erection of the penis.
- C) Contraction of smooth muscles in the male sexual passages causing ejaculation.
- D) Relaxation of the internal urethral and anal sphincters.



51. An injury to the first 4 sacral segments of the spinal cord produces:

- A) Hyperglycemia.
- B) Vasodilatation in the cutaneous blood vessels.
- C) Disturbance of sweat secretion.
- D) Lack of erection of the penis in males & clitoris in females.
- E) Relaxation of the internal anal & urethral sphincters.

52. About the vagus nerve, all the following is true EXCEPT:

- A) It originates mainly from the medulla oblongata.
- B) It contains parasympathetic preganglionic fibers.
- C) It has no direct effect on ventricular contraction.
- D) It causes increased peristaltic movements of the intestinal wall.
- E) It produces micturition.

53. Parasympathetic preganglionic neurons:

- A) Are much shorter than the postganglionic neurons.
- B) Are more numerous than the postganglionic neurons.
- C) Secrete the same transmitter as the postganglionic neurons.
- D) Originate from all the sacral segments of the spinal cord.

54. Vagus parasympathetic nerve supplies all the following structures EXCEPT:

- A) Gall bladder.
- B) Stomach.
- C) Rectum.
- D) Pancreas.

55. Facial nerve:

- A) Stimulation increases the power of the lens of the eye.
- B) Supplies the lachrymal gland.
- C) Supplies the parotid salivary gland.
- D) Contains sympathetic postganglionic fibers.

56. Parasympathetic tone is mostly responsible for:

- A) Decreasing the blood flow to the gut (GIT/alimentary canal).
- B) Maintenance of the systemic arterial blood pressure constant.
- C) Keeping the fasting blood glucose level constant.
- D) Maintaining resting heart rate within normal.
- E) Causing bronchodilatation.

57. Parasympathetic innervation is antagonistic to sympathetic innervation on:

- A) Constrictor pupillae muscle.
- B) Salivary gland secretion.
- C) The act of micturition.
- D) Cutaneous blood vessels.

58. Sympathetic and parasympathetic systems have cooperative effects on:

- A) Cardiac activity.
- B) Sexual activity.
- C) Gastrointestinal activity.
- D) The diameter of the pupil.



59. About acetylcholine, all the following is true EXCEPT:

- A) It is synthesized & released by some sympathetic postganglionic nerve fibers.
- B) It is the chemical transmitter released by all preganglionic nerve fibers.
- C) It is broken down enzymatically by the choline esterase enzyme.
- D) It is liberated by all parasympathetic nerve fibers.
- E) It makes reading difficult because of relaxation of the ciliary muscle.

60. An injection of atropine causes all the following effects EXCEPT:

- A) An increase in the heart rate.
- B) Pupillo-dilatation.
- C) Difficult micturition.
- D) Constriction of the bronchi.
- E) Diminished intestinal motility.

61. About acetylcholine, all the following is true EXCEPT:

- A) Its synthesis is catalyzed by the choline acetyl-transferase enzyme.
- B) Its muscarinic actions can be blocked by atropine.
- C) Its nicotinic actions can be blocked by large doses of nicotine.
- D) It is not secreted by any sympathetic postganglionic nerve fibers.

62. Acetylcholine is secreted at all the following sites EXCEPT:

- A) Postganglionic sympathetic nerve terminals at skeletal muscles.
- B) Postganglionic parasympathetic nerve terminals.
- C) Preganglionic sympathetic & parasympathetic nerve terminals.
- D) Myoneural junctions.
- E) Postganglionic sympathetic noradrenergic nerve terminals.

63. All the following statements are correct EXCEPT:

- A) Stimulation of the greater splanchnic nerve causes adrenomedullary secretion.
- B) Acetylcholine is normally present at low concentrations in the blood.
- C) Intravenous injection of acetylcholine does not cause exactly the same effects as parasympathetic stimulation.
- D) Stimulation of postganglionic nerve fibers arising from the collateral ganglia causes secretion from the adrenal (suprarenal) medulla.

64. Parasympathetic postganglionic cholinergic fibers innervate all the following EXCEPT:

- A) Salivary glands.
- B) Lacrimal glands.
- C) Sweat glands.
- D) Exocrine cells of pancreas.

65. Parasympathetic innervation is antagonistic to sympathetic innervation on:

- A) Constrictor pupillae muscle.
- B) Cutaneous blood vessels.
- C) Salivary gland secretion.
- D) The act of micturition.

66. Acetylcholine:

- A) Binds with nicotinic receptors in cells of autonomic effector tissues.
- B) Stimulates nicotinic receptors, but inhibits muscarinic receptors.



- C) Synthesis is catalyzed by choline esterase enzyme.
- D) Is released in response to Ca^{++} influx into synaptic knobs.

67. Inhibitors of the choline esterase enzyme:

- A) Enhance the effects of parasympathetic stimulation.
- B) Inhibit the effect of parasympathetic stimulation.
- C) Inhibit acetylcholine release by cholinergic nerve terminals.
- D) Block the action of acetylcholine in autonomic ganglia.

68. An abnormally high level of acetylcholine in tissues all over the body is due to:

- A) Exposure to stress.
- B) Widespread physiological activation of the parasympathetic nervous system.
- C) Increased activity of muscarinic receptors.
- D) Decreased activity of cholinesterase enzyme.

69. Administration of a muscarinic receptor antagonist leads to:

- A) Increased sweat secretion.
- B) Decreased sweat secretion.
- C) Increased gastric motility.
- D) Bronchoconstriction.

70. Which of the following visceral tissues receives sympathetic postganglionic neurons only?

- A) Adrenal medulla.
- B) Sweat glands.
- C) Anal sphincters.
- D) Iris.

71. Neurotransmitters released by the parasympathetic preganglionic and postganglionic nerve fibers are:

- A) Acetylcholine and norepinephrine.
- B) Acetylcholine and epinephrine.
- C) Norepinephrine and acetylcholine.
- D) Acetylcholine and acetylcholine.

72. Acetylcholine is the chemical transmitter at all sites EXCEPT:

- A) All neuromuscular junctions in the somatic nervous system.
- B) All postganglionic sympathetic endings.
- C) All autonomic ganglia.
- D) All postganglionic parasympathetic effector endings.
- E) Any sites that are blocked by atropine.

73. The substances that block neuromuscular transmission include:

- A) Neomycin.
- B) Curare.
- C) Epinephrine.
- D) Neostigmine.

74. Drugs which augment parasympathetic activity do NOT include:

- A) Diisopropyl-fluoro-phosphate (DFP).
- B) Isoproterenol.
- C) Pilocarpine.



D) Prostigmine.

E) Muscarine.

75. Regarding acetylcholine:

A) It can be blocked by atropine at the motor end plates.

B) It is released at the parasympathetic ganglia only.

C) It is synthesized from active acetate and the base choline.

D) It is inactivated by the choline esterase enzyme, of which there are 3 types.

76. Blockade of autonomic cholinergic nerve endings causes:

A) An increase in sweat production.

B) An increase in the resting heart rate.

C) Excessive salivation.

D) Over-activity of the intestine.

77. A frightened (afraid) person might present with:

A) Decreased heart rate.

B) Excessive salivation.

C) Bronchoconstriction.

D) Pupil dilatation.

E) Increased intestinal motility.

78. Sympathetic postganglionic fibers:

A) Are very short nerve fibers.

B) Originate at the lateral horn cells of the thoracolumbar region.

C) Secrete epinephrine that acts on target organ adrenergic receptors.

D) Secrete norepinephrine at the myoneural junction.

E) Can release acetylcholine to stimulate sweat secretion.

79. Vagus nerve stimulation leads to:

A) Decrease in pupil size.

B) Vasodilatation in most of the body arterioles.

C) Bronchodilatation.

D) Bradycardia.

E) Urine retention.

80. Adrenal medulla is:

A) Stimulated by the vagus nerve to release acetylcholine.

B) The main source of epinephrine.

C) Receives sympathetic postganglionic adrenergic fibers.

D) Blocked by atropine injection.

E) A parasympathetic ganglion.

81. Parasympathetic nervous system is responsible for:

A) Vasomotor tone.

B) Ability to read a book from a near distance.

C) Glycogenolysis during fasting.

D) Sweat production.

E) Increasing the heart rate.

82. Reversible acetylcholine esterase inhibitors cause:

A) Tachycardia.

B) Dry mouth.

C) Bronchoconstriction.

D) Mydriasis.

E) Constipation.



83. Sympathetic and parasympathetic systems have antagonistic actions in:

- A) Ventricles.
- B) Adrenal medulla.
- C) SA node.
- D) Salivary glands.
- E) Lacrimal gland.

84. Nicotinic action of acetylcholine:

- A) Is blocked by atropine.
- B) Is excitatory to the heart.
- C) Is stimulatory to autonomic ganglia.
- D) Is not affected by curare.
- E) Causes micturition.

85. Injection of atropine is liable to cause:

- A) Excessive salivation.
- B) Diarrhea.
- C) Bronchoconstriction.
- D) Bradycardia.
- E) Constipation.

86. Beta-1 receptor stimulation causes:

- A) Closure of the internal urethral sphincter.
- B) Bronchodilatation.
- C) Tachycardia.
- D) Mydriasis.
- E) Adrenal medulla stimulation.

87. The alpha adrenergic receptors mediate all the following effects EXCEPT:

- A) Vasoconstriction.
- B) Pupil dilatation.
- C) Spleen contraction.
- D) Sphincteric contraction.
- E) Increased cardiac activity.

88. Epinephrine (Adrenaline):

- A) Is the chemical transmitter of sympathetic postganglionic fibers.
- B) Is secreted with norepinephrine from the adrenal medulla at an equal rate.
- C) Helps treatment of bronchial asthma.
- D) Has more vasoconstrictor effect on blood vessels than norepinephrine.
- E) Its effects on alpha receptors are greater than on beta receptors.

1-B, 2-D, 3-B, 4-B&D, 5-A, 6-A, 7-B, 8-C, 9-C, 10-C, 11-B, 12-D, 13-C, 14-D, 15-B, 16-D, 17-D, 18-C, 19-C, 20-C, 21-B, 22-E, 23-C, 24-A, 25-B, 26-B, 27-D, 28-C, 29-B, 30-D, 31-C, 32-A, 33-D, 34-D, 35-B, 36-D, 37-C, 38-B, 39-C, 40-D, 41-C, 42-C, 43-D, 44-B, 45-C, 46-E, 47-D, 48-D, 49-D, 50-D, 51-D, 52-E, 53-C, 54-C, 55-B, 56-D, 57-A, 58-B, 59-E, 60-D, 61-D, 62-E, 63-D, 64-C, 65-A, 66-D, 67-A, 68-D, 69-B, 70-B, 71-D, 72-B, 73-B, 74-B, 75-C, 76-B, 77-D, 78-E, 79-D, 80-B, 81-B, 82-C, 83-C, 84-E, 85-E, 86-C, 87-E, 88-C



Dr. Sanad Questions

1. The major difference between the autonomic & somatic divisions of the nervous system is:

- A) Only the somatic nerves innervate skeletal muscle.
- B) The somatic nerves only originate from the thoraco-lumbar segments of the spinal cord.
- C) The autonomic nerves only originate from the thoraco-lumbar segments of the spinal cord.
- D) Only the autonomic nerves innervate blood vessels.

2. The autonomic nervous system differs from the somatic nervous system in having:

- A) A connector neuron which comes outside the CNS (preganglionic neuron) & a postganglionic neuron which supplies the organ.
- B) Origin from all segments of the spinal cord.
- C) A double innervation to every organ.
- D) A complete control on skeletal muscles.

3. The autonomic ganglia:

- A) Are 5 types.
- B) Are the site of relay of afferent neurons.
- C) Function as distributing centers.
- D) Are located inside the CNS.

4. All of the following about the sympathetic nervous system is true EXCEPT:

- A) All its preganglionic fibers are cholinergic (secreting Ach).
- B) It is involved in stress and emergency conditions.
- C) It has a vasodilator effect on skeletal muscle blood vessels during rest.
- D) Its stimulation increases the cardiac properties.

5. The preganglionic sympathetic nerve that innervates the eye originates from:

- A) Superior colliculi.
- B) Thoracic segments of the spinal cord.
- C) Superior cervical ganglion.
- D) Ciliary ganglion.

6. Postganglionic sympathetic fibers starting from the cervical and upper four thoracic ganglia cause:

- A) Secretion of the adrenomedullary hormones.
- B) Contraction of the splenic capsule.
- C) Dilatation of the coronary vessels.
- D) Narrowing of the respiratory passages.

7. Concerning the adrenal medulla, all the following is true EXCEPT:

- A) It acts only in emergency conditions.
- B) It secretes epinephrine 80% and norepinephrine 20%.
- C) It is a modified sympathetic ganglion.
- D) It receives preganglionic nerve supply from the anterior horn cells of the lower 6 thoracic segments of the spinal cord.

8. The sympathetic division of the autonomic nervous system is characterized by:

- A) Localized actions affecting few systems.
- B) Short postganglionic fibers.
- C) Thoracolumbar outflow from the spinal cord.



D) Adrenergic preganglionic fibers.

9. The sympathetic supply cannot be responsible for:

- A) Constriction of the pupil.
- B) Erection of hair.
- C) Secretion of sweat glands.
- D) Dilatation of skeletal muscle blood vessels.

10. Sympathetic supply to the pelvic viscera causes:

- A) Micturition.
- B) Dilatation of vessels in external genitalia.
- C) Defecation.
- D) Ejaculation.

11. All the following structures are supplied by the sympathetic system only, EXCEPT:

- A) Ventricles of the heart.
- B) Constrictor pupillae muscle.
- C) Adrenal medulla.
- D) Skin.

12. Preganglionic autonomic fibers are of:

- A) Type A (alpha).
- B) Type B.
- C) Type A (delta).
- D) Type C.

13. Stimulation of the cervical sympathetic nerve causes:

- A) Dropping of the upper eyelid.
- B) A small-sized pupil.
- C) Enophthalmos.
- D) Relaxation of the ciliary muscle of the eye.
- E) All of the above.

14. Stimulation of the sympathetic nervous system causes:

- A) Contraction of the ciliary muscle for near vision.
- B) Bronchoconstriction.
- C) Generalized actions affecting many systems.
- D) Decreased glycogenolysis & lipolysis.

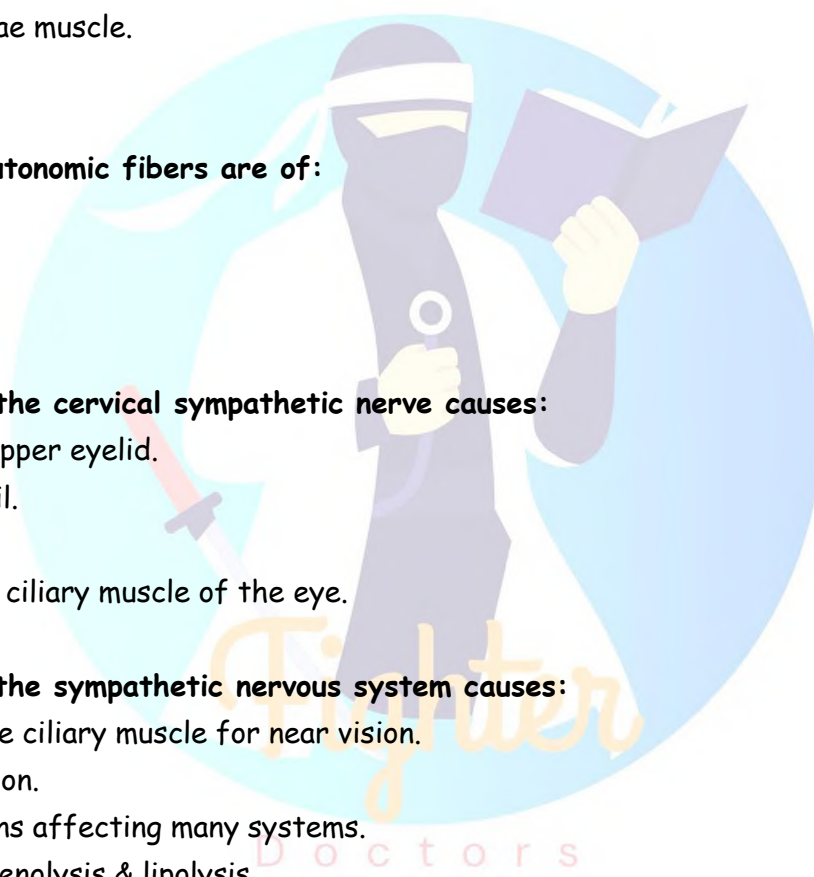
15. The sympathetic supply cannot be responsible for:

- A) Constriction of the pupil.
- B) Erection of hair.
- C) Secretion of eccrine sweat glands.
- D) Dilatation of skeletal muscle blood vessels.

16. A lesion in the cervical sympathetic chain (Horner's syndrome) causes:

- A) Increased sweating.
- B) Dropping of the upper eyelid (= ptosis).
- C) Vasoconstriction of the skin blood vessels (leading to pallor).
- D) Mydriasis (= dilatation of the pupil).

17. Stimulation of the greater splanchnic nerve produces:





- A) Increased motility of the stomach & upper half of the small intestine.
- B) Inhibition of micturition.
- C) Increased blood level of both glucose & catecholamines.
- D) Vasoconstriction of the blood vessels in the erectile tissue of both males & females.

18. Concerning the autonomic ganglia, all the following is true EXCEPT:

- A) They act as distributing centers.
- B) They are 3 types.
- C) Their chemical transmitter is acetylcholine.
- D) Their receptors are specifically blocked by atropine.

19. Which of the following organs receives both sympathetic & parasympathetic innervation?

- A) Adrenal medulla.
- B) Sweat glands.
- C) Pilomotor muscles.
- D) Lungs.

20. Increased activity of the sympathetic nervous system causes:

- A) Penile erection.
- B) Pupillary constriction.
- C) Accommodation for near vision.
- D) Bronchiolar dilatation.

21. All of the following about Horner's syndrome is true EXCEPT:

- A) Manifested by dryness & redness of skin.
- B) Group of signs including ptosis & miosis.
- C) Is due to injury of LHCs present in thoracic 1-2 spinal segments.
- D) Causes pallor of skin due to vasoconstriction.

22. Stimulation of the greater splanchnic nerve produces:

- A) Increased motility of the plain muscles of the stomach.
- B) Excessive secretion of the gastric juice by the stomach.
- C) Decreased blood levels of both glucose & free fatty acids.
- D) Secretion of epinephrine & norepinephrine by the adrenal medulla.

23. Stimulation of the greater splanchnic nerve produces:

- A) Increased blood glucose level.
- B) Decreased blood volume.
- C) Secretion of epinephrine 20% & norepinephrine 80% from the adrenal medulla.
- D) Increased gastrointestinal motility.

24. The sympathetic preganglionic nerve fibers:

- A) Leave the spinal cord in the grey rami communicantes.
- B) Originate at the autonomic ganglia.
- C) Are myelinated nerve fibers belonging to the B group of nerve fibers.
- D) Show minimal divergence in the autonomic ganglia.

25. Stimulation of the lesser splanchnic nerve causes:

- A) Micturition.
- B) Defecation.
- C) Ejaculation.

Doctors



D) Erection.

26. The following reflexes are all autonomic, EXCEPT:

A) Salivary secretion.

B) Flexion of the arm following a painful stimulus.

C) Micturition.

D) Defecation.

27. Activation of the sympathetic nervous system produces the following EXCEPT:

A) Bronchodilatation.

B) Tachycardia.

C) Mydriasis (dilatation of pupil).

D) Contraction of the detrusor muscle of the urinary bladder wall.

28. The autonomic nervous system differs from the somatic nervous system in:

A) Having one efferent neuron.

B) Preganglionic neurons are located in the anterior horn of the spinal cord.

C) Efferent nerves have either excitatory or inhibitory effects on the effector organs.

D) Postganglionic neurons are located in the lateral horn of the spinal cord.

29. Choose the correct statement:

A) Postganglionic fibers starting from the superior cervical ganglion cause constriction of the pupils.

B) The sympathetic nervous system supplies every organ in the body.

C) Sympathetic supply to the lungs causes bronchoconstriction.

D) Stimulation of the greater splanchnic nerve results in adrenomedullary secretion.

30. Activation of the sympathetic system causes:

A) Pupillary constriction.

B) Bradycardia.

C) Sweat secretion.

D) Cutaneous vasodilatation.

31. Among symptoms of Horner's syndrome there is:

A) Increased sweating.

B) Dilatation of pupil.

C) Drooping of the upper eyelid.

D) Vasoconstriction of skin blood vessels.

32. The autonomic ganglia have all the following characters EXCEPT:

A) Are three types: lateral, collateral and terminal.

B) Are sites of relay of autonomic preganglionic and postganglionic fibers.

C) Are all present adjacent to the spinal cord.

D) Are functioning as distributing centers.

33. Which of the following may be produced by parasympathetic stimulation:

A) Acceleration of the heart.

B) Contraction of the urinary bladder wall.

C) Vasodilatation of skeletal muscle blood vessels.

D) Dilatation of the pupil.

34. Stimulation of the pelvic nerve (sacral autonomic) produces:

A) Relaxation of the wall of the rectum & contraction of the internal anal sphincter.



- B) Vasoconstriction of the blood vessels in the external genitalia causing erection of the penis.
- C) Contraction of the urinary bladder wall & relaxation of the internal urethral sphincter.
- D) Contraction of smooth muscles in the male sexual passages causing ejaculation.

35. Concerning the pelvic nerve, all the following is true EXCEPT:

- A) It is the nerve of erection (nervus erigentes).
- B) It is motor to the urinary bladder & distal half of the large intestine.
- C) It is inhibitory to both the internal anal & urethral sphincters.
- D) It is motor to the wall of the gall bladder.

36. Stimulation of the pelvic nerve causes:

- A) Contraction of the bronchial muscle.
- B) Contraction of the internal urethral & rectal (anal) sphincters.
- C) Vasodilatation of the blood vessels in the external genitalia.
- D) Stoppage of prostatic secretion.

37. Vagus parasympathetic nerve supplies all the following structures EXCEPT:

- A) Gall bladder.
- B) Stomach.
- C) Rectum.
- D) Pancreas.

38. All of the following about sacral autonomic outflow is true EXCEPT:

- A) It is the nerve of defecation.
- B) It is the nerve of ejaculation.
- C) It is the nerve of micturition.
- D) It relays in terminal ganglia.

39. Generalized sympathetic stimulation causes:

- A) Inhibition of sweating.
- B) Increased gastric motility.
- C) Decreased blood sugar levels.
- D) Raised arterial blood pressure.

40. Stimulation of the greater splanchnic nerve produces all the following EXCEPT:

- A) Secretion of catecholamines from the suprarenal medulla.
- B) Increased blood glucose level.
- C) Ejaculation.
- D) Decreased gastrointestinal motility.

41. The parasympathetic nerves:

- A) Are important in emergency conditions, e.g. fight & flight reactions.
- B) Tend to have longer postganglionic fibers than preganglionic fibers.
- C) Are catabolic in their actions.
- D) Generally exert opposite effects to those of the sympathetic nerves.

42. The functions of parasympathetic innervation to the salivary glands is:

- A) Vasoconstriction & secretion.
- B) Vasodilatation & secretion.
- C) Vasodilatation & inhibition of secretion.
- D) Secretion of scanty viscous saliva.

**43. Facial nerve:**

- A) Stimulation increases the power of the lens of the eye.
- B) Supplies the lacrimal gland.
- C) Supplies the parotid salivary gland.
- D) Contains sympathetic postganglionic fibers.

44. Concerning the cranial parasympathetic outflow:

- A) Preganglionic outflow in the oculomotor nerve arises from the ciliary ganglion.
- B) Preganglionic outflow in the glossopharyngeal nerve arises from the superior salivary nucleus.
- C) They relax the sphincter of Oddi.
- D) They cause bronchodilatation.

45. Acetylcholine action at the autonomic ganglia is:

- A) Muscarinic-like action.
- B) Nicotine-like action.
- C) Blocked by atropine.
- D) Blocked by muscarine.

46. Nicotinic receptors:

- A) Accept nicotine released by nerve endings near the receptor.
- B) Are not sensitive to acetylcholine.
- C) Are blocked by atropine.
- D) Are found in sympathetic ganglia.

47. About acetylcholine, all the following is true EXCEPT:

- A) It is synthesized & released by some sympathetic postganglionic nerve fibers.
- B) It is the chemical transmitter released by the preganglionic nerve fibers.
- C) It is broken down enzymatically by the cholinesterase enzyme.
- D) It makes reading difficult because of relaxation of the ciliary muscle.

48. Stimulation of the pelvic nerve causes:

- A) Contraction of the internal urethral sphincter.
- B) Contraction of the internal anal sphincter.
- C) Contraction of the bronchial muscle.
- D) Dilatation of the blood vessels of the external genitalia.

49. An injection of atropine causes all the following effects EXCEPT:

- A) An increase in the heart rate.
- B) Pupillo-dilatation.
- C) Difficult micturition.
- D) Constriction of the bronchi.

50. About acetylcholine, all the following is true EXCEPT:

- A) Its synthesis is catalyzed by the choline acetyltransferase enzyme.
- B) Its muscarinic action can be blocked by atropine.
- C) Its nicotinic actions can be blocked by large doses of nicotine.
- D) It is not secreted by any sympathetic postganglionic nerve fibers.

51. Motility of the intestine is increased by:

- A) Adrenaline.
- B) Parasympathetic stimulation.



- C) Sympathetic blockers.
- D) Adrenal medulla secretions.

52. All of the following organs receive their parasympathetic innervation via the vagus nerves EXCEPT:

- A) Gall bladder.
- B) Atria of the heart.
- C) Proximal half of the large intestine.
- D) Rectum.

53. The nicotinic action of acetylcholine:

- A) Is blocked by atropine.
- B) Is excitatory to both intestinal movements & secretion.
- C) Is not affected by curare.
- D) Is stimulatory to the autonomic ganglia.

54. Salivary secretion increases on stimulation of:

- A) Trigeminal nerve.
- B) Glossopharyngeal nerve.
- C) Vagus nerve.
- D) Hypoglossal nerve.

55. Vagal stimulation causes:

- A) Tachycardia.
- B) Dilatation of the bronchi and bronchioles.
- C) Inhibition of plain muscles in the stomach and small intestine.
- D) Increase in gastric and pancreatic secretion.

56. Acetylcholine is secreted at all the following sites EXCEPT:

- A) Postganglionic sympathetic nerve terminals at skeletal muscle.
- B) Postganglionic parasympathetic nerve terminals.
- C) Preganglionic sympathetic & parasympathetic nerve terminals.
- D) Postganglionic sympathetic noradrenergic nerve terminals.

57. The nicotinic-like action is present in:

- A) All preganglionic nerve endings.
- B) The sympathetic postganglionic supply to the blood vessels in skeletal muscles.
- C) The parasympathetic postganglionic supply to the heart.
- D) The sympathetic postganglionic supply to the eccrine sweat glands.

58. About Prostigmine (Neostigmine), all are true EXCEPT:

- A) Acts specifically at the motor end plate.
- B) Inhibits cholinesterase enzyme.
- C) Is used in treatment of myasthenia gravis.
- D) Is an irreversible anticholinesterase.

59. The antidote of eserine is:

- A) Amphetamine.
- B) Noradrenaline.
- C) Ergotoxine.
- D) Atropine.



60. Atropine injection causes all the following EXCEPT:

- A) Pupillo-dilatation.
- B) Increased heart rate.
- C) Relaxation of skeletal muscles.
- D) Diminished salivary secretion.

61. Atropine blocks the action of acetylcholine at the following sites EXCEPT:

- A) Heart.
- B) Adrenal medulla.
- C) Sweat glands.
- D) Constrictor pupillae muscle.

62. Atropine is given before surgical operations to:

- A) Decrease heart rate.
- B) Prevent bronchial secretion.
- C) Relax skeletal muscle.
- D) Induce bronchoconstriction.

63. Muscarine-like action of acetylcholine is present in all the following EXCEPT:

- A) Sympathetic postganglionic supplying the skeletal muscle blood vessels.
- B) Parasympathetic postganglionic nerve endings to the heart.
- C) Sympathetic postganglionic supplying the eccrine sweat glands.
- D) Somatic neuromuscular junction.

64. About acetylcholine, all the following are true EXCEPT:

- A) Is synthesized in the adrenal medulla.
- B) Released by all postganglionic parasympathetic nerves.
- C) Inactivated by cholinesterase enzyme.
- D) Released by the sympathetic cholinergic postganglionic nerve fibers.

65. Which of the following is used in treatment of myasthenia gravis?

- A) Prostigmine.
- B) Eserine.
- C) Atropine.
- D) Curare.

66. Atropine:

- A) Increases the tone & motility of GIT.
- B) Decreases gastric acid secretion.
- C) Stimulates sweat secretion.
- D) Stimulates bronchial secretion.

67. Curare:

- A) Blocks the neuromuscular transmission.
- B) Competes with acetylcholine at autonomic ganglia.
- C) Prevents the release of acetylcholine at the motor end plate.
- D) Causes relaxation of smooth muscles.

68. Administration of physostigmine would be expected to produce all the following EXCEPT:

- A) Increase the secretion of gastric juice.
- B) Increase the rate of epinephrine secretion from the adrenal medulla.



- C) Cause hypotension & bradycardia.
- D) Produce lacrymation & bronchodilatation.

69. Nicotine in small doses:

- A) Acts directly on intestinal smooth muscle, causing it to contract.
- B) Decreases the release of acetylcholine in sympathetic ganglia.
- C) Increases the rate of discharge of postganglionic parasympathetic neurons.
- D) Decreases the rate of discharge of postganglionic sympathetic neurons.

70. Stimulation of alpha adrenergic receptors produces all the following EXCEPT:

- A) Contraction of the dilator pupillae muscle.
- B) Dilatation of skeletal muscle blood vessels.
- C) Constriction of cutaneous blood vessels.
- D) Ejaculation of semen.

71. The following structures respond to beta adrenergic receptor stimulation EXCEPT:

- A) Sinoatrial node.
- B) Skeletal muscle blood vessels.
- C) Dilator pupillae muscle.
- D) Bronchial smooth muscles.

72. Atropine causes:

- A) Contraction of the ciliary muscle.
- B) Dryness of mouth & skin.
- C) Bradycardia during rest.
- D) Bronchodilatation & increased bronchial secretion.

73. Concerning adrenergic receptors, all the following are true EXCEPT:

- A) Stimulation of beta-1 adrenoceptors increases the force & rate of cardiac contraction.
- B) Activation of beta-1 & beta-2 adrenoceptors results in activation of adenylyl cyclase.
- C) Stimulation of alpha adrenoceptors causes bronchodilatation.
- D) Stimulation of beta-1 receptors causes renin secretion & lipolysis.

74. Cholinergic fibers include all the following EXCEPT:

- A) Sympathetic & parasympathetic preganglionic nerve fibers.
- B) Somatic motor nerve fibers.
- C) Sympathetic nerve fibers to the adrenal medulla.
- D) Postganglionic nerve fibers to the dilator pupillae muscle.

75. Catecholamines include all the following EXCEPT:

- A) Epinephrine.
- B) Norepinephrine.
- C) Dopamine.
- D) Ephedrine.

76. Acetylcholine:

- A) Release is blocked by curare at the motor end plate.
- B) Synthesis is catalyzed by cholinesterase enzyme.
- C) Action at the motor end plate is blocked by atropine.
- D) Is the only neurotransmitter secreted by parasympathetic neurons.



77. All the following drugs would be expected to increase sympathetic discharge or mimic the effects of increased sympathetic discharge EXCEPT:

- A) Norepinephrine.
- B) Neostigmine.
- C) Amphetamine.
- D) Tyramine.

78. Noradrenaline:

- A) Increases intestinal motility.
- B) Is the neurotransmitter at the alpha-adrenergic receptors.
- C) Is released by both pre- & postganglionic sympathetic nerve endings.
- D) Is a potent vasodilator.

79. Administration of beta blockers produces all the following EXCEPT:

- A) May cause bronchoconstriction.
- B) Produces bradycardia & lowers the arterial blood pressure.
- C) Decreases the secretion of renin from the kidney.
- D) Leads to bronchodilatation.

80. Concerning epinephrine, all the following are true EXCEPT:

- A) Is synthesized in the adrenal medulla by methylation of norepinephrine.
- B) Is the chemical transmitter of adrenergic postganglionic sympathetic fibers.
- C) Is less potent than norepinephrine as a pressure agent.
- D) Helps in treatment of bronchial asthma.

81. Concerning norepinephrine, all are true EXCEPT:

- A) It contributes 20% of the suprarenal medulla secretion.
- B) It is secreted under stressful conditions.
- C) It is removed from tissues by active reuptake into sympathetic nerve endings.
- D) It is the chemical transmitter at most sympathetic preganglionic endings.

82. A drug which blocks beta adrenoceptors is likely to cause:

- A) Dilatation of bronchi.
- B) Decrease in heart rate.
- C) Ciliary muscle relaxation for far vision.
- D) Reduces blood sugar level due to stimulation of insulin secretion.

83. All of the following statements are true EXCEPT:

- A) Nicotinic receptors of acetylcholine can be blocked by a large dose of nicotine.
- B) Muscarinic receptors of acetylcholine are blocked by atropine.
- C) Alpha adrenergic receptors are blocked by phentolamine & ergotamine.
- D) Beta adrenergic receptors are blocked by guanethidine.

84. Concerning adrenergic receptors, all are true EXCEPT:

- A) Are receptors which respond to norepinephrine secreted by nerve endings & by the adrenal medulla.
- B) Are classified into alpha & beta adrenoceptors.
- C) Trigger the release of an intracellular mediator (2nd messenger) when activated.
- D) When stimulated, always produce excitatory effects on the effector organs.

85. As regards autonomic receptors, all are true EXCEPT:



- A) Sweat glands innervated by sympathetic nerves have receptors that can be blocked by atropine.
- B) Propranolol blocks the action of noradrenaline on alpha receptors.
- C) Nicotinic receptors are blocked following a large dose of nicotine.
- D) Alpha adrenoceptors are blocked by phentolamine or ergotamine.

86. Epinephrine is:

- A) Synthesized in the adrenal medulla by methylation of norepinephrine.
- B) The chemical transmitter of most sympathetic postganglionic nerve fibers.
- C) More powerful as a pressure agent than norepinephrine.
- D) Hydrolyzed by the cholinesterase enzyme.

87. Administration of noradrenaline to a person whose sympathetic system has been surgically removed reacts by developing:

- A) Hypoglycaemia.
- B) Hypotension.
- C) Pupil constriction.
- D) A more marked sympathetic response than before sympathectomy.

88. The alpha-adrenergic receptors mediate all the following effects EXCEPT:

- A) Vasoconstriction.
- B) Pupil dilatation.
- C) Spleen contraction.
- D) Increased cardiac activity.

89. Epinephrine produces all the following effects EXCEPT:

- A) Cutaneous vasoconstriction.
- B) Coronary vasodilatation.
- C) Bronchoconstriction.
- D) Glycogenolysis.

90. All the following drugs augment (increase) sympathetic activity EXCEPT:

- A) Reserpine.
- B) Ephedrine.
- C) Amphetamine.
- D) Tyramine.

91. Removal of norepinephrine at postganglionic sympathetic fibers occurs by:

- A) Reuptake back into the terminal sympathetic nerve endings.
- B) The action of transaminase enzyme.
- C) Transformation into epinephrine.
- D) A demethylation action.

92. The alpha-adrenergic receptors:

- A) Are not stimulated by epinephrine.
- B) When stimulated, they reduce the gastrointestinal (GI) motility.
- C) Are blocked by propranolol.
- D) When stimulated, they accelerate the heart.

93. Stimulation of the lesser splanchnic nerve causes:

- A) Ejaculation.
- B) Defecation.
- C) Micturition.
- D) Erection.

1-A, 2-A, 3-C, 4-C, 5-B, 6-C, 7-D, 8-C, 9-A, 10-D, 11-B, 12-B, 13-E, 14-C, 15-A, 16-B, 17-C, 18-D, 19-D, 20-D, 21-A, 22-D, 23-A, 24-C, 25-C, 26-B, 27-D, 28-C, 29-D, 30-C, 31-C, 32-C, 33-B, 34-C, 35-D, 36-C, 37-C, 38-B, 39-D, 40-C, 41-D, 42-B, 43-B, 44-C, 45-B, 46-D, 47-D, 48-D, 49-D, 50-B, 51-D, 52-D, 53-B, 54-D, 55-D, 56-A, 57-D, 58-D, 59-C, 60-B, 61-B, 62-D, 63-A, 64-A, 65-B, 66-A, 67-B, 68-D, 69-B, 70-C, 71-D, 72-C, 73-D, 74-D, 75-D, 76-B, 77-C, 78-D, 79-B, 80-D, 81-B, 82-D, 83-D, 84-B, 85-C, 86-D, 87-D, 88-C, 89-A, 90-A, 91-B, 92-A

**Practical**

1. When glucose levels in the blood rise, the brain sends a signal to the pancreas. The pancreas releases insulin, which opens channels in cell membranes to allow glucose to enter the cell, lowering blood sugar level. This is an example of a:

- A) Positive feedback response.
- B) Negative feedback loop.

2. Maintaining body temperature is an example of a:

- A) Negative feedback loop.
- B) Positive feedback loop.

3. The two types of feedback systems that help organisms maintain homeostasis are:

- A) Positive and negative.
- B) Receptor and effector.
- C) Static and dynamic.
- D) Minor and major.

4. If the environment gets cold, we will often shiver in order to:

- A) Keep the body temperature the same as the external temperature.
- B) Decrease body temperature.
- C) Increase body temperature.
- D) Regulate blood pressure.

5. Homeostasis:

- A) Allows for a wildly fluctuating internal environment.
- B) Is the maintenance of a relatively stable internal environment and often incorporates a form of feedback regulation.
- C) Is impossible in vertebrates.
- D) Is the maintenance of a relatively stable external environment and often incorporates a form of feedback regulation.

6. Which statement is incorrect?

- A) The term "homeostasis" describes the maintenance of nearly constant conditions in the body.
- B) In most diseases, homeostatic mechanisms are no longer operating in the body.
- C) The body's compensatory mechanisms often lead to deviations from the normal range in some of the body's functions.
- D) Disease is generally considered to be a state of disrupted homeostasis.

7. Moves carbon dioxide and oxygen:

- A) Diffusion.
- B) Osmosis.
- C) Facilitated diffusion.
- D) Active transport.

8. Moves glucose:

- A) Diffusion.
- B) Osmosis.
- C) Facilitated diffusion.
- D) Active transport.

9. Moves from low to high concentration:

- A) Diffusion.
- B) Osmosis.
- C) Facilitated diffusion.
- D) Active transport.



10. Diffusion is the net movement of molecules from a region of (1) to a region of (2):

- A) 1: lower concentration, 2: higher concentration.
- B) 1: higher concentration, 2: lower concentration.
- C) 1: lower concentration, 2: lower concentration.
- D) 1: higher concentration, 2: higher concentration.

11. Simple diffusion and facilitated diffusion share which of the following characteristics:

- A) Can be blocked by specific inhibitors.
- B) Do not require adenosine triphosphate (ATP).
- C) Require transport protein.
- D) Saturation kinetics.
- E) Transport solute against concentration gradient.

12. When the body has to get rid of excess water, the mechanism that is used is:

- A) Sweating.
- B) Diarrhea.
- C) Increased respiratory rate and depth.
- D) Increased urine production.

13. Osmosis is the passage of (1) molecules from a region of their higher concentration to a region of their lower concentration through a (2) permeable membrane:

- A) 1: sugar, 2: completely.
- B) 1: sugar, 2: partially.
- C) 1: water, 2: completely.
- D) 1: water, 2: partially.

14. A red blood cell will swell the most when it is placed in a solution containing:

- A) 100 mmol sodium chloride.
- B) 150 mmol sodium chloride.
- C) 200 mmol sodium chloride.
- D) 300 mmol sodium chloride.

15. The diagram shows two chambers (A and B) with different solute molecules separated by a selectively permeable membrane (Membrane X). What will happen to the water molecules?

- A) Move from B to A due to diffusion.
- B) Move from B to A due to osmosis.
- C) Move from B to A due to active transport.
- D) Move from A to B due to osmosis.

16. A non-carrier mediated transport down the concentration gradient:

- A) Facilitated diffusion.
- B) Simple diffusion.
- C) Active transport.
- D) Osmosis.

17. The cell will shrink if:

- A) The solutes outside the cell are more than inside.
- B) The solutes outside the cell are less than inside.
- C) The solutes outside and inside the cell are the same.
- D) The water outside is more than the water inside.

18. Infusion of an isotonic NaCl solution shows:

- A) Increase in ECF volume.
- B) Decrease in ICF volume.
- C) Increase in ECF osmolarity.
- D) Decrease in ICF osmolarity.



19. Plasma osmolal concentration is:

- A) 270 mOsm/L.
- B) 280 mOsm/L.
- C) 290 mOsm/L.
- D) 300 mOsm/L.
- E) 310 mOsm/L.

20. Which of the following statements is **INCORRECT**?

- A) Osmosis is the movement of solute across a semi-permeable membrane in an attempt to balance osmolal concentration on the 2 sides of the membrane.
- B) Exocytosis & endocytosis tend to be the mode of transport for protein.
- C) Facilitated diffusion makes use of carrier protein & occurs down the concentration gradient across the membrane.
- D) Active transport uses carrier protein & occurs against the concentration gradient across the membrane.

21. Active transport is movement of particles through the cell membrane from a region of (1) to a region of (2) using energy in the form of (3):

- A) 1: lower concentration, 2: higher concentration, 3: NADPH.
- B) 1: higher concentration, 2: lower concentration, 3: NADPH.
- C) 1: higher concentration, 2: lower concentration, 3: ATP.
- D) 1: lower concentration, 2: higher concentration, 3: ATP.

22. The sodium-potassium pump moves sodium ions ____ the neuron:

- A) Into.
- B) Out of.

23. About the $\text{Na}^+ - \text{K}^+$ pump, all the following are correct **EXCEPT**:

- A) Requires high energy phosphate bonds.
- B) Contributes to the creation of resting membrane potential.
- C) Keeps intracellular Na^+ low.
- D) Is explained by facilitated diffusion.

24. What is the threshold voltage that a neuron must reach to fire an action potential:

- A) +50 mV.
- B) -55 mV.
- C) -70 mV.
- D) +35 mV.

25. From the action potential graph (points A, B, C, D, E, F): which of the following is primarily responsible for the membrane potential between points B and D?

- A) Inhibition of the Na^+ , K^+ -ATPase.
- B) Movement of K^+ into the cell.
- C) Movement of K^+ out of the cell.
- D) Movement of Na^+ into the cell.
- E) Movement of Na^+ out of the cell.

26. Which of the following is primarily responsible for the membrane potential between points D and E?

- A) Inhibition of the Na^+ , K^+ -ATPase.
- B) Movement of K^+ into the cell.
- C) Movement of K^+ out of the cell.
- D) Movement of Na^+ into the cell.
- E) Movement of Na^+ out of the cell.



27. Which of the following is an example of counter-transport?

- A) Sodium-Glucose pump.
- B) Sodium-Potassium-Chloride pump.
- C) Sodium-Potassium pump.
- D) Sodium-Amino acid pump.
- E) Sodium-Hydrogen pump.

28. An example of co-transport is:

- A) Na^+-K^+ pump.
- B) Ca^{++} pump.
- C) Na^+-H^+ pump.
- D) Na^+ -glucose transport.

29. Which of the following examples is NOT a primary active transport?

- A) Na^+-K^+ pump.
- B) Active transport of hydrogen.
- C) Co-transport and counter-transport.
- D) Active transport of calcium.

30. Generalized sympathetic stimulation causes all the following EXCEPT:

- A) Increases the heart rate.
- B) Increases sweating.
- C) Relaxes bronchial smooth muscles.
- D) Tightens GIT sphincters.
- E) Constriction of pupil.

31. Generalized sympathetic stimulation causes:

- A) Inhibition of sweating.
- B) Increased gastric motility.
- C) Decreased blood sugar levels.
- D) Raised arterial blood pressure.

32. A lesion in the cervical sympathetic chain (Horner's syndrome) causes:

- A) Increased sweating.
- B) Drooping of the upper eyelid (= ptosis).
- C) Vasoconstriction of the skin blood vessels (leading to pallor).
- D) Mydriasis (= dilatation of the pupil).

33. All of the following about Horner's syndrome is true EXCEPT:

- A) Manifested by dryness & redness of skin.
- B) Group of signs including ptosis & miosis.
- C) Is due to injury of LHCs present in thoracic 1-2 spinal segments.
- D) Causes pallor of skin due to vasoconstriction.

34. Continuous conduction:

- A) Occurs in myelinated nerve fibers.
- B) Occurs by jumping of charges from one node of Ranvier to another.
- C) Is relatively slow, 0.5-2.0 meter/second.
- D) Occurs in the neuro-muscular junction.

35. Saltatory (jumping) conduction of action potential:

- A) Occurs in unmyelinated nerve fibers.
- B) Is slower than the point-to-point conduction.



C) Consumes less energy than point-to-point conduction.

D) None of the above.

36. From the accompanying refractory-period graph, what is period A?

A) Relative refractory period.

B) Absolute refractory period.

C) Supernormal period.

D) Subnormal period.

37. About electrotonus, what is the wrong statement?

A) It results from stimulation of the nerve by a subthreshold galvanic current.

B) A strong catelectrotonus can cause nerve block.

C) Catelectrotonus may be followed by the local response.

38. Catelectrotonus:

A) Is a localized area of hyperpolarization.

B) Results from stimulation by threshold galvanic current.

C) Is associated with increased excitability.

D) Used to produce nerve block.

39. Nerve block is produced by:

A) Severe warming.

B) Strong anelectrotonus.

C) Decreased Ca^{++} ions.

D) Increased Na^{+} ions.

40. All of the following substances may be used to measure ECF EXCEPT:

A) Sucrose.

B) Inulin.

C) Radioactive sodium.

D) Evans blue dye.

41. Infants have the greatest body water content due to:

A) High body fat.

B) High amount of skeletal muscles.

C) Low body fat.

D) High bone mass.

42. Which one of these represents 2/3 of total body fluid?

A) Intracellular fluid.

B) Extracellular fluid.

C) Interstitial fluid.

D) Transcellular fluid compartment.

43. How many % of body weight is the total blood volume?

A) 8%.

B) 15%.

C) 40%.

D) 60%.

E) 5%.

44. Which one is the major anion in extracellular fluid?

A) Sodium.

B) Chloride.

C) Potassium.

D) Phosphate.

45. Hypokalemia is:

A) Decrease in K concentration in the ECF.

B) Increase in Na concentration in ECF.

C) Increase in Na concentration in ICF.

D) Decrease in Na concentration in ECF.

46. Synovial fluid is one of:



- A) Intracellular fluid. B) Extracellular fluid.
C) Interstitial fluid. D) Transcellular fluid compartment.

47. Plasma and interstitial fluid almost have the same composition except for a high concentration in plasma of:

- A) Calcium. B) Potassium.
C) Protein. D) Sodium.

48. Which of the following statements is correct?

- A) Potassium is the chief anion in intracellular fluid.
B) Phosphate is the chief cation in intracellular fluid.
C) Sodium is the chief anion in extracellular fluid.
D) Chloride is the chief anion in extracellular fluid.

49. Hyponatremia means:

- A) Decrease in Na concentration in the ECF.
B) Increase in Na concentration in the ECF.
C) Decrease in Na concentration in the ICF.
D) Increase in Na concentration in the ICF.

50. 1/3 of the body's fluid is:

- A) Intracellular fluid. B) Extracellular fluid.
C) Plasma. D) Interstitial fluid.

51. One of the following is NOT interstitial fluid:

- A) CSF. B) GIT (fluid).
C) Synovial fluid. D) Plasma.

52. What type of feedback loop is more common in the body's reactions?

- A) Positive. B) Neutral.
C) Negative.

1-B, 2-A, 3-A, 4-C, 5-C, 6-B, 7-A, 8-C, 9-D, 10-B, 11-B, 12-D, 13-D, 14-A, 15-B, 16-B, 17-A, 18-A, 19-D, 20-A, 21-D, 22-B, 23-D, 24-B, 25-D, 26-C, 27-E, 28-D, 29-C, 30-E, 31-D, 32-B, 33-D, 34-C, 35-C, 36-B, 37-B, 38-C, 39-D, 40-D, 41-C, 42-A, 43-A, 44-B, 45-D, 46-D, 47-C, 48-D, 49-A, 50-B, 51-D, 52-C

Doctors



Exams

1. ANS exam 1

1. The autonomic ganglia:

- A) Are 5 types.
- B) Are site of relay of afferent neuron.
- C) Located inside the CNS.
- D) Act as distributing centers.

2. Stimulation of the greater splanchnic nerve produces:

- A) Increase motility of plain muscles of stomach.
- B) Decrease levels of blood glucose & free fatty acids.
- C) Secretion of adrenaline & noradrenalin by the adrenal medulla.
- D) Inhibition of micturition.

3. Paralysis of sympathetic to the head causes all the following EXCEPT:

- A) Drooping of the upper eyelid.
- B) Pupillary constriction.
- C) Inability of the eye to accommodate.
- D) Cutaneous vasodilatation.

4. About the suprarenal medulla, all the following is true EXCEPT:

- A) Secretes 80% adrenalin & 20% noradrenalin.
- B) Is a modified sympathetic ganglion.
- C) Receives preganglionic nerve supply from the anterior horn cells of the lower 6 thoracic segments.
- D) No postganglionic fibers arise from it.

5. Concerning the sympathetic nervous system:

- A) Consists only of cranial nerves.
- B) Its efferent pathway is made of one neuron.
- C) All postganglionic sympathetic fibers secrete adrenalin.
- D) Its stimulation causes dilation of the airways.

6. The sympathetic division of the autonomic nervous system:

- A) Generally has short postganglionic fibers.
- B) Has its cell of origin in the lateral portion of the thoracic and lumbar segments of the spinal cord.
- C) Is concerned primarily with energy conservation and secretory activity.
- D) Has its cells originating from the brain stem.

7. Which of the following structures is innervated by preganglionic sympathetic nerve fibers?

- A) Lung.
- B) Heart.
- C) Pancreas.
- D) Adrenal medulla.

8. The preganglionic sympathetic nerve that innervates the eye originates from:

- A) Superior colliculi.
- B) Superior cervical ganglion.
- C) Thoracic segments of the spinal cord.
- D) Ciliary ganglion.



9. A lesion in the cervical sympathetic chain causes:

- A) Increased sweating.
- B) Ptosis.
- C) Vasoconstriction of the skin blood vessels.
- D) Mydriasis.

10. Stimulation of the lesser splanchnic nerve produces:

- A) Increased blood glucose level.
- B) Increased adrenalin secretion from the adrenal medulla.
- C) Shrinkage of external genitalia.
- D) Erection.

11. All of the following are properties of autonomic ganglia EXCEPT:

- A) Conduction is unidirectional.
- B) Chemical transmitter is acetylcholine (nicotinic-like action).
- C) Equal divergence in both lateral & terminal ganglia.
- D) Ganglia can be blocked by a large dose of nicotine.

12. Postganglionic nerves from the superior cervical ganglia cause:

- A) Pupillary constriction (miosis).
- B) Trophic secretion from salivary glands.
- C) Decreased sweat secretion.
- D) Cutaneous vasodilatation.

13. Stimulation of the lesser splanchnic nerve causes:

- A) Micturition.
- B) Defecation.
- C) Ejaculation.
- D) Erection.

14. All of the following about Horner's syndrome is true EXCEPT:

- A) Manifested by dryness & redness of skin.
- B) Group of signs including ptosis & miosis.
- C) Is due to injury of LHCs present in thoracic 1-2 spinal segments.
- D) Causes pallor of skin due to vasoconstriction.

15. The major difference between somatic & autonomic nervous systems is:

- A) Only somatic nerves innervate muscle cells.
- B) Both of them release acetylcholine at effector organs.
- C) Somatic nerves only originate from thoracolumbar segments of the spinal cord.
- D) Only autonomic nerves supply blood vessels.



2. ANS exam 2

1. The parasympathetic nerves:

- A) Are important in emergency conditions, e.g. fight & flight.
- B) Tend to have longer postganglionic fibers than preganglionic fibers.
- C) Are catabolic in their actions.
- D) Generally exert opposite effects to those of the sympathetic nerves.

2. Stimulation of the pelvic nerve causes:

- A) Contraction of the bronchial muscle.
- B) Contraction of the internal urethral & rectal (anal) sphincters.
- C) Vasodilatation of the blood vessels in the external genitalia.
- D) Stoppage of prostatic secretion.

3. Which of the following may be produced by parasympathetic stimulation?

- A) Acceleration of the heart.
- B) Contraction of the urinary bladder wall.
- C) Vasodilatation of skeletal muscle blood vessels.
- D) Dilatation of the pupil.

4. Stimulation of the pelvic nerve (sacral autonomic) produces:

- A) Relaxation of the wall of the rectum & contraction of the internal anal sphincter.
- B) Vasoconstriction of the blood vessels in the external genitalia causing erection of the penis.
- C) Contraction of the urinary bladder wall & relaxation of the internal urethral sphincter.
- D) Contraction of smooth muscles in the male sexual passages causing ejaculation.

5. Concerning the pelvic nerve, all the following is true EXCEPT:

- A) It is the nerve of erection (nervus erigentes).
- B) It is motor to the urinary bladder & distal half of the large intestine.
- C) It is inhibitory to both the internal anal & urethral sphincters.
- D) It is motor to the wall of the gall bladder.

6. Vagus parasympathetic nerve supplies all the following structures EXCEPT:

- A) Gall bladder.
- B) Stomach.
- C) Rectum.
- D) Pancreas.

7. All of the following about sacral autonomic outflow is true EXCEPT:

- A) It is the nerve of defecation.
- B) It is the nerve of ejaculation.
- C) It is the nerve of micturition.
- D) It relays in terminal ganglia.

8. The functions of parasympathetic innervation to the salivary glands is:

- A) Vasoconstriction & secretion.
- B) Vasodilatation & secretion.
- C) Vasodilatation & inhibition of secretion.
- D) Secretion of scanty viscous saliva.

9. Facial nerve:

- A) Stimulation increases the power of the lens of the eye.



- B) Supplies the lacrimal gland.
- C) Supplies the parotid salivary gland.
- D) Contains sympathetic postganglionic fibers.

10. Concerning the cranial parasympathetic outflow:

- A) Preganglionic outflow in the oculomotor nerve arises from the ciliary ganglion.
- B) Preganglionic outflow in the glossopharyngeal nerve arises from the superior salivary nucleus.
- C) They relax the sphincter of the gall bladder.
- D) They cause bronchodilatation.

11. All the following organs receive their parasympathetic innervation via the vagus nerves EXCEPT:

- A) Gall bladder.
- B) Atria of the heart.
- C) Proximal half of the large intestine.
- D) Rectum.

12. Salivary secretion increases on stimulation of:

- A) Trigeminal nerve.
- B) Glossopharyngeal nerve.
- C) Vagus nerve.
- D) Hypoglossal nerve.

13. Vagal stimulation causes:

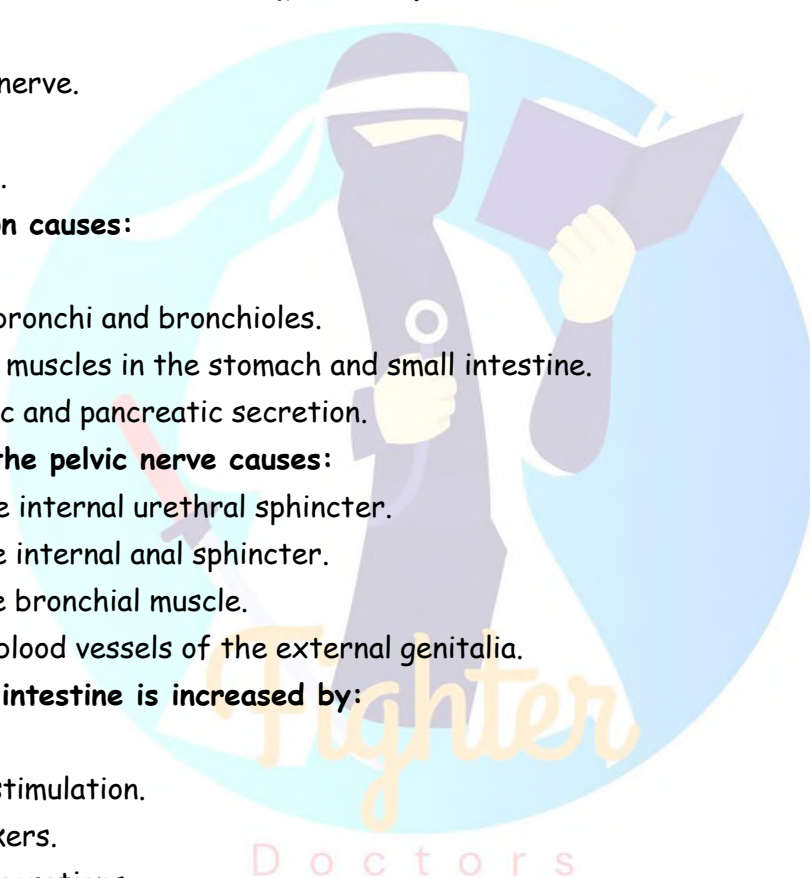
- A) Tachycardia.
- B) Dilatation of the bronchi and bronchioles.
- C) Inhibition of plain muscles in the stomach and small intestine.
- D) Increase in gastric and pancreatic secretion.

14. Stimulation of the pelvic nerve causes:

- A) Contraction of the internal urethral sphincter.
- B) Contraction of the internal anal sphincter.
- C) Contraction of the bronchial muscle.
- D) Dilatation of the blood vessels of the external genitalia.

15. Motility of the intestine is increased by:

- A) Adrenaline.
- B) Parasympathetic stimulation.
- C) Sympathetic blockers.
- D) Adrenal medulla secretions.



**3. formative exam (group A & B)**

1. The first step in nerve impulse initiation is:

- A) Decreased Na^+ permeability.
- B) Membrane depolarization.
- C) Increased K^+ permeability.
- D) Membrane hyperpolarization.

2. The resting potential of a nerve membrane is primarily dependent on the concentration gradient of:

- A) Potassium.
- B) Sodium.
- C) Calcium.
- D) Chloride.

3. Which of the following is actively transported out of the neurons?

- A) K^+ .
- B) HPO_4^- .
- C) Cl^- .
- D) Na^+ .

4. Na^+-K^+ pump:

- A) Requires energy which is taken from GTP.
- B) Couples Na^+ and K^+ with ratio 3:2.
- C) Helps intracellular accumulation of Na^+ .
- D) Is a passive process, needs no energy.

Answer key (1-4): 1-B, 2-S, 3-D, 4-B

4. Formative exam 2020

1. What is chronaxie?

- A) Is a threshold stimulus.
- B) Twice the rheobase.
- C) Can be used as a measure of excitability.
- D) Time needed to excite a nerve using a current strength equal to the rheobase.

2. Which of the following is untrue about action potential?

- A) Firing level is the level at which the spike begins.
- B) Rapid repolarization is due to increased K^+ efflux.
- C) Rapid depolarization is due to increased Na^+ influx.
- D) Absolute refractory period extends from the firing level until repolarization is completed.

3. Which of the following is true concerning the Na^+/K^+ pump?

- A) Pumps 3 Na^+ molecules inside & 2 K^+ molecules outside the nerve.
- B) Membrane needs high energy phosphate bond.
- C) Is the main cause of RMP.
- D) Pumps 3 K^+ molecules inside & 2 Na^+ molecules outside the nerve membrane.

**4. Which statement is true about RMP of a nerve?**

- A) Caused by equal distribution of ions along the nerve membrane.
- B) Caused by selective permeability of the membrane to ions.
- C) Na^+ - K^+ pump has no role.
- D) Caused mainly by inward movement of Na^+ ions.

5. Which statement is untrue about electrotonus?

- A) It results from stimulation of the nerve by a subthreshold galvanic current.
- B) Catelectrotonus is accompanied by increased nerve excitability.
- C) Strong catelectrotonus can lead to nerve block.
- D) Catelectrotonus may be followed by the local response.

6. What is true regarding the absolute refractory period of the nerve fiber (ARP)?

- A) Lasts throughout the action potential.
- B) Follows the relative refractory period.
- C) Occurs during increased membrane permeability to Na^+ .
- D) Occurs mainly during whole repolarization.

7. Which statement is true about repolarization?

- A) Occurs gradually at first then becomes fast.
- B) Represented by the ascending limb of the spike.
- C) Results from closure of Na gates and opening of K^+ gates.
- D) Terminated by sodium efflux.

8. What will happen if Na^+ concentration decreases around a nerve?

- A) Decreasing the resting potential.
- B) Increases the overshoot.
- C) Decreases nerve excitability.
- D) Increases action potential.

9. What is true about nerve excitability among the following statements?

- A) During rest, the nerve membrane is depolarized.
- B) Chronaxie is the time needed to produce a response using the rheobase.
- C) K^+ is the least permeable cation through the nerve membrane.
- D) Resting potential becomes less negative if extracellular K^+ concentration is increased.

10. What is the action potential of a neuron?

- A) Is initiated by efflux of Na^+ ions.
- B) Terminated by efflux of K^+ ions.
- C) Declines in amplitude as it moves along the axon.
- D) Is not associated with excitability changes in the axon.

11. Which one of the following can produce nerve block?

- A) Severe cooling.
- B) Decreased Ca^{++} ions.
- C) Strong catelectrotonus.
- D) Increased Na^+ ions.

12. Which one of the following is true concerning depolarization in the nerve?

- A) Is caused by Na^+ outflow.
- B) Is caused by K^+ outflow.
- C) Is due to opening of inactivation gates of voltage gated Na^+ channels.
- D) During which the nerve cannot respond to any stimulus.



13. Which one of the following is true about sweeping conduction?

- A) More rapid than saltatory conduction.
- B) Occurs in myelinated nerves.
- C) Consumes less energy than saltatory conduction.
- D) Needs more energy than saltatory conduction.

14. Which one of the following is true regarding the local response?

- A) Is characterized by depressed excitability.
- B) Obeys the all-or-none law.
- C) Has an absolute refractory period.
- D) None of the above.

15. What is true about the action potential of a nerve among the following statements?

- A) Depolarization only.
- B) Repolarization only.
- C) Rapid changes in membrane potential by an effective stimulus.
- D) All of the above.

16. Which of the following is correct?

- A) The amplitude of the action potential depends on stimulus strength.
- B) Graded (local) potentials can cause membrane depolarization without summation.
- C) The amplitude of the action potential does not depend on stimulus strength.
- D) Conduction of action potential in the axon occurs only in one direction.

17. Which one of the following causes a decrease in nerve fiber excitability?

- A) When extracellular Ca^{++} is decreased.
- B) When extracellular fluid K^{+} is decreased.
- C) Any condition that increases the membrane permeability for Na^{+} .
- D) During catelectrotonic state.

18. What impulse can a threshold stimulus produce according to the all-or-none law?

- A) Is weaker than that produced by a stronger stimulus.
- B) Is propagated only partially along the cell membrane.
- C) Does not need any energy.
- D) Is not different from that produced by a stronger stimulus.

19. What is untrue concerning excitable tissues?

- A) Spend energy during rest to maintain their membranes at a polarized state.
- B) Become more excitable when partially depolarized.
- C) Become more permeable to Na^{+} ions when membrane potential decreases to the firing level.
- D) Usually respond to stimuli that decrease their membrane Na^{+} permeability.

20. Which one of the following is untrue about nerve impulses?

- A) Conducted in both directions, but in the body, conduction is mainly one way.
- B) They are conducted passively.
- C) Their conduction is an active self-propagating process associated with reversal of polarity across the nerve cell membrane.
- D) Characterized by being constant in amplitude & velocity for a certain nerve fiber.

1-C, 2-D, 3-B, 4-B, 5-C, 6-C, 7-C, 8-C, 9-D, 10-B, 11-A, 12-D, 13-D, 14-D, 15-C, 16-C, 17-B, 18-D, 19-D,
20-B



5. End module 2020 (Group A)

1. Which of the following has the lowest conduction velocity?

- A) A-alpha fibers.
- B) A-beta fibers.
- C) B fibers.
- D) C fibers.

2. Anelectrotonus:

- A) Is a localized area of depolarization.
- B) Results from stimulation by an effective galvanic current.
- C) Is associated with decreased excitability.
- D) Used to stimulate the nerve fibers.

3. In the all-or-none rule:

- A) A minimal (threshold) stimulus produces a maximal response.
- B) The response in a single nerve fiber increases with increased intensity of stimulus.
- C) The nerve trunk either responds maximally or does not respond at all.
- D) Minimal stimulus produces minimal response.

4. Repolarization:

- A) Occurs gradually at first then becomes fast.
- B) Results from closure of sodium gates and opening of potassium gates.
- C) Is represented by the ascending limb of the spike.
- D) Is followed by appearance of a response.

5. RMP of a nerve:

- A) Is caused by equal distribution of ions along both sides of the membrane.
- B) Is caused by selective permeability of the membrane to the ions.
- C) Na⁺-K⁺ pump has no role in RMP.
- D) Is caused mainly by inward movement of Na⁺ ions.

6. Stimulation of the pelvic nerve (sacral autonomic) produces:

- A) Relaxation of the wall of the rectum & contraction of the internal anal sphincter.
- B) Vasoconstriction of the blood vessels in the external genitalia causing erection of the penis.
- C) Contraction of the urinary bladder wall & relaxation of the internal urethral sphincter.
- D) Contraction of smooth muscles in the male sexual passages causing ejaculation.

7. Activation of the sympathetic system causes:

- A) Pupillary constriction.
- B) Sweat secretion.
- C) Bradycardia.
- D) Cutaneous vasodilatation.

8. The autonomic nervous system differs from the somatic nervous system in:

- A) Having one efferent neuron.
- B) Preganglionic neurons are in the anterior horn of the spinal cord.
- C) Efferent nerves have either excitatory or inhibitory effects on effector organs.
- D) Postganglionic neurons are in the lateral horn of the spinal cord.

9. Salivary secretion increases on stimulation of:

- A) Trigeminal nerve.
- B) Glossopharyngeal nerve.
- C) Vagus nerve.
- D) Hypoglossal nerve.

10. Which of the following may be produced by parasympathetic stimulation?

- A) Acceleration of heart.
- B) Contraction of urinary bladder wall.
- C) Vasodilatation of skeletal muscle blood vessels.
- D) Dilatation of pupil.

1-D, 2-C, 3-A, 4-B, 5-B, 6-C, 7-B, 8-C, 9-B, 10-B



6. End module 2020 (Group B)

1. Anelectrotonus:

- A) Is a localized area of depolarization.
- B) Results from stimulation by an effective galvanic current.
- C) Is associated with decreased excitability.
- D) Used to stimulate the nerve fibers.

2. Continuous conduction:

- A) Occurs in myelinated nerve fibers.
- B) Occurs by jumping of charges from one node of Ranvier to another.
- C) Is relatively slow, 0.5-2.0 meter/second.
- D) Occurs in the neuro-muscular junction.

3. The resting membrane potential is caused by:

- A) Diffusion of K⁺ ions outside the nerve fibers.
- B) Diffusion of Na⁺ ions inside the nerve fibers.
- C) Opening of the chemically activated ion channels.
- D) Opening of the voltage activated ion channels.

4. During depolarization:

- A) Voltage activated Na⁺ channels open.
- B) The membrane becomes impermeable to Na⁺.
- C) When membrane potential reaches -55 mV, Na⁺ & K⁺ fluxes occur at the same time.
- D) K⁺ ions diffuse outside.

5. Chronaxie:

- A) Is the minimal stimulus which produces a response after a relatively long time.
- B) Is the minimal time needed by the strong stimulus to produce a response.
- C) Is the time required to stimulate the nerve by a minimal stimulus.
- D) Is the time required to stimulate the nerve by a stimulus which is double the rheobase.

6. Stimulation of the pelvic nerve causes:

- A) Contraction of the bronchial muscle.
- B) Contraction of the internal urethral & rectal (anal) sphincters.
- C) Vasodilatation of the blood vessels in the external genitalia.
- D) Stoppage of prostatic secretion.

7. Activation of the sympathetic system causes:

- A) Pupillary constriction.
- B) Sweat secretion.
- C) Bradycardia.
- D) Cutaneous vasodilatation.

8. Stimulation of the greater splanchnic nerve produces:

- A) Increased blood glucose level.
- B) Decreased blood volume.
- C) Secretion of epinephrine 20% & norepinephrine 80% from the adrenal medulla.
- D) Increased gastrointestinal motility.

9. Stimulation of the lesser splanchnic nerve causes:

- A) Micturition.
- B) Defecation.
- C) Ejaculation.
- D) Erection.

10. Vagal stimulation causes:

- A) Tachycardia.
- B) Dilatation of the bronchi and bronchioles.
- C) Inhibition of plain muscles in the stomach and small intestine.
- D) Increase in gastric and pancreatic secretion.

1-C, 2-C, 3-A, 4-A, 5-D, 6-C, 7-B, 8-A, 9-C, 10-D



7. End Semester 2019 (First Round)

1. What is the effect of stimulation of the greater splanchnic nerve?

- A) Decreased blood volume.
- B) Increased blood glucose level.
- C) Increased gastrointestinal motility.
- D) Secretion of epinephrine 20% & norepinephrine 80% from the adrenal medulla.

2. Which character among the following applies to the autonomic nervous system?

- A) Efferent nerves have either excitatory or inhibitory effects on effector organs.
- B) Having one efferent neuron.
- C) Preganglionic neurons are in the anterior horn of the spinal cord.
- D) Postganglionic neurons are in the lateral horn of the spinal cord.

3. What is the effect of activation of the sympathetic system?

- A) Bradycardia.
- B) Cutaneous vasodilatation.
- C) Pupillary constriction.
- D) Sweat secretion.

4. Which of the following symptoms appear in Horner's syndrome?

- A) Dilatation of pupil.
- B) Drooping of the upper eyelid.
- C) Increased sweating.
- D) Vasoconstriction of skin blood vessels.

5. Which character belongs to parasympathetic nerves?

- A) Are important in emergency conditions, e.g. fight & flight reactions.
- B) Are catabolic in their actions.
- C) Generally exert opposite effects to those of the sympathetic nerves.
- D) Tend to have longer postganglionic fibers than preganglionic fibers.

6. Which of the following functions result from stimulation of parasympathetic innervation to the salivary glands?

- A) Secretion of scanty viscous saliva.
- B) Vasoconstriction & secretion.
- C) Vasodilatation & secretion.
- D) Vasodilatation & inhibition of secretion.

7. What is the effect of stimulation of the facial nerve?

- A) Stimulation increases the power of the lens of the eye.
- B) Supplies the lacrimal gland.
- C) Supplies the parotid salivary gland.
- D) Contains sympathetic postganglionic fibers.

8. Which of the following organs does NOT receive its parasympathetic innervation via the vagus nerves?

- A) Atria of the heart.
- B) Gall bladder.
- C) Proximal half of the large intestine.



D) Rectum.

9. What is the effect of stimulation of the pelvic nerve?

- A) Contraction of the urinary bladder wall & relaxation of the internal urethral sphincter.
- B) Contraction of smooth muscles in the male sexual passages causing ejaculation.
- C) Relaxation of the wall of the rectum & contraction of the internal anal sphincter.
- D) Vasoconstriction of the blood vessels in the external genitalia causing erection of the penis.

10. What is the definition of chronaxie?

- A) Is the minimal stimulus which produces a response after a relatively long time.
- B) Is the minimal time needed by the strong stimulus to produce a response.
- C) Is the time required to stimulate the nerve by a minimal stimulus.
- D) Is the time required to stimulate the nerve by a stimulus which is double the rheobase.

11. What is the cause of the resting membrane potential?

- A) Diffusion of K^+ ions outside the nerve fibers.
- B) Diffusion of Na^+ ions inside the nerve fibers.
- C) Opening of the chemically activated ion channels.
- D) Opening of the voltage activated ion channels.

12. Which of the following events happens with the repolarization phase of the action potential?

- A) Is represented by the ascending limb of the spike.
- B) Is followed by appearance of a response.
- C) Occurs gradually at first then becomes fast.
- D) Results from closure of sodium gates and opening of potassium gates.

1-B, 2-A, 3-D, 4-B, 5-C, 6-C, 7-B, 8-D, 9-A, 10-D, 11-A, 12-D



8. End Semester 2019 (Second Round)

1. What is the effect of stimulation of the lesser splanchnic nerve?

- A) Defecation.
- B) Ejaculation.
- C) Erection.
- D) Micturition.

2. Which of the following refers to the autonomic nervous system?

- A) Having one efferent neuron.
- B) Efferent nerves have either excitatory or inhibitory effects on the effector organs.
- C) Postganglionic neurons are in the lateral horn of the spinal cord.
- D) Preganglionic neurons are in the anterior horn of the spinal cord.

3. What is the effect of generalized sympathetic stimulation?

- A) Decreases blood sugar levels.
- B) Inhibition of sweating.
- C) Increased gastric motility.
- D) Rise in arterial blood pressure.

4. Which of the following symptoms appear in Horner's syndrome?

- A) Dilatation of pupil.
- B) Dropping of the upper eyelid.
- C) Increased sweating.
- D) Vasoconstriction of skin blood vessels.

5. Which of the following effects result from stimulation of the pelvic nerve (sacral autonomic)?

- A) Contraction of the urinary bladder wall & relaxation of the internal urethral sphincter.
- B) Contraction of smooth muscles in the male sexual passages causing ejaculation.
- C) Relaxation of the wall of the rectum & contraction of the internal anal sphincter.
- D) Vasoconstriction of the blood vessels in the external genitalia causing erection of the penis.

6. Salivary secretion increases on stimulation of which of the following nerves?

- A) Hypoglossal nerve.
- B) Glossopharyngeal nerve.
- C) Trigeminal nerve.
- D) Vagus nerve.

7. What is the effect of vagal stimulation?

- A) Dilatation of the bronchi and bronchioles.
- B) Inhibition of plain muscles in the stomach and small intestine.
- C) Increase in gastric and pancreatic secretion.
- D) Tachycardia.

8. Which of the following organs does NOT receive its parasympathetic innervation via the vagus nerves?

- A) Atria of heart.
- B) Gall bladder.
- C) Rectum.
- D) Proximal half of large intestine.

9. What is the effect of stimulation of the pelvic nerve?



- A) Relaxation of the wall of the rectum & contraction of the internal anal sphincter.
- B) Vasoconstriction of the blood vessels in the external genitalia causing erection of the penis.
- C) Contraction of the urinary bladder wall & relaxation of the internal urethral sphincter.
- D) Contraction of smooth muscles in the male sexual passages causing ejaculation.

10. Nerve block is produced by which of the following?

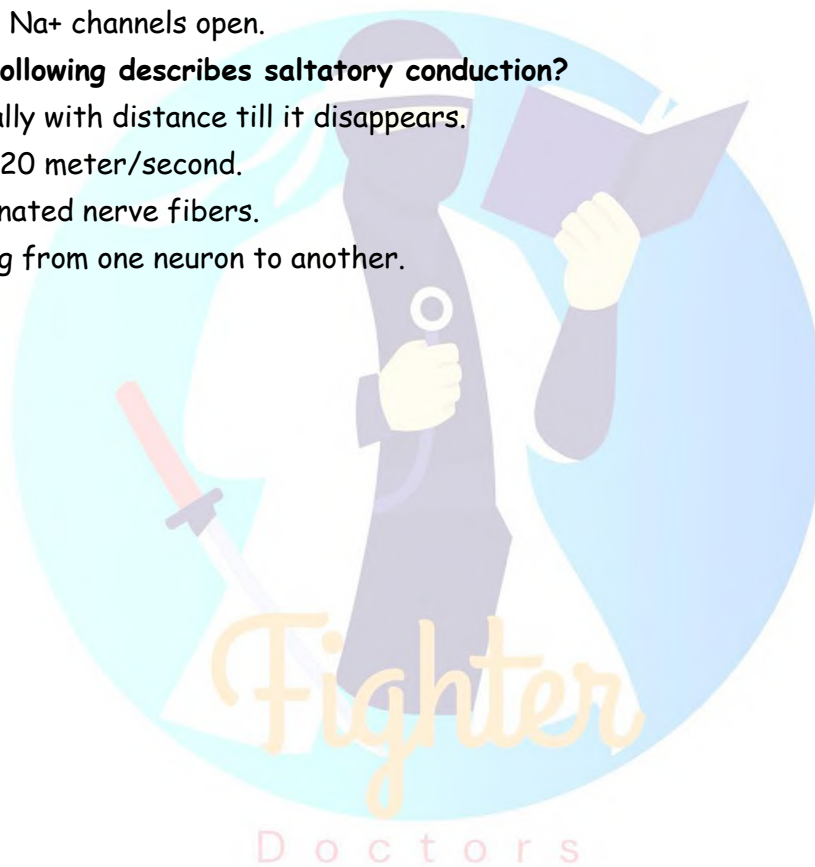
- A) Decrease Ca^{++} ions.
- B) Increased Na^{+} ions.
- C) Severe cooling.
- D) Strong catelectrotonus.

11. Which of the following events occur during depolarization?

- A) K^{+} ions diffuse outside.
- B) The membrane becomes impermeable to Na^{+} .
- C) When membrane potential reaches -55 mV , Na^{+} & K^{+} fluxes occur at the same time.
- D) Voltage activated Na^{+} channels open.

12. Which of the following describes saltatory conduction?

- A) Decreases gradually with distance till it disappears.
- B) May reach up to 120 meter/second.
- C) Occurs in unmyelinated nerve fibers.
- D) Occurs by jumping from one neuron to another.



1-B, 2-B, 3-D, 4-B, 5-A, 6-B, 7-C, 8-C, 9-C, 10-C, 11-D, 12-B



9. End semester 2021

1. Resting membrane potential of a nerve:

- A) Is caused by equal distribution of ions along both sides of the membrane.
- B) Is caused by selective permeability of the membrane to the ions.
- C) Na^+ - K^+ pump has no role in RMP.
- D) Is caused mainly by inward movement of Na^+ ions.

2. During depolarization:

- A) Voltage activated Na^+ channels open.
- B) The membrane becomes impermeable to Na^+ .
- C) When membrane potential reaches -55 mV , Na^+ & K^+ fluxes occur at the same time.
- D) K^+ ions diffuse outside.

3. Repolarization:

- A) Occurs gradually at first then becomes fast.
- B) Results from closure of sodium gates and opening of potassium gates.
- C) Is represented by the ascending limb of the spike.
- D) Is followed by appearance of a response.

4. Saltatory conduction:

- A) Occurs in unmyelinated nerve fibers.
- B) May reach up to 120 meter/second.
- C) Occurs by jumping from one neuron to another.
- D) Decreases gradually with distance till it disappears.

5. Nerve block is produced by:

- A) Severe cooling.
- B) Strong catelectrotonus.
- C) Decreased Ca^{++} ions.
- D) Increased Na^+ ions.

6. Generalized sympathetic stimulation causes:

- A) Inhibition of sweating.
- B) Increased gastric motility.
- C) Decreased blood sugar levels.
- D) Raised arterial blood pressure.

7. The autonomic nervous system differs from the somatic nervous system in:

- A) Having one efferent neuron.
- B) Preganglionic neurons are located in the anterior horn of the spinal cord.
- C) Efferent nerves have either excitatory or inhibitory effects on the effector organs.
- D) Postganglionic neurons are located in the lateral horn of the spinal cord.

8. The chemical transmitter at autonomic ganglia is:

- A) Adrenaline.
- B) Cholinesterase.
- C) Acetylcholine.
- D) Dopamine.

9. The parasympathetic nerves:

- A) Are important in emergency conditions, e.g. fight & flight reactions.
- B) Tend to have longer postganglionic fibers than preganglionic fibers.
- C) Are catabolic in their actions.
- D) Generally exert opposite effects to those of the sympathetic nerves.

10. Motility of the intestine is increased by:

- A) Adrenaline.
- B) Parasympathetic stimulation.
- C) Sympathetic blockers.
- D) Adrenal medulla secretions.

1-B, 2-A, 3-B, 4-B, 5-A, 6-D, 7-C, 8-C, 9-D, 10-B



10. End semester 2022 (Second Round)

1. Comparing between the autonomic & somatic divisions of the nervous system:

- A) Only the autonomic nerves innervate blood vessels.
- B) The somatic nerves only originate from the thoraco-lumbar segments of the spinal cord.
- C) Autonomic nerves only originate from the thoraco-lumbar segments of the spinal cord.
- D) Only the somatic nerves innervate skeletal muscle.

2. During action potential, rapid depolarization of the membrane is due to:

- A) Increased potassium permeability.
- B) Increased calcium permeability.
- C) Increased chloride permeability.
- D) Increased sodium permeability.

3. To measure resting membrane potential:

- A) We put the two electrodes inside the nerve fiber.
- B) We put the two electrodes outside the nerve fiber.
- C) We stimulate the nerve by an effective stimulus.
- D) We use a special voltmeter or CRO.

4. Which of the following effects are produced by pelvic nerve stimulation?

- A) Relaxation of the internal urethral and anal sphincters.
- B) Vasoconstriction of the blood vessels in the external genitalia causing erection of the penis.
- C) Contraction of smooth muscles in the male sexual passages causing ejaculation.
- D) Relaxation of the wall of the rectum & contraction of the internal anal sphincter.

5. The first step in nerve impulse initiation is:

- A) Membrane hyperpolarization.
- B) Membrane depolarization.
- C) Increased K^+ permeability.
- D) Decreased Na^+ permeability.

6. Autonomic ganglia:

- A) Are located inside the CNS.
- B) Are the sites of relay of afferent neurons.
- C) Function as distributing centers.
- D) Are 5 types.

7. As regards the parasympathetic system:

- A) They cause bronchodilatation.
- B) Preganglionic outflow in the glossopharyngeal nerve is from the superior salivary nucleus.
- C) They relax the sphincter of Oddi.
- D) Preganglionic outflow in the oculomotor nerve arises from the ciliary ganglion.

8. Which of the following is NOT supplied by the sympathetic system?

- A) Skin.
- B) Constrictor of pupil muscle.
- C) Adrenal medulla.
- D) Ventricles of the heart.

9. Sympathetic to head & neck produces:

- A) Relaxation of the ciliary muscle of the eye.



- B) A small-sized pupil.
- C) Enophthalmos.
- D) Dropping of the upper eyelid.

10. The effect of sympathetic innervation on the salivary glands is:

- A) Secretion of scanty viscous saliva.
- B) Vasoconstriction & inhibition of secretion.
- C) Vasodilatation & inhibition of secretion.
- D) Vasodilatation & secretion.

11. Concerning the adrenal medulla, all the following is true EXCEPT:

- A) It receives preganglionic nerve supply from the anterior horn cells of the lower 6 thoracic segments of the spinal cord.
- B) It secretes epinephrine 80% & norepinephrine 20%.
- C) It augments and prolongs the effects of sympathetic stimulation.
- D) Its secretion is increased in emergency conditions.

12. Postganglionic sympathetic fibers starting from the cervical and upper four thoracic ganglia cause:

- A) Narrowing of the respiratory passages.
- B) Dilatation of the coronary vessels.
- C) Contraction of the splenic capsule.
- D) Secretion of the adrenomedullary hormones.

13. Sympathetic tone acts on:

- A) Maintenance of basal sleep rhythm.
- B) Maintenance of basal gastrointestinal motility.
- C) Maintenance of normal arterial blood pressure.
- D) Maintenance of basal cardiac rhythm.

14. What kind of substances can't permeate membranes by passive diffusion?

- A) Hydrophilic.
- B) Non-ionized substances.
- C) Hydrophobic substances.
- D) Lipid soluble.

15. The resting potential of a nerve membrane is primarily dependent on the concentration gradient of:

- A) Chloride.
- B) Sodium.
- C) Calcium.
- D) Potassium.

16. Which of the following effects is produced by an injury to the first 4 sacral segments?

- A) Lack of erection of the penis in males & clitoris in females.
- B) Vasodilatation in the cutaneous blood vessels.
- C) Disturbance of sweat secretion.
- D) Hyperglycemia.

17. Which of the following is correct?

- A) Stimulation of the greater splanchnic nerve results in adrenomedullary secretion.



- B) The sympathetic nervous system supplies every organ in the body.
- C) Sympathetic supply to the lungs causes bronchoconstriction.
- D) Postganglionic fibers starting from the superior cervical ganglion cause constriction of the pupils.

18. Generalized sympathetic stimulation causes:

- A) Rise in arterial blood pressure.
- B) Increased gastric motility.
- C) Decreased blood sugar levels.
- D) Inhibition of sweating.

19. Stimulation of the greater splanchnic nerve does NOT cause:

- A) Decreased gastrointestinal motility.
- B) Ejaculation.
- C) Increased blood glucose level.
- D) Secretion of catecholamines from the suprarenal medulla.

20. The effects of sympathetic stimulation on the iris:

- A) Vitreous secretion.
- B) Pupillary constriction.
- C) Pupillary dilation.
- D) Astigmatism.

21. Cooperative (sympathetic & parasympathetic) autonomic effects act on:

- A) The diameter of the pupil.
- B) Gastrointestinal activity.
- C) Sexual activity.
- D) Cardiac activity.

22. The hyperpolarization phase of the action potential is due to:

- A) Slow return of the K⁺ channels to the closed state.
- B) Increased excitability of the membranes.
- C) Net Na⁺ efflux to the exterior.
- D) Inactivation of the Na⁺ channels.

23. Activation of the sympathetic nervous system produces the following EXCEPT:

- A) Contraction of the detrusor muscle of the urinary bladder wall.
- B) Tachycardia.
- C) Mydriasis (dilatation of pupil).
- D) Bronchodilatation.

24. Sympathetic to the thorax produces:

- A) Narrowing of the respiratory passages.
- B) Dilatation of the coronary vessels.
- C) Contraction of the splenic capsule.
- D) Secretion of the adrenomedullary hormones.

25. Antagonistic autonomic innervation acts on:

- A) The act of micturition.
- B) Cutaneous blood vessels.
- C) Salivary gland secretion.
- D) Constrictor pupillae muscle.

26. Example of collateral ganglia is:

- A) The basal ganglia.
- B) Coeliac ganglion.
- C) Paravertebral ganglia.
- D) Adrenal medullary cells.

27. Acetylcholine is NOT a specific neurotransmitter at:

- A) Parasympathetic postganglionic nerve endings.
- B) Sympathetic postganglionic nerve endings.
- C) Parasympathetic ganglia.
- D) Sympathetic ganglia.

1-A, 2-D, 3-D, 4-A, 5-B, 6-C, 7-C, 8-B, 9-A, 10-A, 11-A, 12-B, 13-C, 14-A, 15-D, 16-A, 17-A, 18-A, 19-B, 20-C, 21-C, 22-A, 23-A, 24-B, 25-A, 26-B, 27-B



11. Formative exam 2025

1. Sympathetic supply to the head & neck produces which of the following?
 - A) Marked salivary secretion.
 - B) Dilatation of skin blood vessels.
 - C) Ptosis (drooping of upper eyelid).
 - D) Pupil dilatation.
2. A method of passive transport which utilizes special proteins called carrier proteins is called:
 - A) Protein channel.
 - B) Simple diffusion.
 - C) Active transport.
 - D) Facilitated diffusion.
3. Which of the following is true about the sympathetic division of the ANS?
 - A) Cranio-sacral in origin.
 - B) Acts during fight or flight response.
 - C) Its stimulation causes pupil constriction.
 - D) Relays at terminal ganglia.
4. Which of the following does NOT occur in Horner's syndrome?
 - A) Pallor due to cutaneous vasoconstriction.
 - B) Miosis.
 - C) Anhydrosis.
 - D) Ptosis.
5. Which is correct about RMP of a nerve?
 - A) Due to K^+ diffusion from outside to inside the nerve fibers.
 - B) Na^+/K^+ pump contributes about -4 mV.
 - C) Due to diffusion of intracellular protein to outside the nerve fibers.
 - D) Normally about -10 mV.
6. Which of the following is false about the autonomic nervous system?
 - A) It is an involuntary system.
 - B) Its thoracolumbar division is the sympathetic system.
 - C) Supplies skeletal muscle.
 - D) Its craniosacral division is the parasympathetic system.
7. Which of the following is correct about autonomic ganglia?
 - A) Act as distributing centers.
 - B) Located along the courses of afferent nerves.
 - C) Chemical transmitter is noradrenalin.
 - D) 5 types.
8. Na^+ -glucose transporter is an example of:
 - A) Antiport.
 - B) Symport.
 - C) Simple diffusion.
 - D) Facilitated diffusion.
9. Which of the following best describes the process of diffusion?
 - A) A "downhill" process.



- B) An "uphill" process.
- C) A process by which only large molecular weight proteins may move across biological membranes.
- D) Requires metabolic energy.

10. The autonomic ganglia:

- A) Function as distributing centers.
- B) Are 5 types.
- C) Are located inside the CNS.
- D) Are the site of relay of afferent neurons.

11. In resting neurons, potassium ions (K⁺) tend to:

- A) Move out of the cells according to the concentration gradient.
- B) Move into the cells according to the electrical gradient.
- C) Move out of the cells against the concentration gradient.
- D) Move into the cells according to the concentration gradient.

12. Saltatory conduction refers to:

- A) The slow conduction of action potentials in dendrites.
- B) Continuous propagation of action potentials along unmyelinated axons.
- C) The jumping of action potentials from one node of Ranvier to the next in myelinated axons.
- D) The initiation of action potentials at the axon hillock.

13. Which of the following best describes the "all-or-none" principle of action potentials?

- A) Action potentials occur only if the stimulus reaches a certain threshold.
- B) Action potentials decrease in strength as they propagate.
- C) Action potentials vary in amplitude based on stimulus strength.
- D) Action potentials can be partial, depending on conditions.

14. The rapid depolarization phase of the action potential is primarily due to:

- A) Efflux of sodium ions.
- B) Efflux of calcium ions.
- C) Influx of sodium ions.
- D) Influx of potassium ions.

15. During the repolarization phase of an action potential, which event occurs?

- A) Chloride channels open.
- B) Calcium channels open.
- C) Sodium channels open.
- D) Potassium channels open.

16. During the relative refractory period, a neuron:

- A) Is undergoing depolarization.
- B) Cannot generate another action potential.
- C) Can generate another action potential with a stronger-than-normal stimulus.
- D) Is at its resting membrane potential.

17. In a typical neuron, the action potential is initiated at the:

- A) Synaptic terminal.
- B) Dendrites.
- C) Axon hillock.
- D) Node of Ranvier.



18. What is the threshold potential typically required to initiate an action potential in neurons?

- A) -70 mV.
- B) -30 mV.
- C) -55 mV.
- D) -90 mV.

19. Which of the following best describes the role of myelin in action potential conduction?

- A) It increases membrane capacitance.
- B) It provides electrical insulation, increasing conduction velocity.
- C) It decreases the speed of action potential propagation.
- D) It facilitates continuous conduction along the entire axon.

20. The "undershoot phase" of an action potential is primarily due to:

- A) Activation of chloride channels.
- B) Inactivation of calcium channels.
- C) Prolonged opening of sodium channels.
- D) Delayed closing of potassium channels.

21. Which of the following ions has the greatest influence on the resting membrane potential of a neuron?

- A) Potassium (K^+).
- B) Sodium (Na^+).
- C) Chloride (Cl^-).
- D) Calcium (Ca^{2+}).

22. The sympathetic supply cannot be responsible for:

- A) Erection of hair.
- B) Secretion of sweat glands.
- C) Dilatation of skeletal muscle blood vessels.
- D) Constriction of the pupil.

23. Vagus parasympathetic nerve supplies all the following structures EXCEPT:

- A) Stomach.
- B) Pancreas.
- C) Rectum.
- D) Gall bladder.

24. Which neurotransmitter is released by preganglionic sympathetic neurons?

- A) Serotonin.
- B) Dopamine.
- C) Acetylcholine.
- D) Norepinephrine.

25. What effect does the parasympathetic system have on heart rate?

- A) Increases it.
- B) Decreases it.
- C) No effect.
- D) Irregular effect.

26. The adrenal medulla releases:

- A) Acetylcholine and norepinephrine.



- B) Epinephrine and norepinephrine.
- C) Serotonin and acetylcholine.
- D) Norepinephrine and dopamine.

27. Vagal stimulation causes:

- A) Dilatation of the bronchi and bronchioles.
- B) Increase in gastric and pancreatic secretion.
- C) Inhibition of plain muscles in the stomach and small intestine.
- D) Tachycardia.

28. A frightened (afraid) person might present with:

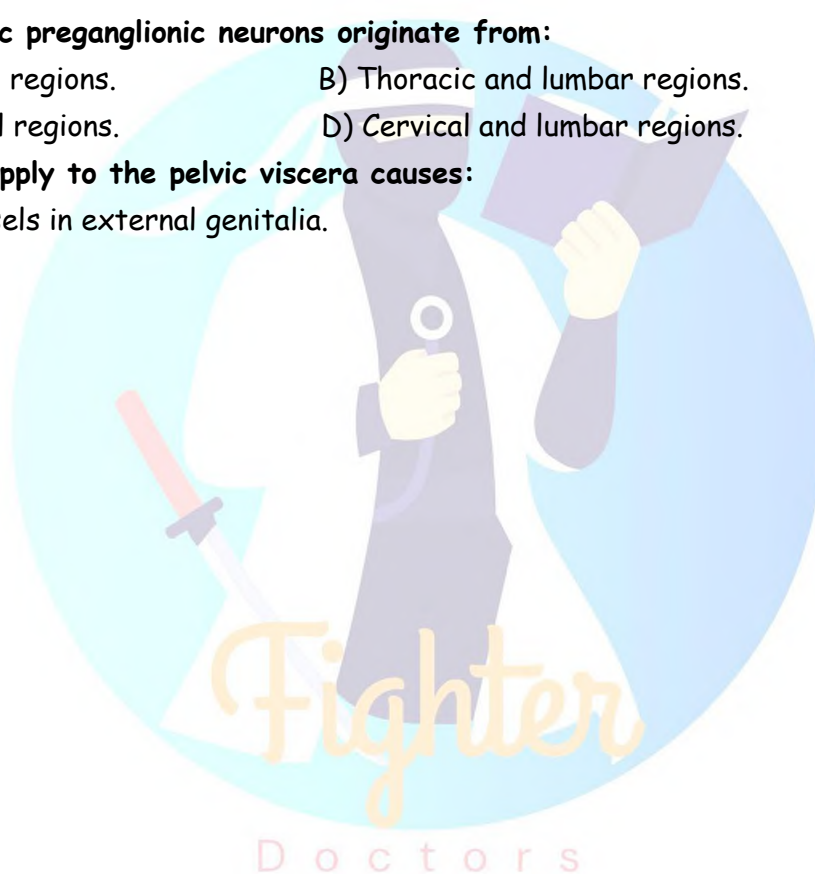
- A) Pupil dilatation.
- B) Decreased heart rate.
- C) Bronchoconstriction.
- D) Excessive salivation.

29. Parasympathetic preganglionic neurons originate from:

- A) Cranial and sacral regions.
- B) Thoracic and lumbar regions.
- C) Lumbar and sacral regions.
- D) Cervical and lumbar regions.

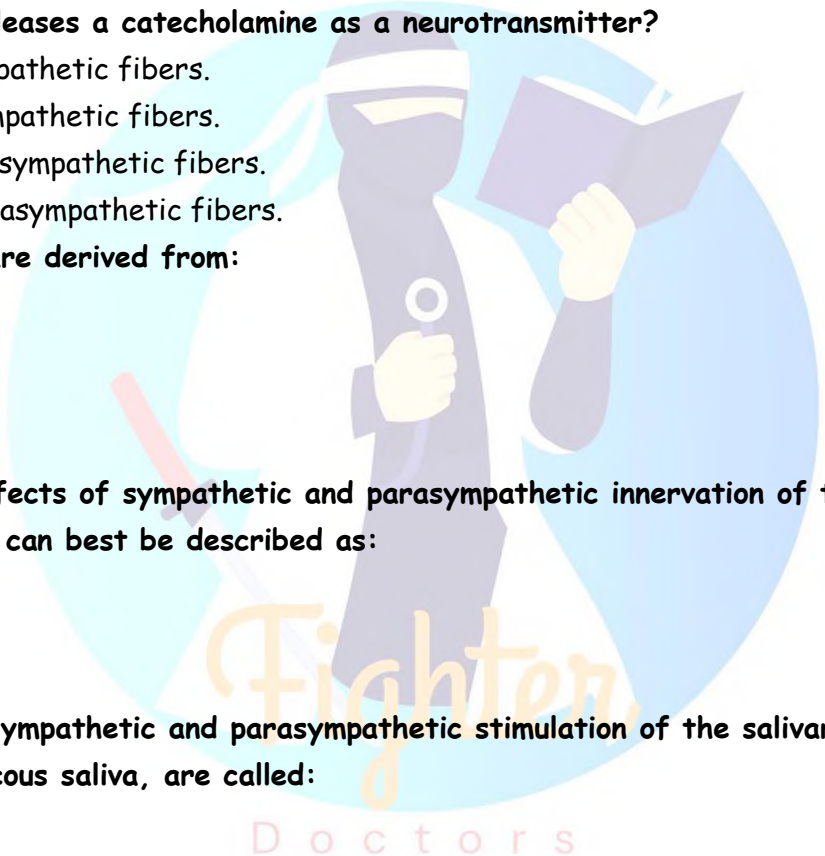
30. Sympathetic supply to the pelvic viscera causes:

- A) Dilatation of vessels in external genitalia.
- B) Defecation.
- C) Ejaculation.
- D) Micturition.





12. End Module Exam 2025

1. Which of the following is innervated by postganglionic fibers arising from the cervical ganglia?
 - A) The eyes.
 - B) The digestive system.
 - C) The urinary system.
 - D) The reproductive system.
 2. Which of the following cranial nerves does NOT contain preganglionic parasympathetic fibers?
 - A) Oculomotor (III).
 - B) Facial (VII).
 - C) Optic (II).
 - D) Glossopharyngeal (IX).
 3. Which neuron releases a catecholamine as a neurotransmitter?
 - A) Preganglionic sympathetic fibers.
 - B) Postganglionic sympathetic fibers.
 - C) Preganglionic parasympathetic fibers.
 - D) Postganglionic parasympathetic fibers.
 4. Catecholamines are derived from:
 - A) Acetylcholine.
 - B) Tyrosine.
 - C) Tryptophan.
 - D) Epinephrine.
 5. The separate effects of sympathetic and parasympathetic innervation of the pacemaker region of the heart can best be described as:
 - A) Antagonistic.
 - B) Complementary.
 - C) Cooperative.
 6. The effects of sympathetic and parasympathetic stimulation of the salivary glands, producing a thicker, more viscous saliva, are called:
 - A) Antagonistic.
 - B) Inhibitory.
 - C) Synergistic.
 7. Which of the following target tissues is innervated only by neurons from the sympathetic nervous system?
 - A) Adrenal medulla.
 - B) Lacrimal glands.
 - C) Bronchioles.
 - D) Pupil of the eye.
 8. The vagus nerve regulates major elements of which part of the nervous system?
 - A) Parasympathetic nervous system.
 - B) Enteric nervous system.
 - C) Sympathetic nervous system.
- 



D) Central nervous system.

9. Which of these is expected if the sympathetic division is activated?

- A) Tear production increases.
- B) Glucose release from the liver decreases.
- C) Secretion of watery saliva increases.
- D) Air passageways dilate.

10. Which of these functions is NOT affected by parasympathetic stimulation?

- A) Secretion of tears.
- B) Secretion of sweat.
- C) Secretion of digestive glands.
- D) Secretion of thin, watery saliva.

11. The effect of parasympathetic stimulation on the eye is to ____ the ciliary muscle and to ____ the pupil:

- A) Contract, constrict.
- B) Contract, dilate.
- C) Relax, constrict.
- D) Relax, dilate.

12. The effect of sympathetic stimulation on the urinary bladder is to ____ the wall and ____ the sphincter:

- A) Contract, contract.
- B) Contract, relax.
- C) Relax, contract.
- D) Relax, relax.

13. The effector organs for the somatic motor nervous system are:

- A) Cardiac muscle.
- B) Smooth muscle.
- C) Glands.
- D) Skeletal muscle.

14. The sympathetic division of the autonomic nervous system normally controls target cells with ____ receptors:

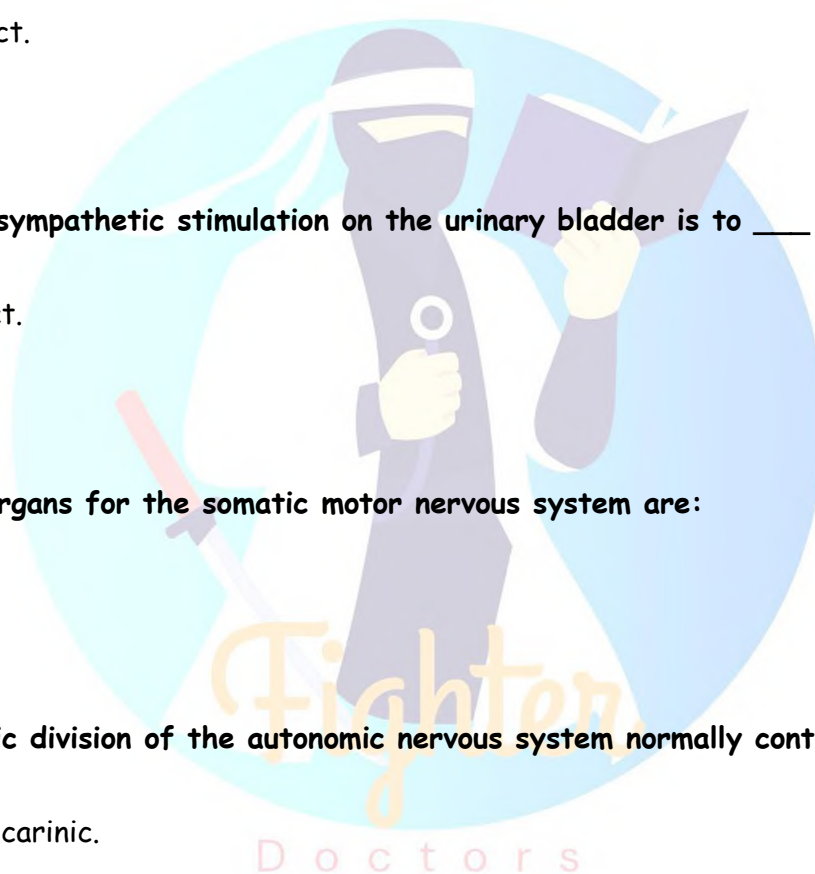
- A) Nicotinic and muscarinic.
- B) All alpha and beta.
- C) Theta and delta.
- D) Nm and M.

15. Which of the following is a sympathetic effect?

- A) Constriction of the pupils of the eyes.
- B) Constriction of the bronchioles in the lungs.
- C) Contraction of the urinary bladder.
- D) Increased heart rate.

16. The excitability of the nerve is zero during:

- A) The supernormal phase of excitability.
- B) The relative refractory period.
- C) The subnormal phase of excitability.





D) The absolute refractory period.

17. All the following structures are supplied by the sympathetic system only, EXCEPT:

- A) Ventricles of the heart.
- B) Constrictor pupillae muscle.
- C) Adrenal medulla.
- D) Skin.

18. Increased activity of the sympathetic nervous system causes:

- A) Pupil constriction.
- B) Peristaltic movements.
- C) Penile erection.
- D) Piloerection.

19. A lesion in the cervical spinal area (above T1 level) is most likely to result in:

- A) Constriction of the eye pupil.
- B) Dilation of the eye pupil.
- C) Drooping of the upper eyelid (ptosis).
- D) Hidrosis (sweating) of the face.

20. Stimulation of the cervical sympathetic nerve causes:

- A) Constriction of the pupil.
- B) Relaxation of the ciliary muscle of the eye.
- C) Miosis (small-sized pupil).
- D) Drooping of the upper eyelid.

21. The sympathetic supply cannot be responsible for:

- A) Erection of hair.
- B) Dilation of skeletal muscle blood vessels.
- C) Secretion of salivary glands.
- D) Constriction of the pupil.

22. Which of the following organs receives both sympathetic and parasympathetic innervation?

- A) Adrenal medulla.
- B) Sweat glands.
- C) Lungs.
- D) Pilomotor muscles.

23. Stimulation of the sympathetic supply to the pelvis causes:

- A) Micturition.
- B) Ejaculation.
- C) Defecation.
- D) Erection.

24. The following reflexes are all autonomic, EXCEPT:

- A) Salivary secretion.
- B) Micturition.
- C) Flexion of the arm following a painful stimulus.
- D) Defecation.



25. During stress, the sympathetic division of the ANS would cause each of the following EXCEPT:

- A) Bronchodilation.
- B) Dilatation of arteries in skeletal muscles and the heart.
- C) Contraction of the urinary bladder.
- D) Constriction of arteries to the gastrointestinal tract.

26. Excitability of the nerve is increased by:

- A) Hypercalcemia.
- B) Hypokalemia.
- C) Hypocalcemia.
- D) Acidosis.

27. All of the following organs receive their parasympathetic innervation via the vagus nerve EXCEPT:

- A) Rectum.
- B) Atria of the heart.
- C) Gall bladder.
- D) Proximal half of the large intestine.

28. Salivary secretion increases on stimulation of:

- A) Trigeminal nerve.
- B) Glossopharyngeal nerve.
- C) Vagus nerve.
- D) Hypoglossal nerve.

29. The sympathetic system originates from:

- A) T1 to S2.
- B) T2 to L1.
- C) T1 to L2.
- D) C1 to L2.

30. Which of the following structures is innervated by preganglionic sympathetic nerve fibers?

- A) Lung.
- B) Heart.
- C) Adrenal medulla.
- D) Pancreas.

31. Which of the following is true about continuous conduction?

- A) Occurs in myelinated nerve fibers.
- B) Occurs by jumping of charges from one node of Ranvier to another.
- C) Is relatively slow, 0.5-2.0 meter/second.
- D) Occurs in the neuro-muscular junction.

32. Stimulation of the parasympathetic nervous system leads to an increase of:

- A) Sweating.
- B) Blood pressure.
- C) Heart rate.
- D) Gastrointestinal motility.

33. Excitability of nerve fibers:

- A) Is increased by decreased temperature.
- B) Is increased by decreased Na^+ ions.
- C) Is decreased by decreased Ca^{++} ions.
- D) Is completely lost by local anaesthetic drugs.



34. The neurotransmitter at autonomic ganglia is:

- A) Acetylcholine.
- B) Adrenaline.
- C) Noradrenaline.
- D) Dopamine.

35. A term used when there is a stimulus which is not sufficient enough to produce an action potential response (local effect):

- A) Threshold.
- B) Subthreshold.
- C) Suprathreshold.

36. The most important ion during the resting membrane potential is:

- A) Na⁺.
- B) K⁺.
- C) Cl⁻.
- D) Ca⁺⁺.

37. During an action potential, depolarization of the nerve cell membrane is brought about by:

- A) Increased sodium permeability.
- B) Increased calcium permeability.
- C) Increased chloride permeability.
- D) Increased potassium permeability.

38. Descending limb of the spike potential in the nerve:

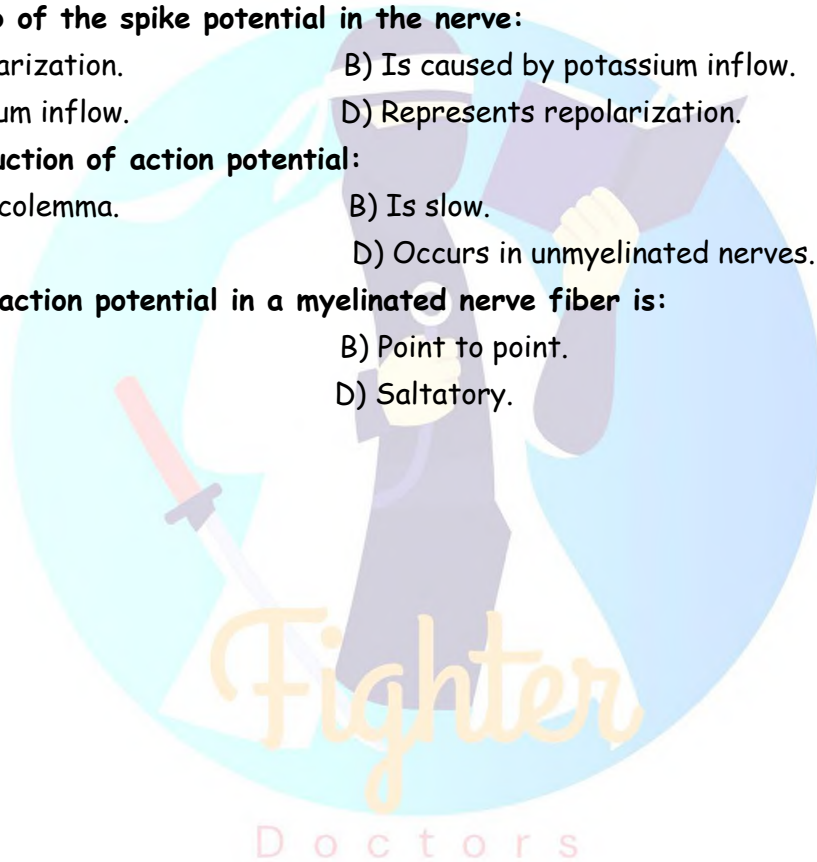
- A) Represents depolarization.
- B) Is caused by potassium inflow.
- C) Is caused by sodium inflow.
- D) Represents repolarization.

39. Saltatory conduction of action potential:

- A) Occurs in the sarcolemma.
- B) Is slow.
- C) Is fast.
- D) Occurs in unmyelinated nerves.

40. Propagation of action potential in a myelinated nerve fiber is:

- A) Slow.
- B) Point to point.
- C) Passive.
- D) Saltatory.



1-A, 2-C, 3-B, 4-B, 5-A, 6-C, 7-A, 8-A, 9-D, 10-B, 11-A, 12-C, 13-D, 14-B, 15-D, 16-D, 17-B, 18-D, 19-B, 20-B, 21-D, 22-C, 23-B, 24-C, 25-C, 26-C, 27-A, 28-B, 29-C, 30-C, 31-B, 32-D, 33-D, 34-A, 35-B, 36-B, 37-A, 38-D, 39-C, 40-D



13. End semester Exam 2025

1. The cell bodies of the preganglionic neuron (autonomic nervous system) are located in:

- A) Autonomic ganglia.
- B) Lateral horn cells.
- C) Blood vessels.
- D) Lower abdomen.

2. The parasympathetic division is activated during:

- A) Exercise.
- B) Fear.
- C) Rest and digest.
- D) Excitement.

3. Preganglionic fibers from the cranial outflow are carried by:

- A) 3rd, 5th, 11th, 8th cranial nerves.
- B) 2nd, 4th, 7th, 9th cranial nerves.
- C) 3rd, 5th, 8th, 10th cranial nerves.
- D) 3rd, 7th, 9th, 10th cranial nerves.

4. Preganglionic fibers of the sacral outflow are carried by:

- A) Ciliary ganglion.
- B) Submandibular ganglion.
- C) Pelvic nerve.
- D) Peripheral ganglion.

5. Which of the following is a parasympathetic effect?

- A) Decreased secretion of lacrimal gland.
- B) Miosis.
- C) Dilated iris of the eye.
- D) Dilated bronchi.

6. During the absolute refractory period of a spike in an axon:

- A) Another AP can be elicited by a subthreshold stimulus.
- B) Another AP can be produced by a threshold stimulus.
- C) Another AP may be initiated by a stimulus bigger than threshold.
- D) It is impossible to initiate another AP even if we give a suprathreshold stimulus.

7. The nerve action potential:

- A) Starts with opening of the voltage-gated Na channels.
- B) Starts by potassium influx into the cell.
- C) Depends to a great extent on the state of chloride channels.
- D) Starts with hyperpolarization of the cell membrane.

8. Reestablishment (restoration) of Na^+/K^+ distribution after an action potential is mainly due to:

- A) Na^+ outflux.
- B) K^+ influx.
- C) Na^+-K^+ pump.
- D) Na^+/K^+ leak channel.

9. Stimulation of the sympathetic supply to the pelvis causes:

- A) Micturition.
- B) Defecation.
- C) Ejaculation.
- D) Erection.

10. Which one of the following effects is NOT produced by sympathetic stimulation?

- A) Bronchodilatation.
- B) Tachycardia.
- C) Mydriasis.
- D) Contraction of the urinary bladder wall.

11. Sympathetic to the thorax causes:

- A) Secretion of the adrenomedullary hormones.
- B) Dilatation of the coronary vessels.
- C) Contraction of the splenic capsule.
- D) Narrowing of the respiratory passages.



12. Which one of the following is wrong about the adrenal medulla?

- A) It is a sympathetic ganglion.
- B) Secretes more noradrenaline than adrenaline.
- C) It releases epinephrine into circulating blood.
- D) It's innervated by sympathetic cholinergic fibers.

13. Parasympathetic tone is mostly responsible for:

- A) Decreasing the blood flow to the gut (GIT/alimentary canal).
- B) Maintenance of the systemic arterial blood pressure constant.
- C) Keeping the fasting blood glucose level constant.
- D) Maintaining resting heart rate within normal.

14. The resting potential of a nerve membrane is primarily dependent on the concentration gradient of:

- A) Potassium.
- B) Sodium.
- C) Calcium.
- D) Chloride.

15. Na^+-K^+ pump:

- A) Requires energy which is taken from GTP.
- B) Couples Na^+ and K^+ with ratio 3:2.
- C) Helps intracellular accumulation of Na^+ .
- D) Is a passive process, needs no energy.

16. Which of the following is actively transported out of the neurons?

- A) K^+
- B) HPO_4^- .
- C) Cl^-
- D) Na^+ .

1-B, 2-C, 3-D, 4-C, 5-B, 6-D, 7-A, 8-C, 9-C, 10-D, 11-C, 12-B, 13-D, 14-A, 15-B, 16-D

Essay Questions

1. List five factors affecting the rate of diffusion.
2. Enumerate the importance of the Na^+-K^+ pump.
3. Conduction in myelinated nerve fibers.
4. Components of the autonomic reflex are?
5. Explain the underlying mechanism of the following:
 - After-depolarization (negative after-potential):
 - After-hyperpolarization (positive after-potential):
6. Mention the effect of the following on nerve excitability:
 - Electrotonus
 - Extracellular fluid calcium
7. Orbeli phenomenon:
8. Functions of sympathetic supply to the thorax?
9. Denervation hypersensitivity:
10. Interaction between sympathetic & parasympathetic systems:



14. End module 2023

1. The function of rough endoplasmic reticulum is to synthesize:

- a. Lipid.
- b. Protein that will be secreted by the cell.
- c. Cytoplasmic protein necessary for the cells' own existence.
- d. Carbohydrates.

2. Smooth endoplasmic reticulum:

- a. Sometimes has ribosomes attached to its membrane
- b. Is rarely found in skeletal muscle cells
- c. Areas of localized basophilic indicate its presence
- d. Is present in cells where drug detoxification is taking place

3. The Golgi apparatus is:

- a. The site of packaging of secretory products of the cell.
- b. The site of protein synthesis.
- c. Continuous with smooth endoplasmic reticulum.
- d. Capable of providing energy.

4. Multivesicular bodies are:

- a. Primary lysosomes.
- b. Formed as a result of digestion of solid material.
- c. Formed as a result of digestion of fluid material.
- d. Autophagic vacuoles.

5. Microtubules have important role in:

- a. Carbohydrate metabolism.
- b. Oxidative phosphorylation.
- c. Water balance.
- d. Formation of the mitotic spindle during mitosis.

6. DNA could be found in:

- a. Ribosomes.
- b. Golgi apparatus.
- c. Mitochondria.
- d. Lysosomes.

7. Heterochromatin is:

- a. Present in the nuclei of metabolically active cells.
- b. Special type of stain.
- c. Present in the nuclei of metabolically inactive cells
- d. Type of cell inclusions.



**8. The nucleolus is formed of:**

- a. Heterochromatin and euchromatin.
- b. Nucleolonema and pars amorpha.
- c. Nucleolus associated and peripheral chromatin.
- d. Condensed chromatin.

9. The nuclear pore

- a. Is at regular intervals.
- b. Is bridged by a unit membrane.
- c. Is a transient structure.
- d. Allows for communication between the nucleus and the cytoplasm

10. Glycocalyx is:

- a. The protein component of the cell membrane.
- b. The hydrophilic end of lipid.
- c. The carbohydrate component of the cell membrane.
- d. The hydrophobic end of lipid.

11. Steriocilia are:

- a. Long motile cilia.
- b. True cilia.
- c. Long motile microvilli.
- d. Long microvilli.

12. Endothelium provides lining of:

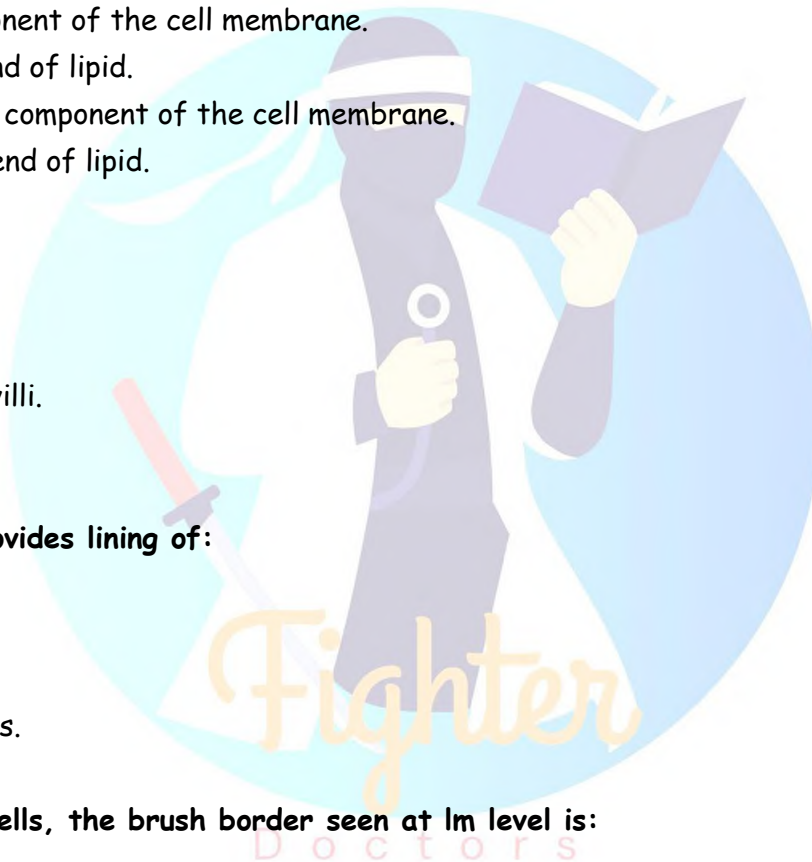
- a. Peritoneum.
- b. Pleural cavity.
- c. The heart.
- d. Alveoli of the lungs.

13. In absorptive cells, the brush border seen at lm level is:

- a. Streocilia.
- b. True cilia.
- c. Microvilli.
- d. Basal infoldings.

14. Stratified squamous epithelium:

- a. Consists of a single layer of flattened cells.
- b. Is protective rather than secretory or absorptive.
- c. Has a rich supply of capillaries running between its cell layers.
- d. May be ciliated.

15. Desmosomes are:



- a. Macula occludens.
- b. Macula adherens.
- c. Zonula occludens.
- d. Zonula adherens.

16. In zonula occludens the distance between two cell membranes is:

- a. 0 nm
- b. 2 nm
- c. 20 nm
- d. 30 nm

17. Compound exocrine glands:

- a. Have unbranched secretory duct
- b. Have branched secretory ducts
- c. Secrete hormones
- d. Secrete into the blood.

18. Which of the following not included in the classical description of a junctional complex:

- a. The fascia occludes
- b. The zonula occludens
- c. The zonula adherens
- d. The macula adherens

19. The main ultrastructural feature (EM) of histiocyte is the presence of:

- a. Microvilli.
- b. Lysosomes.
- c. Smooth endoplasmic reticulum.
- d. Rough endoplasmic reticulum.

20. Plasma cells:

- a. Have well developed smooth endoplasmic reticulum
- b. Have phagocytic ability
- c. Are commonly found in peripheral blood
- d. Show all EM features of protein synthesizing cells.

21. Metachromatic granules are present in:

- a. Mast cells.
- b. Plasma cells.
- c. Fat cells.
- d. Reticular cells.

22. Which of the following would be best suited to differentiate collagen fibers from other fibers?

- a. Hematoxylin and eosin stain



- b. Sudan stain
- c. Silver
- d. Mallory trichrome stain

23. What type of tissue is cornea composed of?

- a. Mucous connective tissue
- b. Mesenchyme
- c. Dense irregular connective tissue
- d. Dense regular connective tissue

24. What is the shape of fat cells present in the unilocular adipose tissue?

- a. Clock face
- b. Signet ring
- c. Horse shoe
- d. Kidney

25. Which of the following is the only example of a unicellular exocrine gland?

- a. Sweat gland
- b. Intestinal gland
- c. Goblet cell
- d. Sebaceous glands

26. Which of the following statements is incorrect?

- a. Microtubules are involved in ciliary movement and intracellular transport.
- b. The cis (forming) face of Golgi apparatus is where secretory granules are formed.
- c. The microtubules of the centrioles and in the basal body of cilia have a 9x3 arrangement.
- d. Ribosomes, made of rrna and protein, are the site of protein synthesis.

27. Pseudostratified columnar ciliated epithelium can be found in:

- a. Fallopian tube
- b. Blood vessels
- c. Respiratory passages
- d. Urinary passage

28. Which of the following glands is considered as an apocrine gland:

- a. Pancreas
- b. Mammary glands
- c. Salivary glands
- d. Sebaceous glands

29. Which of the following statements is incorrect regarding neuroepithelium?

- a. It is modified type of epithelium to act as receptor.
- b. Formed of sensory and supporting cells.
- c. It covers skin and forms glands.



d. Present in taste buds.

30. Which of the following form stroma of the organs?

- a. Loose areolar connective tissue.
- b. Reticular connective tissue.
- c. Unilocular adipose connective tissue.
- d. Muroid connective tissue.

31. Which of the following is one of the characteristics of anatomical position?

- a. The arms are abducted
- b. The palm face anterior
- c. The lower limbs are abducted
- d. The toes directed to the left.

32. Which of the following anatomical planes divided the body into anterior and posterior parts?

- a. Horizontal plane
- b. Coronal plane
- c. Midsagittal plane
- d. Parasagittal plane

33. Where does the superficial palmar arch lie?

- a. In the anterior surface of the hand
- b. In the posterior surface of the hand
- c. In the superior surface of the foot
- d. In the inferior surface of the foot

34. Which of the following movement of the limb upward?

- a. Flexion
- b. Elevation
- c. Extension
- d. Depression

35. Which of the following cartilage disappears in the adult?

- a. Growing plate
- b. Articular cartilage
- c. Costal cartilage
- d. Cartilage of the auricle

36. There are two forms of bone, which of the following is the character of spongy bone?

- a. It consists of number of cylindrical units called haversion system
- b. It is formed of trabeculae
- c. It forms the surface layer of all bones
- d. It provides strength for weight bearing

37. Which of the following arteries supplies the bone marrow?



- a. Periosteal artery
- b. The articular artery
- c. Epiphyseal artery
- d. Nutrient artery

38. Which of the following is fibrous joint?

- a. Elbow joint
- b. Shoulder joint
- c. Inferior tibiofibular joint
- d. Intervertebral disc

39. What is the type of costochondral junction?

- a. Fibrous joint
- b. Primary cartilaginous joint
- c. Secondary cartilaginous joint
- d. Synovial joint

40. Presence of joint cavity is one of the characters of which of the following joint?

- a. Fibrous joint
- b. Primary cartilaginous joint
- c. Secondary cartilaginous joint
- d. Synovial joint

41. Which of the following is the character of the smooth muscle?

- a. it is voluntary non striated muscle
- b. it is attached to the skeleton.
- c. it is non striated involuntary muscle
- d. it is connected by intercalated disc

42. When the muscle fibers are attached to one side of the tendon, it is called which of the following?

- a. triangular
- b. unipennate
- c. bipennate
- d. spiral

43. When the muscle removes unwanted movement in the proximal joint and the prime mover act on the distal, it is called

- a. antagonist
- b. prime mover
- c. synergist
- d. fixator

44. In portal circulation, which of the following ends in the inferior vena cava?

- a. portal vein
- b. hepatic vein
- c. left gastric vein
- d. superior pancreatico duodenal vein



45. Which of the following vessel is fenestrated?

- a. large artery
- b. small artery
- c. vein
- d. blood capillary

46. Which structure does not take place in the late blastocyst?

- a. Zona pellucida
- b. Trophoblast
- c. Embryoblast
- d. Blastocoele

47. Which statement about the neural crest is NOT true?

- a. Pia mater can develop from the neural crest
- b. Neural crest gives rise to the cortex of the suprarenal gland
- c. Melanocytes originate from the neural crest
- d. Ganglionic cells originate from neural crest cells

48. Which statement is true about the trophoblast?

- a. It differentiates into outer cytotrophoblast and inner syncytiotrophoblast
- b. It differentiates into inner cytotrophoblast and outer syncytiotrophoblast
- c. Structure called "trophoblast" does not exist during the development of human
- d. It does not differentiate further

49. Where originates the extraembryonic mesoderm?

- a. It is a result of the migration of cells from the neuroectoderm
- b. In a site between trophoblast and embryoblast
- c. In a site between hypoblast and epiblast
- d. In a site between endoderm and ectoderm

50. Where the fertilization occurs usually?

- a. In the infundibular part of the Fallopian tube
- b. In the ampullary part of the Fallopian tube
- c. Horn of the uterus (cornua uteri)
- d. Cavity of the uterus

51. What is a capacitation?

- a. Capacitation is a process in which the glycoprotein layer over the acrosome is dissolved
- b. Capacitation is a process in which sperm penetrates zona pellucida
- c. Capacitation is a process in which the tail of the sperm is inactivated
- d. Capacitation is a process in which the tail of the sperm is activated

52. Primitive streak originates:

- a. In lateral parts of hypoblast
- b. In lateral parts of epiblast
- c. In the medial line of hypoblast
- d. In the medial line of epiblast

53. When starts the implantation?

- a. Immediately after fertilization
- b. 3rd day after fertilization
- c. 6th day after fertilization
- d. 9th day after fertilization



54. How to describe the origin of the intraembryonic mesoderm?

- a. Cells from the primitive node migrate between ectoderm and endoderm
- b. Cells from the wall of the amnionic sack migrate between ectoderm and endoderm
- c. Cells from the wall of the yolk sack migrate between ectoderm and endoderm
- d. Cells from primitive streak migrate between ectoderm and endoderm

55. Which statement about 'formation of somites is correct?

- a. Formation of somites is gradual: Cranial somites start, caudal somites continue later on
- b. Formation of somites is gradual: Caudal somites start, cranial somites continue later on
- c. Timing of formation of somites is less or more chaotic
- d. All somites are formed at the same time

56. The most common site for implantation in ectopic pregnancy is:

- a. internal os of the uterus
- b. mesentery
- c. ovary
- d. uterine tube

57. The first week of human development is characterized by formation of the:

- a. inner cell mass
- b. trophoblast
- c. blastocyst
- d. all of the above

58. The ovulated mammalian oocyte is arrested at:

- a. prophase of meiosis I
- b. metaphase of meiosis I
- c. prophase of meiosis II
- d. metaphase of meiosis II

59. All but one of following statements about the cleavage are true. EXCEPT?

- a. Cleavage occurs when the embryo travels through the Fallopian tube
- b. During cleavage, the amount of DNA inside the embryo increases
- c. Embryo is closed inside zona pellucida during cleavage
- d. Cleavage continues after implantation

60. Which embryonic structure gives rise to cells in our bodies?

- a. epiblast and hypoblast
- b. Hypoblast
- c. Epiblast
- d. Extraembryonic mesoderm

1:b 2:d 3:a 4:c 5:d 6:c 7:c 8:b 9:d 10:c

11:d 12:c 13:c 14:b 15:b 16:a 17:b 18:a 19:b 20:d

21:a 22:d 23:d 24:b 25:c 26:b 27:c 28:b 29:c 30:b

31:b 32:b 33:a 34:b 35:a 36:b 37:d 38:c 39:b 40:d

41:c 42:b 43:c 44:b 45:d 46:a 47:b 48:b 49:b 50:b

51:a 52:d 53:c 54:d 55:a 56:d 57:d 58:d 59:d 60:c



15. End Module 2024

1. Which of the following is innervated by postganglionic fibers arising from the cervical ganglia?
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 - b. The digestive system
 - c. The urinary system
 - d. The reproductive system
2. Which of the following cranial nerves does not contain preganglionic parasympathetic fibers?
 - a. Oculomotor (III)
 - b. Facial (VII)
 - c. Optic (II)
 - d. Glossopharyngeal (IX)
3. Which neuron does release a catecholamine as a neurotransmitter?
 - a. Preganglionic sympathetic fibers
 - b. Postganglionic sympathetic fibers
 - c. Preganglionic parasympathetic fibers
 - d. Postganglionic parasympathetic fibers
4. Catecholamines are derived from
 - a. Acetylcholine
 - b. Tyrosine
 - c. Tryptophan
 - d. Epinephrine
5. The separate effects of sympathetic and parasympathetic innervation of the pacemaker region of the heart can best be described as
 - a. Antagonistic
 - b. Complementary
 - c. Cooperative
6. The effects of sympathetic and parasympathetic stimulation of the salivary glands, producing a thicker, more viscous saliva, are called
 - a. Antagonistic
 - b. inhibitory
 - c. synergistic
7. Which of the following target tissues is innervated by only neurons from the sympathetic nervous system?
 - a. Adrenal medulla
 - b. Lacrimal glands
 - c. Bronchioles
 - d. Pupil of the eye
8. The vagus nerve regulates major elements of which part of the nervous system?
 - a. Parasympathetic nervous system
 - b. Enteric nervous system
 - c. Sympathetic nervous system
 - d. Central nervous system
9. Which of these is expected if the sympathetic division is activated?
 - a. Secretion of watery saliva increases



- b. Tear production increases
- c. Air passageways dilate
- d. Glucose release from the liver decreases

10. Which of these functions is NOT affected by parasympathetic stimulation?

- a. Secretion of tears
- b. Secretion of sweat
- c. Secretion of digestive glands
- d. Secretion of thin, watery, saliva

11. The effect of parasympathetic stimulation on the eye is to _____ the ciliary muscle and to _____ the pupil.

- a. Contract, constrict
- b. Contract, dilate
- c. Relax, constrict
- d. Relax, dilate

12. The effect of sympathetic stimulation on the urinary bladder is to _____ the wall and _____ the sphincter.

- a. Contract, contract
- b. Contract, relax
- c. Relax, contract
- d. Relax, relax

13. The effector organs for the somatic motor nervous system are

- a. Cardiac muscle
- b. Smooth muscle
- c. Glands
- d. Skeletal muscle

14. The sympathetic division of the autonomic nervous system normally controls target cells with _____ receptors.

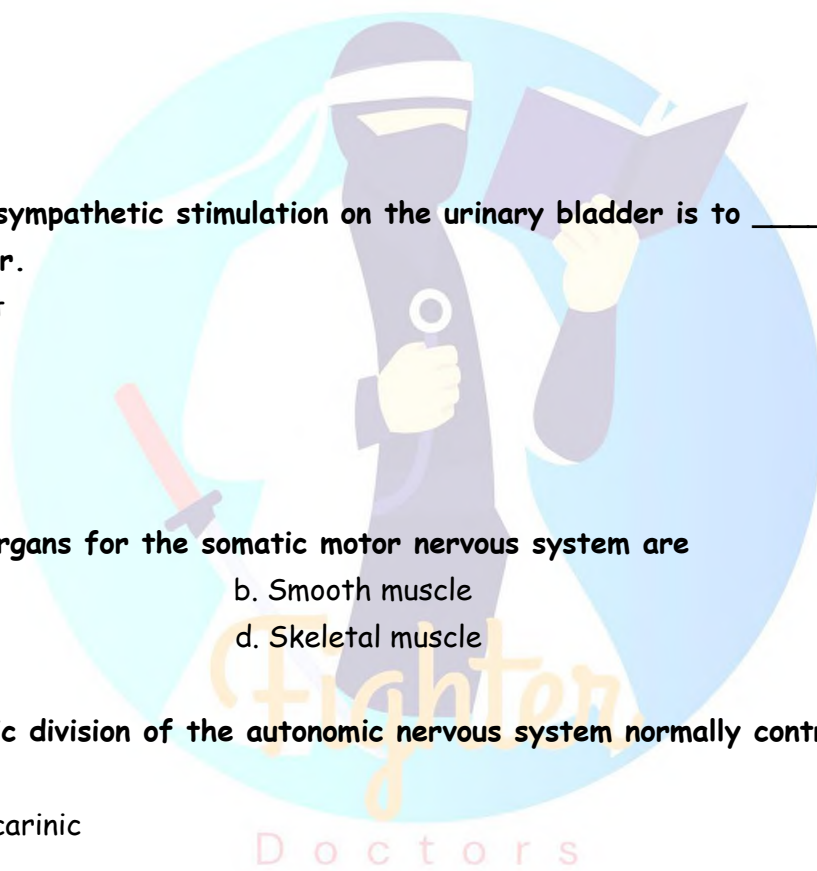
- a. Nicotinic and muscarinic
- b. All alpha and beta
- c. theta and delta
- d. Nm and M

15. Which of the following is a sympathetic effect?

- a. Constriction of the pupils of the eyes
- b. Constriction of the bronchioles in the lungs
- c. Contraction of the urinary bladder
- d. Increased heart rate

16. The excitability of the nerve is zero during:

- a. super normal phase of excitability.





- b. relative refractory period.
- c. sub normal phase of excitability.
- d. absolute refractory period.

17. All the following structures are supplied by sympathetic system only EXCEPT:

- a. Ventricles of the heart.
- b. Constrictor of pupillae muscle.
- c. Adrenal medulla.
- d. Skin.

18. Increased activity of the sympathetic nervous system causes:

- a. Penile erection.
- b. Pupillary constriction.
- c. Accommodation for near vision.
- d. Bronchiolar dilatation.

19. A lesion in the cervical symp. chain (Horner's syndrome) causes:

- a. Increased sweating.
- b. Dropping of the upper eyelid (=ptosis).
- c. Vasoconstriction of the skin blood vessels (leading to pallor).
- d. Mydriasis (= dilatation of the pupil).

20. Stimulation of the cervical sympathetic nerve causes:

- a. Dropping of the upper eyelid.
- b. A small-sized pupil.
- c. Enophthalmos.
- d. Relaxation of the ciliary muscle of the eye

21. The sympathetic supply cannot be responsible for:

- a. Constriction of the pupil.
- b. Erection of hair.
- c. Secretion of eccrine sweat glands.
- d. Dilatation of skeletal muscle blood vessels.

22. Which of the following organs receive both sympathetic & parasympathetic:

- a. Adrenal medulla.
- b. Sweat glands.
- c. Pilomotor muscles.
- d. Lungs.

23. Stimulation of sympathetic to pelvis causes:

- a. Micturition.
- b. Defecation.
- c. Ejaculation.



d. Erection.

24. The following reflexes are all autonomic EXCEPT:

- a. Salivary secretion.
- b. Micturition.
- c. Flexion of the arm following a painful stimulus.
- d. Defecation.

25. During stress the sympathetic division of the ANS would cause each of the following EXCEPT

- a. Bronchodilation
- b. Constriction of arteries to the gastrointestinal tract
- c. Contraction of the urinary bladder
- d. Dilation of arteries in skeletal muscles and the Heart

26. Excitability of the nerve is increased by:

- a. hypercalcemia.
- b. hypokalemia.
- c. acidosis.
- d. hypocalcemia.

27. All of the following organs receive their parasympathetic innervation via the vagus nerves EXCEPT:

- a. Gall bladder.
- b. Atria of the heart.
- c. Proximal half of large intestine.
- d. Rectum.

28. Salivary secretion increases on stimulation of:

- a. Trigeminal nerve.
- b. Glossopharyngeal nerve.
- c. Vagus nerve
- d. Hypoglossal nerve.

29. The sympathetic system originates from:

- a. T1 to L2.
- b. T2 to L1.
- c. C1 to L2.
- d. T1 to S2.

30. Which of the following structure is innervated by preganglionic sympathetic nerve fibers?

- a. lung
- b. heart
- c. pancreas.
- d. adrenal medulla.

31. Which of the following is true about Continuous conduction?:

- a. occurs in myelinated nerve fibers
- b. occurs by jumping of charges from one node of Ranvier to another
- c. is relatively slow 0.5-2.0 meter / second
- d. occurs in the neuromuscular junction



32. Stimulation of the parasympathetic neurons leads to an increase of:

- a. GIT motility.
- b. heart rate.
- c. blood pressure.
- d. sweating.

33. Excitability of nerve fibers:

- a. Is increased by decreased temperature
- b. Is increased by decreased Na^+ .
- c. Is decreased by decrease Ca^{++} ions.
- d. Is completely lost by local anesthetic drugs.

34. The neurotransmitter at autonomic ganglia is:

- a. noradrenaline
- b. Ach
- c. adrenaline.
- d. Dopamine.

35. A term used when there is a stimulus which is not sufficient enough to produce an action potential response (local effect)

- a. Threshold
- b. Subthreshold
- c. suprathreshold.

36. The most important ion during resting membrane potential is:

- a. Na^+
- b. K^+ .
- c. Ca^{++} .
- d. Cl^- .

37. During an action potential, depolarization of nerve cell membrane is brought about by:

- a. increased sodium permeability.
- b. increased calcium permeability.
- c. increased chloride permeability.
- d. increased potassium permeability.

38. Descending limb of the spike potential in the nerve:

- a. represents depolarization.
- b. is caused by potassium inflow.
- c. is caused by sodium inflow.
- d. represents Repolarization.

39. Saltatory conduction of action potential:

- a. occurs in the unmyelinated nerves.
- b. is slow.
- c. occurs in the sarcolemma.
- d. is fast.

40. Propagation of action potential in myelinated nerve fibre is:

- a. slow
- b. point to point.
- c. saltatory.
- d. passive.

1:a 2:c 3:b 4:b 5:a 6:c 7:a 8:a 9:c 10:b
 11:a 12:c 13:d 14:b 15:d 16:d 17:b 18:d 19:b 20:d
 21:a 22:d 23:c 24:c 25:c 26:d 27:d 28:b 29:a 30:d
 31:c 32:a 33:d 34:b 35:b 36:b 37:a 38:d 39:d 40:c



16. End semester 2024

1. The cell bodies of Preganglionic neuron (Autonomic Nervous System) are located in:

- a. Autonomic Ganglia
- b. Lateral horn cells
- c. Blood Vessels
- d. Lower Abdomen

2. The Parasympathetic division is activated during:

- a. Exercise
- b. Fear
- c. rest and digest
- d. Excitement

3. Preganglionic fibers from cranial outflow are carried by:

- a. 3rd,5th,11th,8th cranial nerves
- b. 2nd,4th,7th,9th cranial nerves
- c. 3rd,5th,8th,10th cranial nerves
- d. 3rd,7th,9th,10th cranial nerves

4. preganglionic fibers of the Sacral outflow are carried by:

- a. Ciliary ganglion
- b. Submandibular ganglion
- c. Pelvic nerve
- d. Peripheral ganglion

5. Which of the following is a Parasympathetic effect:

- a. Decreased secretion of lacrimal gland
- b. Miosis
- c. Dilated iris of the eye
- d. Dilated bronchi

6. During the absolute refractory period of a spike P in an axon:

- a. Another AP can be elicited by a subthreshold stimulus.
- b. Another AP can be produced by a threshold stimulus.
- c. Another AP may be initiated by a stimulus bigger than threshold.
- d. It is impossible to initiate another AP even if we give a suprathreshold stimulus.

7. The nerve action potential:

- a. Start with opening of the voltage-gated Na channels.
- b. Start by potassium influx into the cell.
- c. Depends to a great extent on the state of chloride channels.
- d. Start with hyperpolarization of the cell-membrane.

8. Reestablishment (restoration) of Na^+/K^+ distribution after action potential is mainly due to:

- a. Na^+ outflux
- b. K^+ outflux
- c. Na^+/K^+ pump
- d. Na^+/K^+ leak channel

9. Stimulation of Sympathetic to pelvis causes:

- a. Micturition.
- b. Defecation.
- c. Ejaculation.
- d. Erection.

10. Which one of the following effects is not produced by Sympathetic stimulation?

- a. Bronchodilatation.
- b. Tachycardia.
- c. Mydriasis.
- d. Contraction of urinary bladder wall.



11. Sympathetic to thorax causes:

- Secretion of the adrenomedullary hormones.
- Dilatation of the coronary vessels.
- Contraction of the splenic capsule.
- Narrowing of respiratory passages.

12. which one of the following is wrong about adrenal medulla?

- It is sympathetic ganglion
- Secretes more noradrenaline than adrenaline
- It releases epinephrine into circulating blood
- it's innervated by sympathetic cholinergic fibres

13. Parasympathetic tone is mostly responsible for:

- Decreasing the blood flow to the gut (GIT or alimentary canal).
- Maintenance of the systemic arterial blood pressure constant.
- Keeping the fasting blood glucose level constant.
- Maintaining resting heart rate within normal.

14. The resting potential of a nerve membrane is primarily dependent on the concentration gradient of:

- Potassium.
- Sodium.
- Calcium.
- Chloride.

15. Na⁺-K⁺ pump:

- Requires energy which is taken by GTP.
- Couples Na⁺ and K⁺ with ratio 3:2.
- Helps intracellular accumulation of Na⁺.
- Is a passive process needs no energy.

16. Which of the following is actively transported out of the neurons?

- K⁺.
- HPO₄²⁻.
- Cl⁻.
- Na⁺

1:b 2:c 3:d 4:c 5:b 6:d 7:a 8:c 9:c 10:d 11:b 12:b 13:d 14:a 15:b 16:d

Short notes (20 marks)

- List five factors affecting the rate of diffusion? (2 marks)
- Enumerate Importance of Na⁺-K⁺ pump? (2 marks)
- Conduction in myelinated nerve fibers: (2 marks)
- Components of autonomic reflex arc? (2 marks)
- Explain the underlying mechanism of the following? (2 marks)
- Mention the effect of the following on nerve excitability? (2 marks)
- Orbelli phenomenon: (2 marks)
- Functions of Sympathetic to the thorax? (2 marks)
- Denervation hypersensitivity: (2 marks)
- Interaction between sympathetic & parasympathetic system. (2 marks)



17. End-module 2025 اهليه

1. As regard resting membrane potential, which of the followings are correct?

- a. It is the potential difference across the fibre membrane with negatively charged ions inside and positively charged ions outside.
- b. Its decrease is accompanied with decreased excitability of nerve fibre.
- c. It is due to K^+ diffusion from outside to inside the nerve fibres.
- d. It is due to diffusion of intracellular protein to outside the nerve fibre.

2. Which of the following is actively transported out of the neurones?

- a. K^+ .
- b. Cl^- .
- c. HPO_4^{--}
- d. Na^+ .

3. Which is correct about resting membrane potential of a nerve fibre?

- a. Is mainly due to sodium influx.
- b. Is mainly due to potassium influx.
- c. Is mainly due to chloride influx.
- d. Is due to the presence of negative charges on the inner surface in relation to its outer surface.

4. When a nerve fibre is stimulated, depolarization phase stops at +35 mV. This is because:

- a. Activation of the Na^+-K^+ pump that prevent accumulation of Na^+ ions inside the nerve membrane.
- b. Closing of the inactivation gates is faster the opening of activation gates.
- c. The inactivation gates rapidly close to prevent further entry of Na^+ ions.
- d. The activation gates and inactivation gates close at the same time.

5. In a nerve axon, which phase of the action potential is caused by the inactivation of the Na^+ channels?

- a. Upstroke.
- b. Absolute refractory period.
- c. Down-stroke.
- d. Relative refractory period

6. What is the first step in nerve impulse initiation is?

- a. Decreased Na^+ permeability.
- b. Membrane depolarization.
- c. Increased K^+ permeability.
- d. Membrane hyperpolarization.

7. Which of the followings is characteristic of an action potential?

- a. Maximal response to threshold stimulus.
- b. Slight response to threshold stimulus.
- c. No response to threshold stimulus.
- d. Increased response to increased stimulus.

8. Regarding the voltage gated Na^+ channels, which of the followings is correct?

- a. It has two gates, activation gate near the inside of the membrane and inactivation gate near the outside.
- b. At resting state, the activation gates are closed to prevent influx of Na^+ ions inside the nerve fibre.



- c. At resting state, the inactivation gates are also closed.
- d. At the threshold (firing) level, the inactivation gates open.

9. The membrane potential will depolarize by the greatest amount if the membrane permeability increases for:

- a. Potassium.
- b. Potassium & chloride.
- c. Sodium & potassium.
- d. Sodium.

10. The ascending limb of spike of nerve fibre action potential is caused by:

- a. Rapid sodium influx.
- b. Closure of activation gates of voltage gated sodium channels.
- c. Rapid potassium efflux.
- d. Calcium influx.

11. The resting membrane potential of a typical mammalian neuron is approximately:

- a. +90 mV
- b. -70 mV
- c. +30 mV
- d. 0 mV

12. The sodium-potassium pump (Na^+/K^+ ATPase) primarily contributes to the RMP by:

- a. Directly generating the negative potential.
- b. Pumping three K^+ ions in for every two Na^+ ions out.
- c. Pumping three Na^+ ions out for every two K^+ ions in.
- d. Creating equilibrium potential for sodium.

13. At the potassium equilibrium potential (E_K), which of the following is true?

- a. The chemical gradient for K^+ is zero.
- b. The electrical gradient for K^+ is zero.
- c. The net movement of K^+ across the membrane is zero.
- d. The membrane potential is 0 mV.

14. If the extracellular concentration of K^+ increases significantly, the resting membrane potential will:

- a. Become more negative (hyperpolarize).
- b. Become less negative (depolarize).
- c. Remain unchanged.
- d. Oscillate rapidly.

15. The negative charge inside the cell at rest is primarily due to:

- a. An abundance of intracellular sodium ions.
- b. Large, negatively charged intracellular proteins and anions.
- c. The continuous influx of chloride ions.
- d. The positive charge of the nucleus.



16. The threshold potential is defined as the point at which:

- a. The action potential reaches its peak.
- b. Voltage-gated K^+ channels begin to open.
- c. Voltage-gated Na^+ channels open and an all-or-none response is initiated.
- d. The membrane is at its most negative value.

17. The rapid depolarization phase of the action potential is caused by the:

- a. Rapid closure of K^+ channels.
- b. Rapid influx of Na^+ ions through voltage-gated channels.
- c. Efflux of K^+ ions.
- d. Activation of the sodium-potassium pump.

18. The "all-or-none" principle of the action potential states that:

- a. All stimuli, regardless of strength, will generate an action potential.
- b. Once the threshold is reached, the AP is generated to its full amplitude and cannot be larger.
- c. An action potential will either travel the entire length of the axon or not at all.
- d. All neurons in a nerve fire simultaneously.

19. The hyperpolarization (after-hyperpolarization) seen after an action potential is due to:

- a. Na^+ channels remaining open.
- b. Voltage-gated K^+ channels remaining open, allowing K^+ to continue to leave the cell.
- c. The sodium-potassium pump becoming hyperactive
- d. An influx of Cl^- ions.

20. A stimulus that causes the membrane potential to become more negative than the RMP is called:

- a. Depolarization.
- b. Hyperpolarization.
- c. Repolarization.
- d. Threshold.

21. If you block voltage-gated potassium channels with Tetraethyl ammonium (TEA), what is the most likely effect on the action potential?

- a. The depolarization phase will not occur.
- b. The repolarization phase will be significantly slowed or absent.
- c. The threshold will be much more positive.
- d. The axon will become hyper excitable and fire spontaneously.

22. The initial segment of the axon (axon hillock) is often the site of action potential initiation because it:

- a. Has the lowest density of voltage-gated Na^+ channels.
- b. Has the highest density of voltage-gated Na^+ channels, giving it the lowest threshold.
- c. Is the farthest point from the synaptic inputs.
- d. Is covered in myelin.



23. The term "overshoot" in an action potential refers to the part of the curve where:

- a. The membrane potential is more negative than the RMP.
- b. The membrane potential is positive (above 0 mV).
- c. The membrane is at the potassium equilibrium potential.
- d. The membrane is repolarizing.

24. The stability of the resting membrane potential is most directly dependent on the continuous activity of:

- a. Voltage-gated Na⁺ channels.
- b. The Na⁺/K⁺ ATPase pump.
- c. Ligand-gated channels.
- d. Leak K⁺ channels.

25. Diffusion of oxygen across the cell membrane occurs by:

- a. Simple diffusion
- b. Facilitated diffusion
- c. Active transport
- d. Endocytosis

26. Facilitated diffusion differs from simple diffusion by:

- a. Requiring energy
- b. Using carrier proteins
- c. Moving against gradient
- d. Transporting only water

27. Secondary active transport uses energy from:

- a. ATP hydrolysis directly
- b. Electrochemical gradient of ions
- c. Vesicle formation
- d. Osmotic pressure

28. Which of the following is NOT a sympathetic effect?

- a. Increase secretion of sweat glands
- b. watery secretion of salivary glands
- c. Relaxes of the ciliary muscle of the eye
- d. Contracts of the erector pili muscles

29. Which of the following effects occur due to sympathetic innervation?

- a. Release of epinephrine from adrenal medulla
- b. Increase in digestive tract motility
- c. Decrease in strength of heart contractions
- d. Increases strength of contractions in uterine wall



30. What is the term for a reflex that does not include a CNS component?

- a. long reflex
- b. visceral reflex
- c. somatic reflex
- d. local axon reflex

31. Sympathetic division stimulation causes._____

- a. decreased blood glucose, increased GI peristalsis, and increased heart rate and blood pressure
- b. increased blood glucose, increased GI peristalsis, and decreased heart rate and blood pressure
- c. increased blood glucose, decreased GI peristalsis, and increased heart rate and blood pressure
- d. decreased blood glucose, increased GI peristalsis, and decreased heart rate and

32. Generalized sympathetic stimulation causes all the following except:

- a. Increases the heart rate
- b. Relaxes bronchial smooth muscles
- c. Tightens GIT sphincters
- d. Constriction of pupil

33. Cell bodies of preganglionic neurons of the sympathetic nervous system are found in:

- a. The dorsal root ganglion
- b. The sympathetic chain
- c. The dorsal horn cells of the spinal cord
- d. The lateral horn cells of the spinal cord

34. Regarding the sympathetic nervous system:

- a. Preganglionic fibers release noradrenalin
- b. Postganglionic fibers release noradrenalin
- c. Release of acetylcholine by the pre-ganglionic fibers
- d. Has longer pre-ganglionic fibers than post-ganglionic fibers

35. The sympathetic nerves supply:

- a. The external sphincters of the gastrointestinal & urinary tracts
- b. Smooth muscle of the constrictor pupillae
- c. Smooth muscle of the bronchioles
- d. Vasodilator fibers to skin vessels

36. Generalized sympathetic activity is characterized by....?

- a. Contraction of the radial muscle in the iris
- b. Increased urinary excretion of catecholamines
- c. Relaxation of sphincteric smooth muscle in the alimentary tract
- d. Decreased conduction rate in the atrio-ventricular bundle

37. Targets of the autonomic nervous system include all of the following EXCEPT

- a. Cardiac muscle
- b. Skeletal muscle
- c. Exocrine glands
- d. smooth muscle



38. Somatic motor neurons have cell bodies located ____ the CNS that project axons only to __; and are usually under __control.

- a. Outside; skeletal muscle; involuntary
- b. Inside; the viscera; voluntary
- c. Inside; the viscera; involuntary
- d. Inside; skeletal muscle; voluntary

39. Activation of the sympathetic nervous system leads to the 'fight and flight' response. Which of these is not part of that process?

- a. Vasodilatation in skeletal muscle
- b. Sweating
- c. Bladder relaxation
- d. Increased gut motility

40. Sympathetic supply to the pelvic viscera cause:

- a. Micturition.
- b. Dilatation of vessels in external genitalia.
- c. Defecation.
- d. Ejaculation.

41. Postganglionic sympathetic starting from cervical and upper four thoracic ganglia causes:

- a. Secretion of the adrenomedullary hormones.
- b. Dilatation of the coronary vessels.
- c. Contraction of the splenic capsule.
- d. Narrowing of the respiratory passages.

42. A lesion in the cervical symp. chain (Horner's syndrome) causes :

- a. Increased sweating.
- b. Drooping of the upper eyelid (=ptosis).
- c. Vasoconstriction of the skin blood vessels (leading to pallor).
- d. Mydriasis (= dilatation of the pupil).

43. Stimulation of the greater splanchnic nerve produces:

- a. Increased motility of the stomach & upper half of the small intestine.
- b. Inhibition of micturition.
- c. Increased blood level of both glucose & catecholamines.
- d. V.C. of the blood vessels in the erectile tissue of both males & females.

44. During a "fight-or-flight" response, how does the sympathetic nervous system primarily alter blood flow distribution?

- a. Increases flow to the skin and digestive organs.
- b. Increases flow to the skeletal muscles and heart while reducing flow to the skin and digestive tract.
- c. Directs all available blood to the brain.
- d. Prioritizes flow to the kidneys to maintain filtration.



45. What is the overall metabolic goal of sympathetic activation in the liver and adipose tissue?

- a. To promote energy storage as glycogen and triglycerides.
- b. To increase the availability of fuel in the bloodstream (glucose and fatty acids).
- c. To reduce metabolic rate to conserve energy for a long-term threat.
- d. To stimulate the synthesis of proteins for tissue repair.

46. If the sympathetic nervous system is globally activated, what is its net effect on the processes of digestion and absorption?

- a. It strongly stimulates motility and secretion to process food for energy.
- b. It has no significant effect on the digestive system.
- c. It inhibits motility and secretion, delaying digestive processes.
- d. It specifically increases large intestine activity for water conservation.

47. How does sympathetic activation directly support intense physical activity?

- a. By constricting the airways to prevent inhalation of pollutants.
- b. By dilating the airways to maximize air flow.
- c. By stimulating mucus production in the respiratory tract.
- d. By suppressing the cough reflex.

48. During a stressful event, how does the sympathetic nervous system affect the sweat glands, and what is the functional purpose?

- a. It inhibits sweating to conserve body water.
- b. It stimulates sweating to dissipate heat generated by increased metabolism.
- c. It produces an oily secretion to protect the skin.
- d. It has no functional control over sweat glands.

49. What is the net effect of sympathetic stimulation on the urinary system?

- a. It promotes urination by contracting the bladder.
- b. It promotes urination by relaxing the sphincters.
- c. It inhibits urination by relaxing the bladder and contracting sphincters.
- d. It has no direct influence on urinary function.

50. The "adrenaline rush" from the adrenal medulla is best described as:

- a. A localized, targeted response to a specific organ.
- b. A redundant backup for parasympathetic nervous system failure.
- c. A hormone-amplified version of the sympathetic neural signal, affecting the entire body.
- d. A mechanism primarily for stimulating digestion under stress.

51. From a systems physiology perspective, what is the ultimate purpose of the sympathetic "fight-or-flight" response?

- a. To maintain long-term homeostasis by promoting growth and restoration.
- b. To facilitate the absorption of nutrients from a recent meal.
- c. To enable the body to expend energy immediately to survive a perceived threat.



d. To lower the core body temperature and metabolic rate.

52. How does sympathetic activation alter mental alertness and sensory processing?

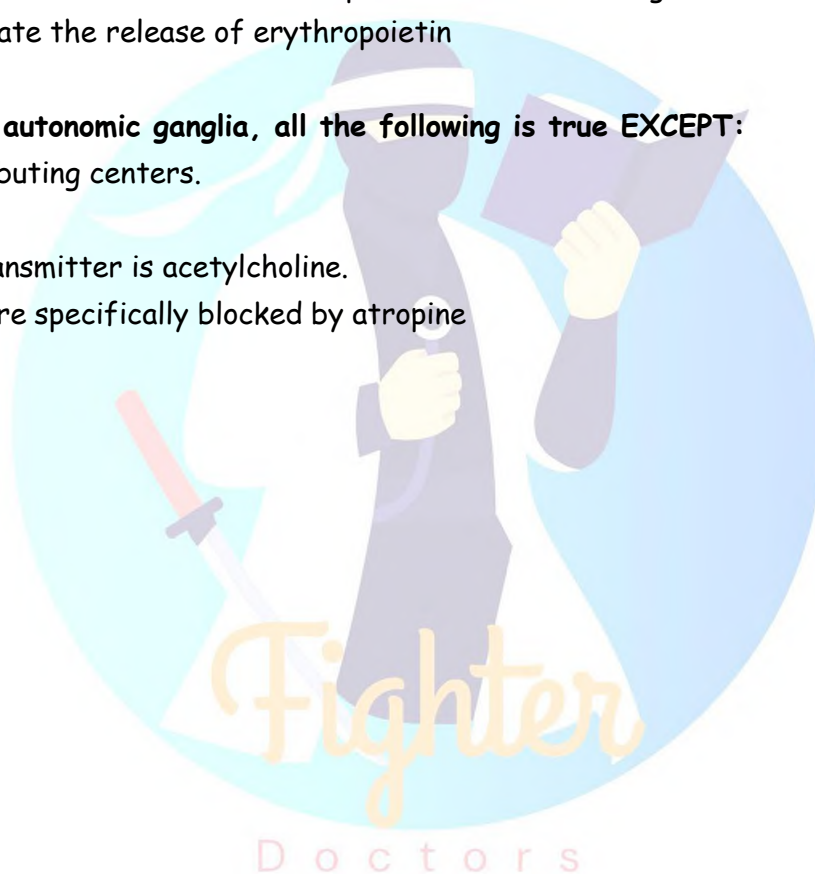
- a. It induces drowsiness and focuses attention inward.
- b. It increases alertness and can sharpen sensory acuity for external threats.
- c. It primarily impairs short-term memory formation.
- d. It has no known effect on central nervous system function.

53. During sympathetic activation, what is the primary goal of redirecting blood flow away from the kidneys?

- a. To increase urine production and flush out toxins.
- b. To protect the kidneys from high blood pressure.
- c. To conserve water and redirect cardiac output to more critical organs like muscles.
- d. To directly stimulate the release of erythropoietin

54. Concerning the autonomic ganglia, all the following is true EXCEPT:

- a. They act as distributing centers.
- b. They are 3 types.
- c. Their chemical transmitter is acetylcholine.
- d. Their receptors are specifically blocked by atropine



1:a 2:d 3:d 4:c 5:c 6:b 7:a 8:b 9:d 10:a
11:b 12:c 13:c 14:b 15:b 16:c 17:b 18:b 19:b 20:b
21:b 22:b 23:b 24:d 25:a 26:b 27:b 28:b 29:a 30:d
31:c 32:d 33:d 34:c 35:c 36:a 37:b 38:d 39:d 40:d
41:b 42:b 43:c 44:b 45:b 46:c 47:b 48:b 49:c 50:c
51:c 52:b 53:c 54:d



18. End semester exam KSU 2026

1. Which is an example of negative feedback?

- a. Blood clotting
- b. Parturition
- c. Temperature regulation
- d. Action potential depolarization

2. Which is true about homeostasis?

- a. Maintaining constant external environment
- b. Maintaining stable internal environment
- c. Maintaining equal intracellular and extracellular fluid
- d. Preventing any change

3. Stimulation of sympathetic to pelvis causes:

- a. Micturition
- b. Defecation
- c. Ejaculation
- d. Erection

4. Which one of the following effects is not produced by sympathetic stimulation?

- a. Bronchodilatation
- b. Tachycardia
- c. Mydriasis
- d. Contraction of urinary bladder wall

5. Sympathetic to thorax causes:

- a. Secretion of the adrenomedullary hormones
- b. Dilatation of coronary vessels
- c. Contraction of the splenic capsule
- d. Narrowing of respiratory passages

6. Which one of the following is wrong about adrenal medulla?

- a. It is sympathetic ganglion
- b. Secretes more noradrenaline than adrenaline
- c. It releases epinephrine into circulating blood
- d. It is innervated by sympathetic cholinergic fibers

7. Parasympathetic tone is mostly responsible for:

- a. Decreasing the blood flow to the gut (GIT or alimentary canal)
- b. Maintenance of the systemic arterial blood pressure constant
- c. Keeping the fasting blood glucose level constant
- d. Maintaining resting heart rate within normal

8. The resting potential of a nerve membrane is primarily dependent on:

- a. Potassium
- b. Sodium
- c. Calcium
- d. Chloride



9. Na^+/K^+ pump:

- a. Requires energy which is taken by GTP
- b. Uses Na^+ and K^+ with ratio 3:2
- c. Is a passive process needs no energy
- d. Helps intracellular accumulation of K^+

10. Which of the following is actively transported out of the neuron?

- a. Na^+
- b. K^+
- c. Cl^-
- d. Ca^{2+}

11. Preganglionic neuron (autonomic nervous system) are located in:

- a. Autonomic ganglia
- b. Lateral horn cells
- c. Blood vessels
- d. Lower motor neurons

12. The parasympathetic division is activated during:

- a. Exercise
- b. Fear
- c. Excitement
- d. Rest and digest

13. Preganglionic fibers from cranial outflow are carried by:

- a. 3rd, 5th, 11th, 8th cranial nerves
- b. 3rd, 7th, 9th, 10th cranial nerves
- c. 5th, 7th, 10th cranial nerves
- d. 3rd, 7th, 9th cranial nerves

14. Preganglionic fibers of the sacral outflow are carried by:

- a. Ciliary ganglion
- b. Submandibular ganglion
- c. Pelvic nerve
- d. Peripheral ganglion

1:c 2:b 3:c 4:d 5:b 6:b 7:d 8:a 9:d 10:a 11:b 12:d 13:b 14:c

Short notes (22 marks)

1. Denervation hypersensitivity (2.5 marks) Def, Examples, Mechanism

2. Cholinergic receptors (4 marks)

Muscarinic

- Sites
- Parasympathetic postganglionic
- Sympathetic (sweat glands)
- Effects on smooth muscle, glands, blood vessels

Nicotinic

- Autonomic ganglia
- Neuromuscular junction
- Adrenal medulla